# CHAPTER 6

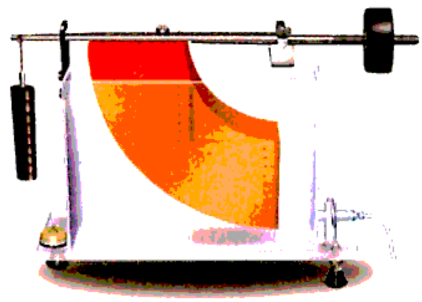
# ADDITIONAL APPLICATIONS OF THE INTEGRAL

*The only way to learn mathematics is to do mathematics.*

Paul Halmos
*Hilbert Space Problem Book*

## PREVIEW

In Chapter 5, we found that area and average value can be expressed in terms of the definite integral. The goal of this chapter is to consider various other applications of integration such as computing volume, arc length, surface area, work, hydrostatic force, centroids of planar regions, and applications to business, economics, and life sciences.




**Apparatus for measuring hydrostatic pressure**

## PERSPECTIVE

What is the volume of the material that remains when a hole of radius $0.5R$ is drilled into a sphere of radius $R$? The "center" of a rectangle is the point where its diagonals intersect, but what is the center of a quarter circle? If a reservoir is filled to the top of a dam shaped like a parabola, what is the total force of the water on the face of the dam? If a worker is carrying a bag of sand up a ladder and the bag is leaking sand through a hole in such a way that all the sand is gone when the worker reaches the top, how much work is done by the worker? An investor asks a question seeking the future value of an income flow, or a physician asks how to measure the flow of blood through an artery. These questions are typical of the kind we shall answer in this chapter.

## CONTENTS

# 6.1 AREA BETWEEN TWO CURVES

**IN THIS SECTION:** *Area between curves, area using vertical strips, area using horizontal strips*
We have seen how the area between the curves defined by $y = f(x)$ and the $x$-axis can be computed by the integral $\int_a^b f(x)\,dx$ on an interval $a \le x \le b$ where $f(x) \ge 0$. In this section, we find that integration can be used to find the area of more general regions between curves.

## Area Between Curves

In some practical problems, you may have to compute the area between two curves. Suppose $f$ and $g$ are functions such that $f(x) \ge g(x)$ on the interval $[a, b]$, as shown in Figure 6.1. We do not insist that both $f$ and $g$ be nonnegative functions, but we begin by showing that case in Figure 6.1.

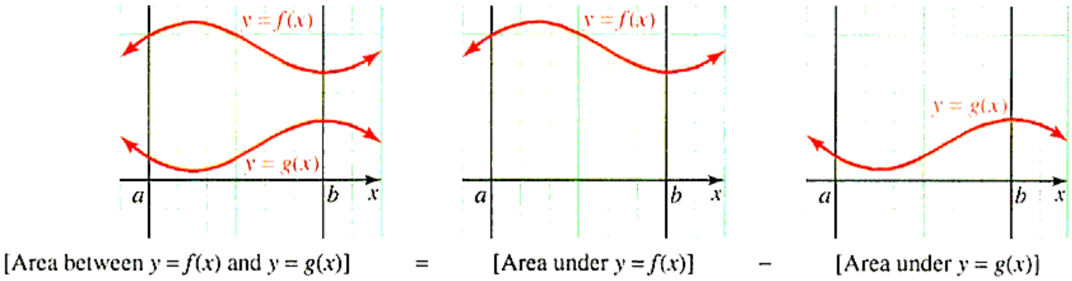



**Figure 6.1** Area between two curves

To find the area of the region $R$ between the curves defined by $y = f(x)$ and $y = g(x)$ from $x = a$ to $x = b$, we subtract the area between the lower curve $y = g(x)$ and the $x$-axis from the area between the upper curve $y = f(x)$ and the $x$-axis; that is,

$$\text{AREA OF } R = \int_a^b f(x)\,dx - \int_a^b g(x)\,dx = \int_a^b \left[f(x) - g(x)\right] dx$$

This formula seems obvious in the situation where both $f(x) \ge 0$ and $g(x) \ge 0$, as shown in Figure 6.1. However, the following derivation requires only that $f$ and $g$ be continuous and satisfy $f(x) \ge g(x)$ on the interval $[a, b]$. We wish to find the area between the curves $y = f(x)$ and $y = g(x)$ on this interval. Choose a partition

$$\{x_0 = a, x_1, x_2, \cdots, x_n = b\}$$

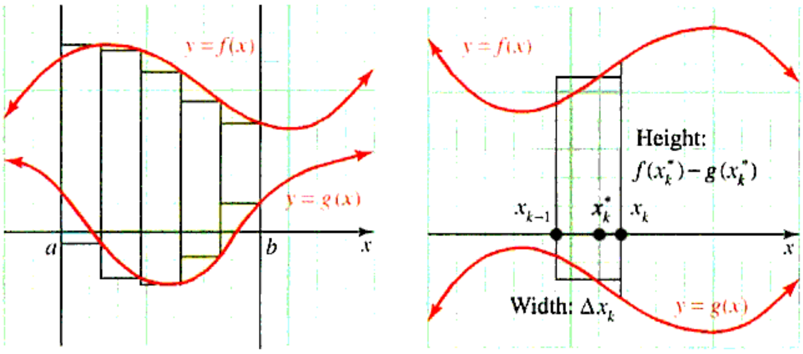
of the interval $[a, b]$ and a representative number $x_k^*$ from each subinterval $\left[x_{k-1}, x_k\right]$. Next, for each index $k$, with $k = 1, 2, \cdots, n$, construct a rectangle of width

$$\Delta x_k = x_k - x_{k-1}$$

and height

$$f(x_k^*) - g(x_k^*)$$

equal to the vertical distance between the two curves at $x = x_k^*$. A typical approximating rectangle is shown in Figure 6.2. We refer to this approximating rectangle as a **vertical strip**.




**a.** Approximating rectangles **b.** A typical vertical strip

**Figure 6.2** Using Riemann sums to find the area between two curves

The representative rectangle has area

$$\Delta A_k = \left[f(x_k^*) - g(x_k^*)\right] \Delta x_k$$

and the total area between the curves $y = f(x)$ and $y = g(x)$ can be estimated by the Riemann sum

$$A_n = \sum_{k=1}^{n} \left[f(x_k^*) - g(x_k^*)\right] \Delta x_k$$

It is reasonable to expect this estimate to improve if we increase the number of subdivision points in the partition $P$ in such a way that the norm $\|P\|$ approaches 0. Thus, the region between the two curves has area

$$A = \lim_{\|P\| \to 0} \sum_{k=1}^{n} \left[f(x_k^*) - g(x_k^*)\right] \Delta x_k$$

which we recognize as the integral of the function $f(x) - g(x)$ over the interval $[a, b]$. These observations may be used to define the area between the curves.

**AREA BETWEEN TWO CURVES** If $f$ and $g$ are continuous and satisfy $f(x) \geq g(x)$ on the closed interval $[a, b]$, then the **area between the two curves** $y = f(x)$ and $y = g(x)$ is given by

$$A = \int_a^b \left[f(x) - g(x)\right] dx$$

What this says To find the area between two curves on a given closed interval $[a, b]$ use the formula

$$A = \int_a^b [\text{TOP CURVE} - \text{BOTTOM CURVE}]\, dx$$

It is no longer necessary to require either curve to be above the $x$-axis. In fact, we will see later in this section that the curves might cross somewhere in the domain so that one curve is on top for part of the interval and the other curve is on top for the rest.

**Example 1 Area between two curves**

Find the area of the region between the curves $y = x^3$ and $y = x^2 - x$ on the interval $[0, 1]$.

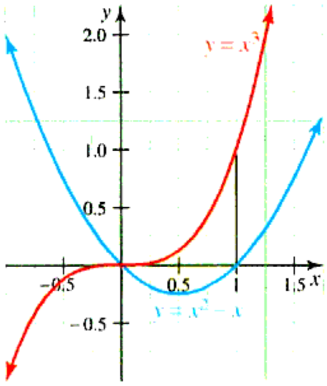



**Figure 6.3** Area between curves

***Solution*** The region is shown in Figure 6.3.
We need to know which curve is the *top curve* on $[0, 1]$. Solve

$$x^3 = x^2 - x \qquad \text{or} \qquad x(x^2 - x + 1) = 0$$

to find the points of intersection. The only real root is

$$x = 0$$

($x^2 - x + 1 = 0$ has no real roots). Thus, by the intermediate value theorem, the same curve is on top throughout the interval $[0, 1]$. To see which curve is on top, take some representative value in $[0, 1]$, such as $x = 0.5$, and note that because $(0.5)^3 > 0.5^2 - 0.5$, the curve $y = x^3$ must be above $y = x^2 - x$. Thus, the required area is given by

$$A = \int_0^1 \Big[\underset{\substack{\uparrow \\ \text{Top curve}}}{x^3} - \underbrace{(x^2 - x)}_{\text{Bottom curve}}\Big]\,dx = \left[\frac{1}{4}x^4 - \frac{1}{3}x^3 + \frac{1}{2}x^2\right]_0^1 = \frac{5}{12}$$

In Section 5.2, we defined area for a continuous function $f$ with the restriction that $f(x) \geq 0$. Example 1 shows us that when considering the area between two curves, we no longer need to be concerned with the nonnegative restriction but only with whether $f(x) \geq g(x)$. We now use this idea to find the area for a function whose graph is below the $x$-axis.

**Example 2 Area with a function whose graph is below the $x$-axis**

Find the area of the region bounded by the curve $y = e^{2x} - 3e^x + 2$ and the $x$-axis.

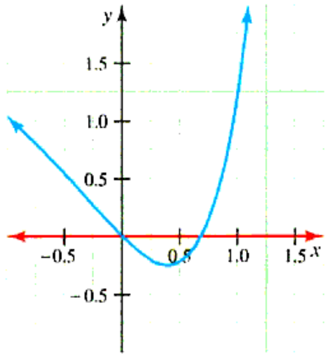


**Figure 6.4** Graph of $f$ with area shaded

***Solution*** We find the points of intersection of the curve and the $x$-axis:

$$\begin{aligned} e^{2x} - 3e^x + 2 &= 0 && \textit{The curve intersects the x-axis where } y = 0. \\ (e^x - 1)(e^x - 2) &= 0 && \textit{Factor} \\ x &= 0, \ln 2 && \textit{Solve } e^x = 1 \textit{ and } e^x = 2. \end{aligned}$$

The graph of $f(x) = e^{2x} - 3e^x + 2$ is shown in Figure 6.4. Notice the graph is below the $x$-axis.

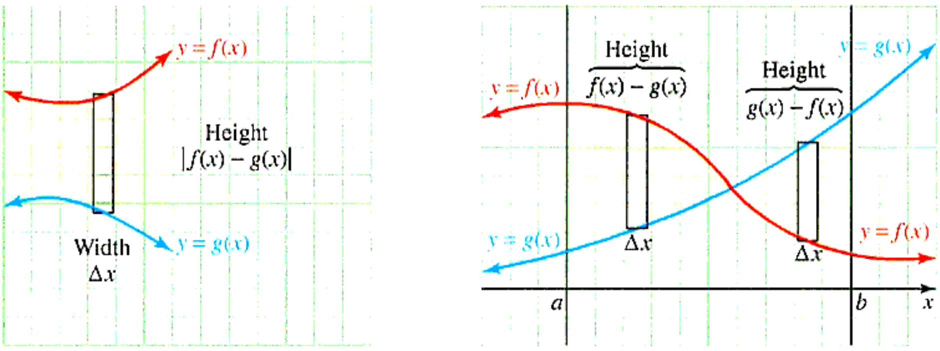
We see that on the interval $[0, \ln 2]$, $f(x) \leq 0$, but we can find the area of the given region by considering the area between the curves defined by equations $y = 0$ and $y = e^{2x} - 3e^x + 2$.

$$\begin{aligned} A &= \int_0^{\ln 2} [\underbrace{0}_{\text{Top curve}} - \underbrace{(e^{2x} - 3e^x + 2)}_{\text{Bottom curve}}]\,dx \\ &= \left[-\frac{e^{2x}}{2} + 3e^x - 2x\right]_0^{\ln 2} \\ &= \frac{3}{2} - 2\ln 2 \end{aligned}$$

The approximate value of this area can be found using a calculator: $A \approx 0.1137$.

## Area Using Vertical Strips

Although the only mathematically correct way to establish a formula involving integrals is to form Riemann sums and take the limit according to the definition of the definite integral, we can simulate this procedure with approximating strips. This simplification is especially useful for finding the area of a complicated region formed when two curves intersect one or more times, as shown in Figure 6.5.




**a.** A typical vertical strip **b.** Approximation by strips

**Figure 6.5** Area by vertical strips

Note that the vertical strip has height $f(x) - g(x)$ if $y = f(x)$ is above $y = g(x)$, and height $g(x) - f(x)$ if $y = g(x)$ is above $y = f(x)$. In either case, the height can be represented by $|f(x) - g(x)|$, and the area of the vertical strip is

$$\Delta A = |f(x) - g(x)|\,\Delta x = |f(x) - g(x)|\,dx$$

Thus, we have a new integration formula for area; namely,

$$A = \int_a^b |f(x) - g(x)|\,dx$$

*You cannot use the formula $A = \int [f - g]\,dx$ directly here because the hypothesis $f \geq g$ is not satisfied. In order to use $A = \int |f - g|\,dx$ over the entire interval, you must remember that $|f - g|$ might be $f - g$ over part of the interval and $g - f$ over another part (see Figure 6.5b). Make sure you check to see which curve is on top. Because the curves cross in Example 2, the interval must be subdivided accordingly.*

**Example 3 Area using vertical strips**

Find the area of the region bounded by the line $y = 3x$ and the curve $y = x^3 + 2x^2$.

***Solution*** The region between the curve and the line is the shaded portion of Figure 6.6.

Part of the process of graphing these curves is to find which is the top curve and which is the bottom. To do this we need to find where the curves intersect:

$$x^3 + 2x^2 = 3x$$
$$x^3 + 2x^2 - 3x = 0$$
$$x(x+3)(x-1) = 0$$

The points of intersection occur at $x = -3, 0$, and $1$. In the subinterval $[-3, 0]$, labeled $A_1$ in Figure 6.6, the curve $y = x^3 + 2x^2$ is on top (test a typical point in the subinterval, such as $x = -1$), and on $[0, 1]$, the region labeled $A_2$, the curve $y = 3x$ is on top. The representative vertical strips are shown in Figure 6.6, and the area between the curve and the line is given by the sum

$$A = \int_{-3}^{0} [(x^3 + 2x^2) - (3x)]\,dx + \int_0^1 [(3x) - (x^3 + 2x^2)]\,dx$$
$$= \left[\frac{1}{4}x^4 + \frac{2}{3}x^3 - \frac{3}{2}x^2\right]_{-3}^{0} + \left[\frac{3}{2}x^2 - \frac{1}{4}x^4 - \frac{2}{3}x^3\right]_0^1$$

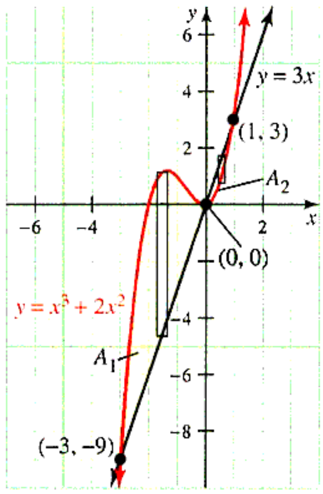



**Figure 6.6 Interactive** Area between curves

$$= 0 - \left(\frac{81}{4} - \frac{54}{3} - \frac{27}{2}\right) + \left(\frac{3}{2} - \frac{1}{4} - \frac{2}{3}\right) - 0$$
$$= \frac{71}{6}$$

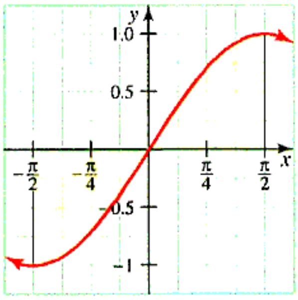



**Figure 6.7** Area of a symmetric region

*A common mistake is to work this problem by evaluating the integral*

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin x\, dx = (-\cos x)\Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$$
$$= -0 + 0$$
$$= 0$$

*forgetting that the integral does not give the area unless the function is nonnegative.*

**Example 4 Area using symmetry**

Find the area bounded by the curve $y = \sin x$ and the $x$-axis between $x = -\frac{\pi}{2}$ and $x = \frac{\pi}{2}$.

***Solution*** We sketch the region as shown in Figure 6.7.

The area we seek is found by

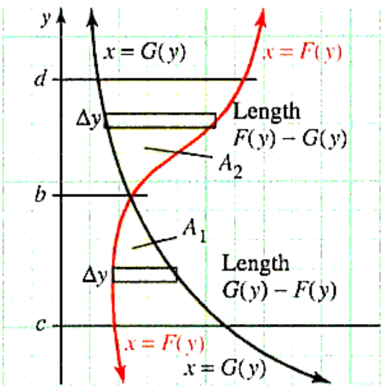
$$\int_{-\frac{\pi}{2}}^{0} (0 - \sin x)\, dx + \int_{0}^{\frac{\pi}{2}} (\sin x - 0)\, dx = \cos x\Big|_{-\frac{\pi}{2}}^{0} + (-\cos x)\Big|_{0}^{\frac{\pi}{2}}$$
$$= (1 - 0) + (0 + 1)$$
$$= 2$$

There is an easier procedure using symmetry of the curve $y = \sin x$ with respect to the origin. Symmetry uses the fact that the area bounded by the lines $x = -\frac{\pi}{2}$ and $x = 0$ is the same as the area bounded by the lines $x = 0$ and $x = \frac{\pi}{2}$. As a result, we often work this type of problem as shown here:

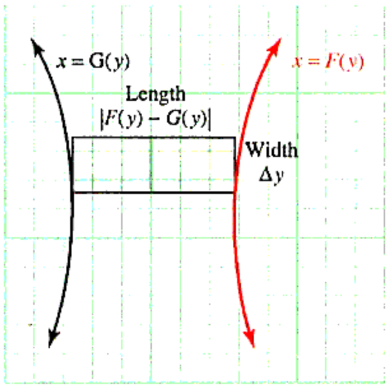
$$A = 2\int_{0}^{\frac{\pi}{2}} (\sin x - 0)\, dx \qquad \textit{By symmetry}$$
$$= 2(-\cos x)\Big|_{0}^{\frac{\pi}{2}}$$
$$= 2(0 + 1)$$
$$= 2$$

## Area Using Horizontal Strips

For many regions, it is easier to form horizontal strips rather than vertical strips. The procedure for horizontal strips is the same as the procedure for vertical strips. If we want to find the area between two curves of the form $x = F(y)$ and $x = G(y)$ on the interval $[c, d]$ on the $y$-axis, we form horizontal strips. In Figure 6.8, such a region and its typical horizontal approximating rectangles, or **horizontal strips,** are illustrated. Note that the width of the horizontal strip is denoted by $\Delta y$.




**a.** Approximation by horizontal strips of width $\Delta y$




**b.** A typical horizontal strip

**Figure 6.8** Area by horizontal strips

Regardless of which curve is "ahead" or "behind," the horizontal strip has length $|F(y) - G(y)|$ and area

$$\Delta A = |F(y) - G(y)|\,\Delta y$$

However, in practice, you must make sure to find the points of intersection of the curves and divide the integrals so that in each region one curve is the *leading curve* ("right curve") and the other is the *trailing curve* ("left curve"). Suppose the curves intersect where $y = b$ for $b$ on the interval $[c, d]$, as shown in Figure 6.8. Then

$$A = \int_c^b \underbrace{[G(y) - F(y)]}_{G \text{ leading } F}\,dy + \int_b^d \underbrace{[F(y) - G(y)]}_{F \text{ leading } G}\,dy$$

**Example 5 Area using horizontal strips**

Find the area of the region, $R$, between the parabola $x = 4y - y^2$ and the line $x = 2y - 3$.

***Solution*** Figure 6.9 shows the region between the parabola and the line, together with a typical horizontal strip.

To find where the line and the parabola intersect, solve

$$\begin{aligned} 4y - y^2 &= 2y - 3 \\ y^2 - 2y - 3 &= 0 \\ (y+1)(y-3) &= 0 \\ y &= -1, 3 \end{aligned}$$

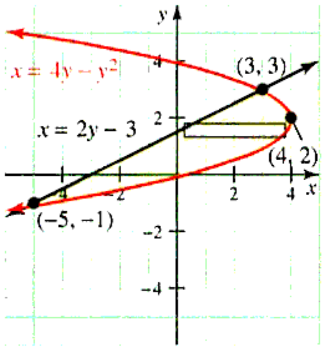



**Figure 6.9 Interactive** Area of $R$ using horizontal strips

Throughout the interval $[-1, 3]$ the parabola is to the right of the line (test a typical point between $-1$ and 3, such as $y = 0$). Thus, the horizontal strip has area

$$\Delta A = [\underbrace{(4y - y^2)}_{\text{Right curve}} - \underbrace{(2y - 3)}_{\text{Left curve}}]\,\Delta y$$

and the area between the parabola and the line is given by

$$\begin{aligned} A &= \int_{-1}^{3} \left[(4y - y^2) - (2y - 3)\right] dy \\ &= \int_{-1}^{3} (3 + 2y - y^2)\,dy \\ &= \left[3y + y^2 - \frac{1}{3}y^3\right]_{-1}^{3} \\ &= (9 + 9 - 9) - \left(-3 + 1 + \frac{1}{3}\right) \\ &= \frac{32}{3} \end{aligned}$$

In Example 5, the area can also be found by using vertical strips, but the procedure is more complicated. Note in Figure 6.10 that on the interval $[-5, 3]$, a representative vertical strip would extend from the bottom half of the parabola $y^2 - 4y + x = 0$ to the line $y = \frac{1}{2}(x + 3)$, whereas on the interval $[3, 4]$, a typical vertical strip would extend from the bottom half of the parabola $y^2 - 4y + x = 0$ to the top half.

Thus, the area is given by the sum of two integrals, the second of which requires solving the equation $y^2 - 4y + x = 0$ for $y$ using the quadratic formula. With some effort, it can be shown that the computation of area by vertical strips gives the same result as that found by horizontal strips in Example 5.

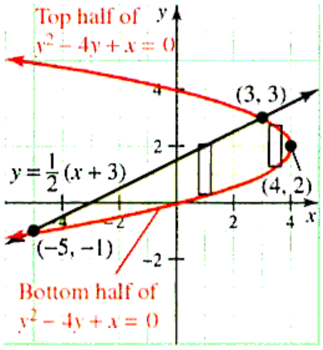



**Figure 6.10** Area of $R$ using vertical strips

## PROBLEM SET 6.1

**Level 1**

*Write integrals to express the area of the shaded regions in Problems 1-12. Both positive regions (pink) and negative regions (blue) should be included. Do not use absolute value symbols if possible.*

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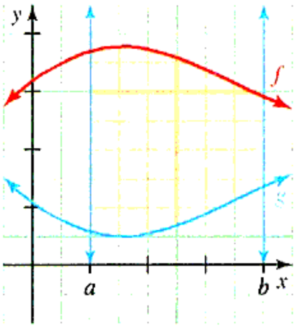
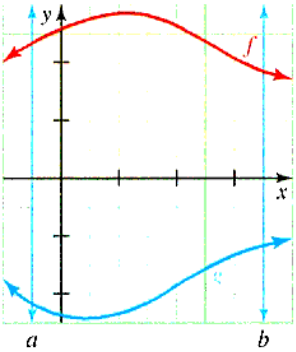
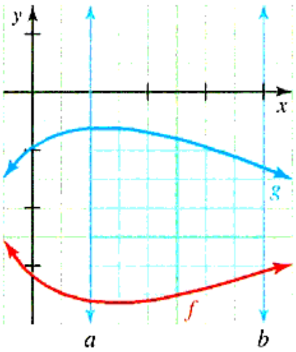


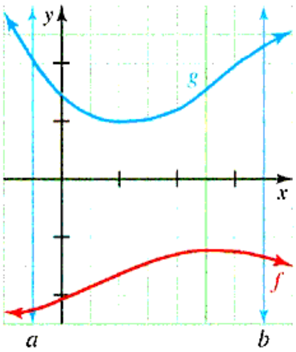

**2.**



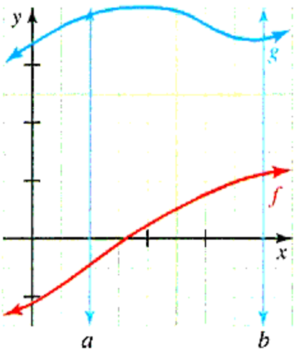

**3.**



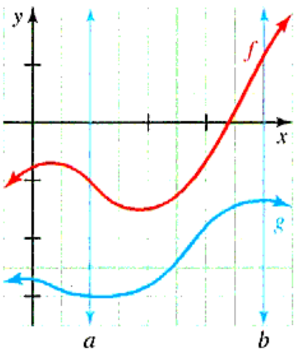

**4.**




**5.**




**6.**

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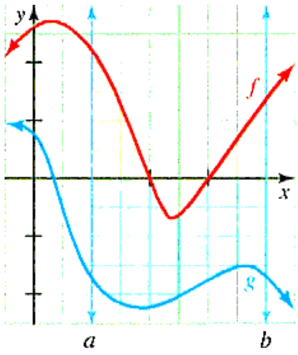


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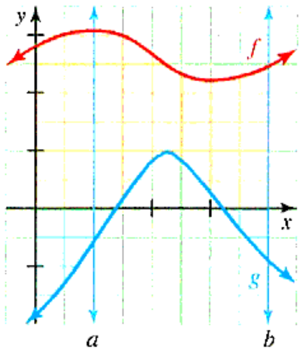


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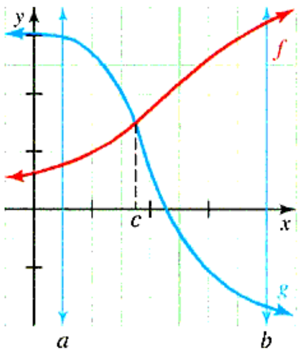


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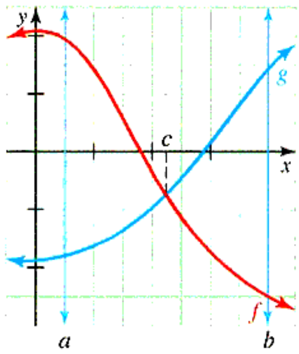
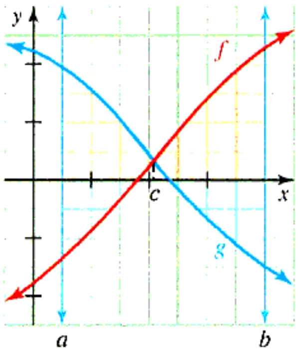
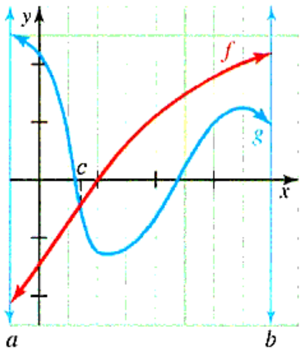



**11.**

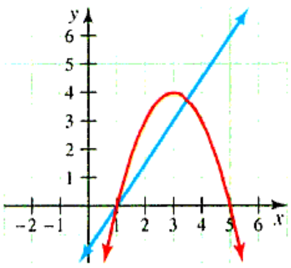



**12.**

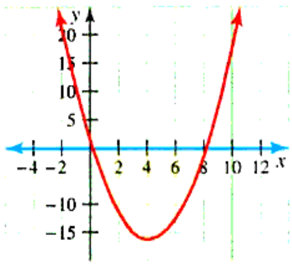


*Sketch a representative vertical or horizontal strip and find the area of the given regions bounded by the specified curves in Problems* 13-18.

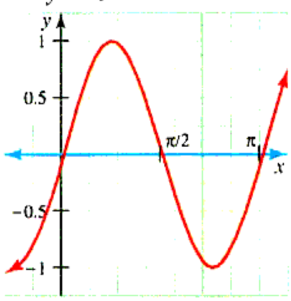
**13.** $y = \frac{3}{2}x - \frac{3}{2}$,
$y = -x^2 + 6x - 5$



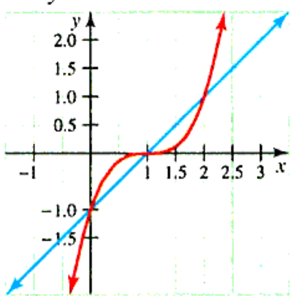

**14.** $y = x^2 - 8x$,
$y = 0$



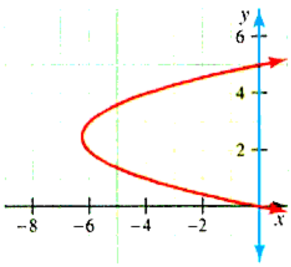

**15.** $y = \sin 2x$ on $[0, \pi]$
$y = 0$



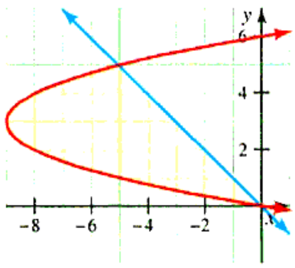

**16.** $y = (x - 1)^3$,
$y = x - 1$




**17.** $x = y^2 - 5y$
$x = 0$




**18.** $x = y^2 - 6y$
$x = -y$




*Sketch the region bounded between the given curves and then find the area of each region in Problems* 19-43.

**19.** $y = x^2$, $y = x$, $x = -1$, $x = 1$
**20.** $y = 6x^2$, $y = 0$, $x = 1$, $x = 5$.
**21.** $y = 4x^3$, $y = 0$, $x = 1$, $x = 4$.
**22.** $y = x^3$, $y = x$, $x = -1$, $x = 1$
**23.** $y = x^{-1}$ and the $x$-axis on $[0.5, 1]$
**24.** $y = x^{-1}$ and the $x$-axis on $[0.1, 1]$
**25.** $y = x^2$, $y = x^3$
**26.** $y = x^2$, $y = \sqrt[3]{x}$
**27.** $y = x^2 - 1$, $x = -1$, $x = 2$, $y = 0$
**28.** $y = 4x^2 - 9$, $y = 0$, between $x = 0$ and $x = 3$
**29.** $y = x^4 - 3x^2$, $y = 6x^2$
**30.** $x = 8 - y^2$, $x = y^2$
**31.** $x = 2 - y^2$, $x = y$
**32.** $y = x^2 + 3x - 5$, $y = -x^2 + x + 7$
**33.** $y = 2x^3 + x^2 - x - 1$, $y = x^3 + 2x^2 + 5x - 1$
**34.** $y = \sin x$, $y = \cos x$, $x = 0$, $x = \frac{\pi}{4}$
**35.** $y = \sin x$, $y = \sin 2x$, $x = 0$, $x = \pi$
**36.** $y = |x|$, $y = x^2 - 6$
**37.** $y = |4x - 1|$, $y = x^2 - 5$, $x = 0$, $x = 4$
**38.** $y = e^x$, the $x$-axis on $[-1, 2]$
**39.** $y = e^{-x}$, the $x$-axis on $[-2, 1]$
**40.** $x$-axis, $y = x^3 - 2x^2 - x + 2$
**41.** $y$-axis, $x = y^3 - 3y^2 - 4y + 12$
**42.** $y = e^x$, $y = \frac{1}{2}e^x + \frac{1}{2}$, $x = -2$, $x = 2$
**43.** $y = \dfrac{1}{\sqrt{1 - x^2}}$, $y = \dfrac{2}{x + 1}$, $y$-axis

## Level 2

**44.** ■ What does this say? When finding the area between two curves, discuss criteria for deciding between vertical and horizontal strips.

**45.** Find the number $k$ (correct to two decimal places) so that the line $y = k$ bisects the area under the curve $y = \sin^{-1} x$ for $0 \le x \le 1$.

**46.** Find the area of the region that contains the origin and is bounded by the lines $2y = 11 - x$ and $y = 7x + 13$ and the parabola $y = x^2 - 5$.

**47.** Show that the region defined by the inequalities $x^2 + y^2 \le 8$, $x \ge y$, and $y \ge 0$ has area $\pi$.

**48.** Find the area of the region bounded by the curve $\sqrt{x} + \sqrt{y} = 1$ and the coordinate axes.

**49.** The Sorite Corporation has found that production costs are determined by the function

$$C(x) = \frac{100}{\sqrt{x + 25}} + 50$$

where $C(x)$ is the cost to produce the $x$th item. Use integration to find the total cost of producing 2,000 items.

**50.** The EPA (Environmental Protection Agency) estimates that pollutants are being dumped into the Russian River according to the formula

$$P(t) = \frac{1{,}000}{\sqrt[3]{(t + 5)^2}}$$

where $t$ is the year and $P(t)$ is pollutants in tons. What is the total amount (to the nearest ton) of pollutants that will be dumped into the river in the next 25 years?

**51.** The marginal revenue and marginal cost (in dollars) per day are given in terms of the $t$th month by the formulas

$$R'(t) = 500e^{0.01t} \quad \text{and} \quad C'(t) = 50 - 0.1t$$

where $R(0) = C(0) = 0$. Find the total profit for the first year.

52. A chemical spill is known to kill fish according to the formula

$$N(t) = \frac{100{,}000}{\sqrt{t} + 250}$$

where $N$ is the number of fish killed on the $t$th day since the spill. What is the approximate total number of fish killed in 30 days?

53. A culture is growing at a rate of

$$R'(t) = 200e^{0.05t} \qquad \text{for } 0 \le t \le 10$$

per hour. Find the area between the graph of this function and the $t$-axis. What do you think this area represents?

54. The rate of change of the demand for a product with respect to time (in weeks) is given by the function

$$D'(t) = 20 + 0.012t^2 \qquad \text{for } 0 \le t \le 25$$

Find the area between the graph of this function and the $t$-axis. What do you think this area represents?

55. Suppose the accumulated cost of a piece of equipment is $C(t)$ and the accumulated revenue is $R(t)$, where both of these are measured in thousands of dollars and $t$ is the number of years after the piece of equipment was installed. If it is known that

$$C'(t) = 1 \qquad \text{and} \qquad R'(t) = 2e^{-0.1t}$$

find the area (to the nearest unit) in the bounded region between the graphs of $C'$ and $R'$ (do not forget $t \ge 0$). What do you think this area represents?

56. Suppose the accumulated cost of a piece of equipment is $C(t)$ and the accumulated revenue is $R(t)$, where both of these are measured in thousands of dollars and $t$ is the number of years after the piece of equipment was installed. It is known that

$$C'(t) = 18 \qquad \text{and} \qquad R'(t) = 21e^{-0.01t}$$

Find the area (to the nearest unit) in the bounded region between the graphs of $C'$ and $R'$ (do not forget $t \ge 0$). What do you think this area represents?

**Level 3**

**Modeling Problems** *Imagine a cylindrical fuel tank of length L lying on its side, where the ends are circular with radius b. Determine the amount of fuel in the tank for a given level by completing Problems 57-59.*

57. Explain why the volume of the tank may be modeled by

$$V = 2L\int_{-b}^{b} \sqrt{b^2 - y^2}\,dy$$

58. Explain why the volume of fuel at level $h$ $(-b \le h \le b)$ may be modeled by

$$V(h) = 2L\int_{-b}^{h} \sqrt{b^2 - y^2}\,dy$$

59. Finally, for $b = 4$ and $L = 20$, numerically compute $V(h)$ for $h = -3, -2, \cdots, 4$. *Note:* $V(0)$ and $V(4)$ will serve as a check on your work.

60. The curve with the equation

$$(x^2 + y^2)^3 = x^2$$

has a loop for $0 \le x \le 1$. Find the area of the loop. *Hint:* Use the substitution $u = x^{1/3}$ in the area integral.

## 6.2 Volume

**IN THIS SECTION:** *Method of cross sections, method of disks and washers, method of cylindrical shells*
We have seen how to compute area by integration, and now we are ready to investigate several methods for computing volume.

The methods used in Section 6.1 to compute area by integration can be modified to compute the volume of a solid region. We shall develop methods for computing volume of several kinds of solids in this section, beginning with the case where the solid has a known cross section.

### Method of Cross Sections

Let $S$ be a solid and suppose that for $a \le x \le b$, the cross section of $S$ that is perpendicular to the $x$-axis at $x$ has area $A(x)$.

To find the volume of $S$, we first take a partition $x_0 = a, x_1, x_2, x_3, \cdots, x_n = b$ of the interval $[a, b]$ and choose a representative number $x_k^*$ in each subinterval $[x_{k-1}, x_k]$. Think of cutting the solid with a knife at $x = x_k^*$ and removing a very thin slab whose face has area $A(x_k^*)$ and whose thickness is $\Delta x_k$, as shown in Figure 6.11.

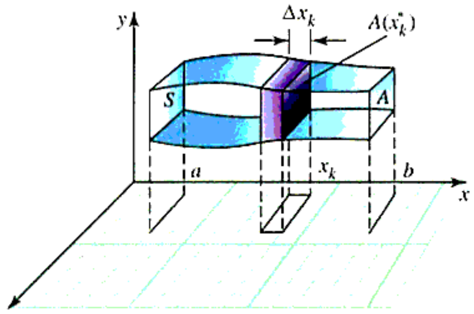



**Figure 6.11** Volume of solids with known cross sections

Thus, we consider a slab with cross-sectional area $A(x_k^*)$ and width

$$\Delta x_k = x_k - x_{k-1}$$

This slab has volume

$$\Delta V_k = A(x_k^*)\Delta x_k$$

By adding up the volumes of all such slabs, we obtain an approximation to the volume of the solid $S$:

$$V_n = \sum_{k=1}^{n} A(x_k^*)\Delta x_k$$

The approximation improves as the number of partition points increases, and it is reasonable to *define* the volume $V$ of the solid $S$ as the limit of $V_n$ as the norm of the partition $\|P\|$ tends to 0. That is,

$$V = \lim_{\|P\|\to 0} \sum_{k=1}^{n} A(x_k^*)\Delta x_k$$

which we recognize as the definite integral $\int_a^b A(x)\,dx$. To summarize:

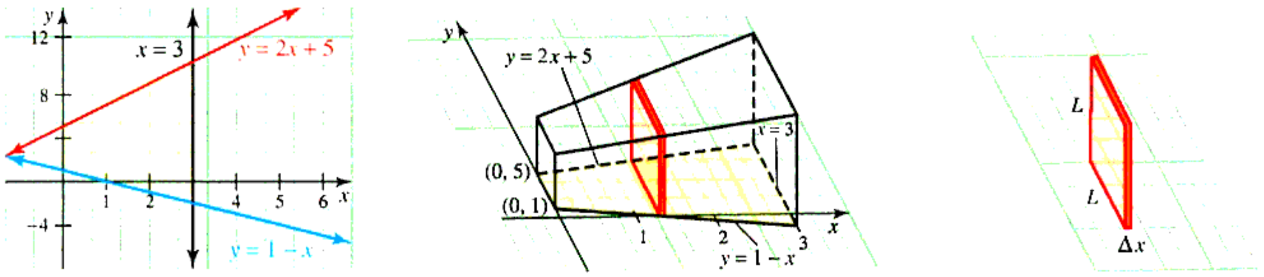
**VOLUME OF A SOLID WITH KNOWN CROSS-SECTIONAL AREA**
A solid $S$ with cross-sectional area $A(x)$ perpendicular to the $x$-axis at each point on the interval $[a, b]$ has **volume**

$$V = \int_a^b A(x)\,dx$$

**Example 1 Volume of a solid using square cross sections**

The base of a solid is the region in the $xy$-plane bounded by the $y$-axis and the lines $y = 1 - x$, $y = 2x + 5$, and $x = 3$. Each cross section perpendicular to the $x$-axis is a square. Find the volume of the solid.

***Solution*** The solid resembles a tapered brick, and it may be constructed by "gluing" together a number of thin slabs with square cross sections, like the one shown in Figure 6.12c. We begin by drawing the base in two dimensions and then find the volume of the $k$th slice.




**a.** Two-dimensional graph of the base

**b.** Three-dimensional solid

**c.** Cross-sectional representative element $\Delta V = L^2 \Delta x$

**Figure 6.12** A solid with a square cross section

To model this construction mathematically, we subdivide the interval $[0, 3]$, form a vertical approximating rectangle on each resulting subinterval, and then construct a slab with square cross section on each approximating rectangle. If we choose the thickness of a typical slab to be $\Delta x$ and the height and width to be $L$, the slab will have volume $\Delta V$, where

$$\begin{aligned}\Delta V &= L^2 \Delta x\\ &= L^2(x)\Delta x\\ &= [(2x+5)-(1-x)]^2\,\Delta x\\ &= (3x+4)^2 \Delta x\end{aligned}$$

The volume of the entire solid is obtained by integrating to "add up" all the volumes $\Delta V$, and we find that the volume of the solid is

$$\begin{aligned}V &= \int_0^3 (3x+4)^2 dx\\ &= \int_0^3 (9x^2+24x+16)\,dx\\ &= \left[3x^3+12x^2+16x\right]_0^3\\ &= 237\end{aligned}$$

**Example 2 Volume of a regular pyramid with a square base**

A regular pyramid has a square base of side $L$ and its apex located $H$ units above the center of its base. Derive the formula $V = \frac{1}{3}HL^2$.

***Solution*** The pyramid is shown in Figure 6.13.

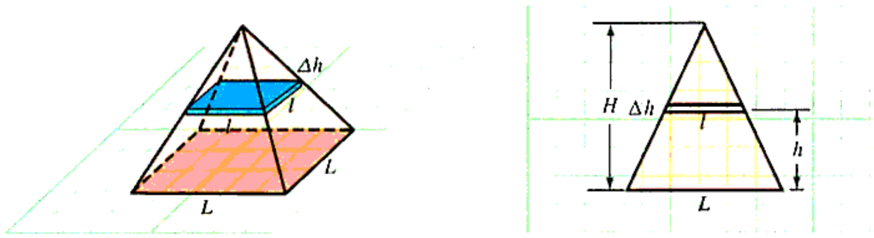



**Figure 6.13** The volume of a pyramid

This pyramid can be constructed by stacking a number of thin square slabs. Suppose that a representative slab has side $\ell$ and thickness $\Delta h$, and that it is located $h$ units above the base of the pyramid as shown in Figure 6.13. By creating a proportion from corresponding parts of similar triangles, we see that

$$\frac{\ell}{L} = \frac{H-h}{H} \quad \text{so that} \quad \ell = \left(1 - \frac{h}{H}\right) L$$

Therefore, the volume of the representative slab is

$$\begin{aligned} \Delta V &= \ell^2 \Delta h \\ &= \left(1 - \frac{h}{H}\right)^2 L^2 \Delta h \end{aligned}$$

To compute the volume $V$ of the entire pyramid, we integrate with respect to $h$ from the base of the pyramid ($h = 0$) to the apex ($h = H$). Thus,

$$\begin{aligned} V &= \int_0^H \left(1 - \frac{h}{H}\right)^2 L^2 dh \\ &= L^2 \int_0^H \left(1 - \frac{2}{H}h + \frac{1}{H^2}h^2\right) dh \\ &= L^2 \left[h - \frac{h^2}{H} + \frac{h^3}{3H^2}\right]_0^H \\ &= L^2 \left(H - \frac{H^2}{H} + \frac{H^3}{3H^2}\right) \\ &= \frac{1}{3}HL^2 \end{aligned}$$

Other volume formulas can be found in a similar fashion (see Problems 59 and 60).

## Method of Disks and Washers

A **solid of revolution** is a solid figure $S$ obtained by revolving a region $D$ in the $xy$-plane about a line $L$ (called the **axis of revolution**) that lies outside or on the boundary of $D$. Some familiar solids of revolution are shown in Figure 6.14.

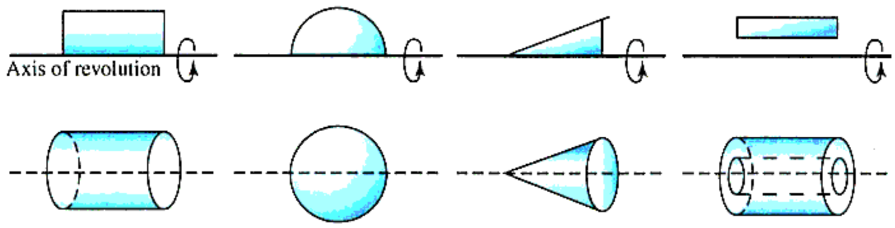
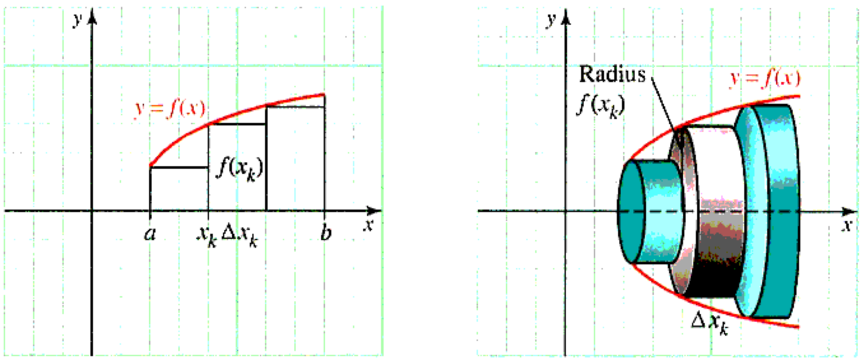



**Figure 6.14** Solids of revolution

Note that such a solid $S$ may be thought of as having circular cross sections in the direction perpendicular to $L$.

Suppose the function $f$ is continuous and satisfies $f(x) \geq 0$ on the interval $[a, b]$, and suppose we wish to find the volume of the solid $S$ generated when the region $D$ under the curve $y = f(x)$ on $[a, b]$ is revolved about the $x$-axis. That is, *the axis of revolution is horizontal and it is a boundary of the region* $D$. Our strategy will be to form vertical strips and revolve them about the $x$-axis, generating what are called **disks** (that is, thin right circular cylinders) that approximate a portion of the solid of revolution $S$, as shown in Figure 6.15.




**a.** A representative vertical strip has height $f(x_k)$ and width $\Delta x_k$

**b.** The representative disk is formed by revolving the representative strip about the $x$-axis

**Figure 6.15 Interactive** The disk method

Now we can compute the total volume of $S$ by using integration to sum the volumes of all the approximating disks. Recall that the formula for the volume of a cylinder of height $h$ and cross-sectional area $A$ is $Ah$. Figure 6.15**a** shows a typical vertical strip with height $f(x_k)$ and width $\Delta x_k$. You might notice that the width of the strip is the same as the thickness of the disk. The solid of revolution can be thought of as having cross sections perpendicular to the $x$-axis that are circular disks of volume

$$\Delta V_k(x) = \underbrace{\pi[f(x_k)]^2}_{\text{Area of circular cross section}} \Delta x_k$$

The total volume may be found by integration:

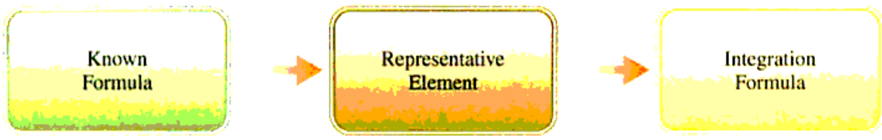
$$V = \int_a^b \overbrace{A(x)}^{\text{Area of cross section}} \underbrace{dx}_{\text{Thickness}} = \int_a^b \pi\left[f(x)\right]^2 dx$$

This procedure may be summarized as follows:

**THE DISK METHOD** The **disk method** is used to find a volume generated when a region $D$ is revolved about an axis $L$ that is *perpendicular* to a typical approximating strip in $D$. Suppose $D$ is the region bounded by the curve $y = f(x)$, the $x$-axis, and the vertical lines $x = a$ and $x = b$. Then if $D$ is revolved about the $x$-axis, it generates a solid with volume

$$V = \int_a^b \pi y^2 dx = \int_a^b \pi \left[f(x)\right]^2 dx$$

What this says The following diagram may help you remember the key ideas behind the disk method.




Volume of disk: $V = Bh = \pi r^2 h$

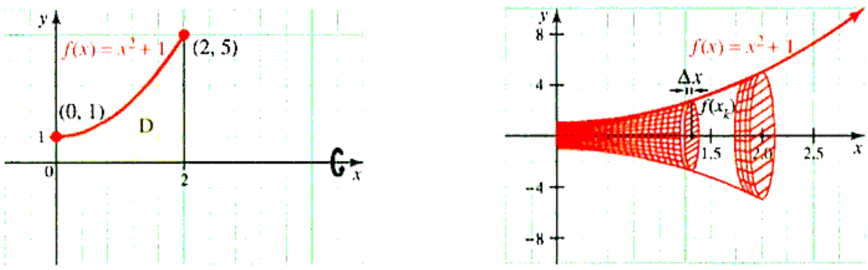
$\Delta V = \pi y^2 \Delta x$

$V = \pi \int_a^b \left[f(x)\right]^2 dx$

**Example 3 Volume by disks**

Find the volume of the solid $S$ formed by revolving the region $D$ under the curve $y = x^2 + 1$ on the interval $[0, 2]$ about the $x$-axis.

***Solution*** The region is shown in Figure 6.16.




**a.** The region, $D$ **b.** The solid of revolution, $S$

**Figure 6.16 Interactive** Volume of a solid of revolution: The disk method

$$\begin{aligned} V &= \pi \int_0^2 (x^2 + 1)^2 dx \\ &= \pi \int_0^2 (x^4 + 2x^2 + 1)\, dx \\ &= \pi \left[\frac{1}{5}x^5 + \frac{2}{3}x^3 + x\right]_0^2 \\ &= \frac{206}{15}\pi \end{aligned}$$

The volume is about 43 cubic units.

With a small modification of the disk method, we can find the volume of a solid figure generated by revolving about the $x$-axis the region between two curves $y = f(x)$ and $y = g(x)$ where $f(x) \geq g(x) \geq 0$ for $a \leq x \leq b$. When a typical vertical strip is revolved about the $x$-axis, a "washer" with cross-sectional area $\pi([f(x)]^2 - [g(x)]^2)$ is formed, as shown in Figure 6.17. This can be remembered by $\pi R^2 - \pi r^2$, where $R$ is the outer radius and $r$ is the inner radius.

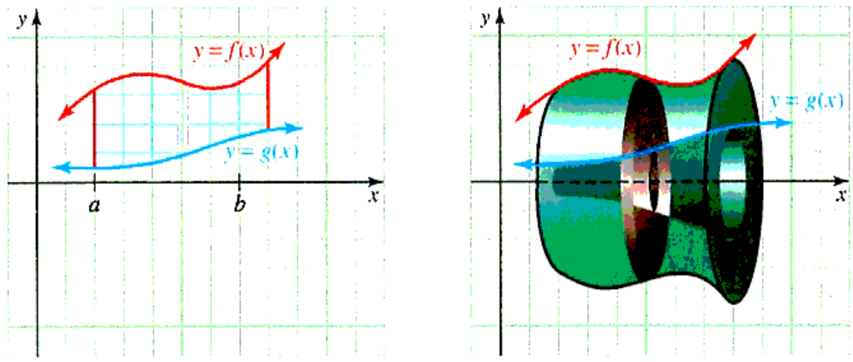



**Figure 6.17** The washer method

The volume of the solid of revolution is found by the formula given in the following box.

**THE WASHER METHOD** The **washer method** is used to find volume when a region between two curves is revolved about an external axis perpendicular to the approximating strip. In particular, suppose $f$ and $g$ are continuous functions on $[a, b]$ with $f(x) \geq g(x) \geq 0$. If $R$ is the (outer) curve $y = f(x)$ and $r$ is the (inner) curve $y = g(x)$, and the region bounded by $y = f(x)$, $y = g(x)$, $x = a$, and $x = b$ is revolved about the $x$-axis, then the volume thus formed is

$$V = \int_a^b \pi\left(\underbrace{[f(x)]^2}_{\text{Outer radius}} - \underbrace{[g(x)]^2}_{\text{Inner radius}}\right) dx$$

$f^2 - g^2 \neq (f - g)^2$

**What this says** In practice, what this means is that the "inner" solid is "subtracted" from the "outer" solid, much like coring an apple.

The disk method and the washer method also apply when the axis of revolution is a line other than the $x$-axis. In Example 4, we consider what happens when a particular region $D$ is revolved not only about the $x$-axis, but also about other axes parallel to a coordinate axis.

**Example 4 Volume by washers**

Let $D$ be the solid region bounded by the parabola $y = x^2$ and the line $y = x$. Find the volume of the solid generated when $D$ is revolved about the

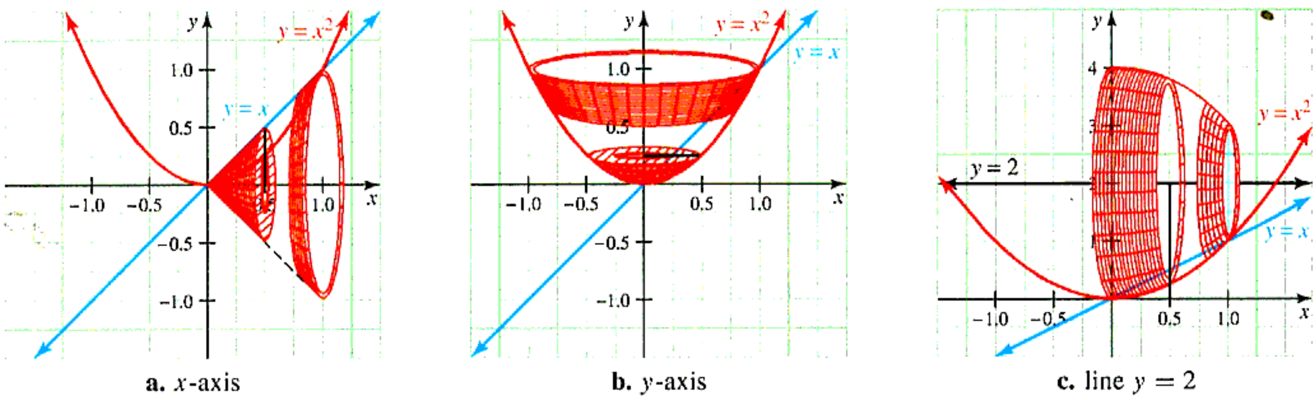
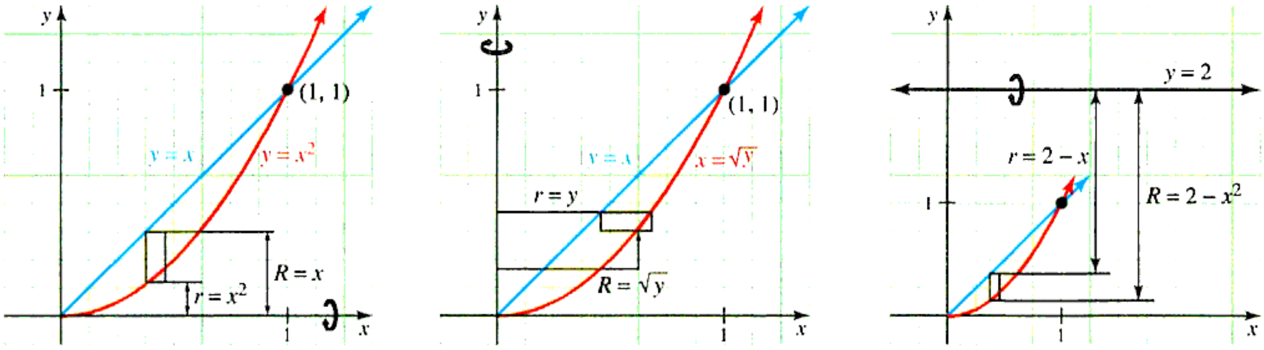



**Figure 6.18 Interactive** Volume by washers

***Solution*** The region $D$ to be rotated is shown in Figure 6.19. Note that since $x = x^2$ at $x = 0$ and at $x = 1$, the line and parabola intersect at the origin and at $(1, 1)$.

**a.** The line is always above the parabola on the interval $[0, 1]$, so when we form a washer to approximate the volume of revolution, the outer radius is $R = x$ and the inner radius is $r = x^2$, as shown in Figure 6.19**a**. Thus, the required volume is

$$\begin{aligned} V &= \pi \int_0^1 \left[x^2 - (x^2)^2\right] dx \\ &= \pi \int_0^1 (x^2 - x^4)\, dx \\ &= \pi \left[\frac{1}{3}x^3 - \frac{1}{5}x^5\right]_0^1 \\ &= \frac{2\pi}{15} \end{aligned}$$




**a.** Vertical strip rotated about the $x$-axis

**b.** Horizontal strip rotated about the $y$-axis

**c.** Vertical strip rotated about the line $y = 2$

**Figure 6.19** Volume obtained by revolving a region about different lines

*For the disk method or the washer method, make sure that the approximating strips are perpendicular to the axis of revolution.*

**b.** Because we are revolving $D$ about the $y$-axis, we use *horizontal* strips to approximate the solid of revolution, as shown in Figure 6.19**b**. Note that the parabola $x = \sqrt{y}$ is to the right of the line $x = y$ on the interval $[0, 1]$, so the approximating washer has outer radius $R = \sqrt{y}$ and inner radius $r = y$; thus,

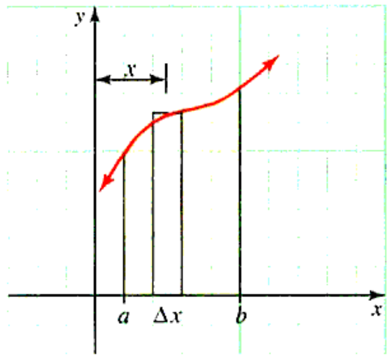
$$\begin{aligned} V &= \pi \int_0^1 \left[(\sqrt{y})^2 - (y)^2\right] dy \\ &= \pi \int_0^1 (y - y^2)\, dy \\ &= \pi \left[\frac{y^2}{2} - \frac{y^3}{3}\right]_0^1 \\ &= \frac{\pi}{6} \end{aligned}$$

**c.** Since the axis of rotation is the line $y = 2$, the outer radius of a typical approximating washer is $R = 2 - x^2$, and the inner radius is $r = 2 - x$, as shown in Figure 6.19**c**. Thus, the volume of revolution is

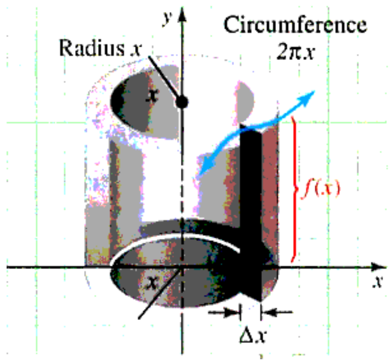
$$
\begin{aligned}
V &= \pi \int_0^1 \left[(2-x^2)^2 - (2-x)^2\right] dx \\
&= \pi \int_0^1 (x^4 - 5x^2 + 4x)\, dx \\
&= \pi \left[\frac{x^5}{5} - \frac{5x^3}{3} + \frac{4x^2}{2}\right]_0^1 \\
&= \frac{8\pi}{15}
\end{aligned}
$$

## The Method of Cylindrical Shells

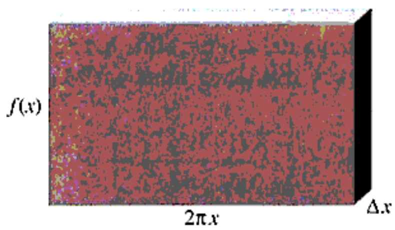
Sometimes it is easier (or even necessary) to compute a volume by taking the approximating strip parallel to the axis of rotation instead of perpendicular to the axis as in the disk and washer methods. Figure 6.20**a** shows a region $D$ under the curve $y = f(x) \geq 0$, on the interval $[a, b]$, together with a representative vertical strip. When this strip is revolved about the $y$-axis, it forms a solid called a **cylindrical shell** of height approximately $f(x)$ and thickness $\Delta x$, as shown in Figure 6.20**b**.



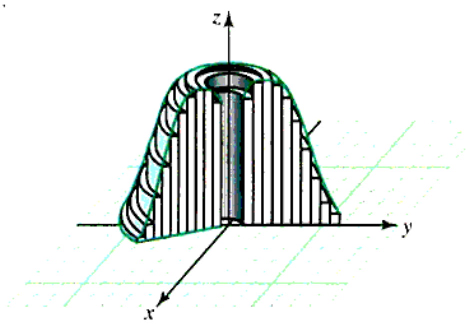

**a.** Vertical strip




**b.** When the strip (part **a**) is revolved about the $y$-axis, a shell is generated




**c.** The "unwrapped" flattened shell has volume $\Delta V = 2\pi x f(x) \Delta x$




**d.** Approximating a solid of revolution using cylindrical shells

**Figure 6.20** Method of cylindrical shells

Because the strip is $x$ units from the axis of rotation and is assumed to be very thin, the cross section of the shell (perpendicular to the $y$-axis) will be a circle of radius $x$ and circumference $2\pi x$. If we imagine the shell to be cut and flattened out, it is seen to be approximately a rectangular slab of volume

$$\Delta V = \underbrace{2\pi x f(x)}_{\textit{Area of the rectangular slab}} \cdot \overbrace{\Delta x}^{\textit{Thickness}}$$

(See Figure 6.20c.) Thus, the total volume of the solid is given by the integral

$$V = \int_a^b 2\pi x\, f(x)\, dx$$

The approximating shells are shown in Figure 6.20**d** and the formula is repeated in the following box.

**METHOD OF CYLINDRICAL SHELLS** The **shell method** is used to find a volume generated when a region $D$ is revolved about an axis $L$ *parallel* to a typical approximating strip in $D$.
In particular, if $D$ is the region, as shown in Figure 6.21, bounded by the curve $y = f(x) \geq 0$, the $x$-axis, and the vertical lines $x = a$ and $x = b$ where $0 \leq a \leq b$, then the solid generated by revolving $D$ about the $y$-axis has volume

$$V = \int_a^b 2\pi x f(x)\, dx$$

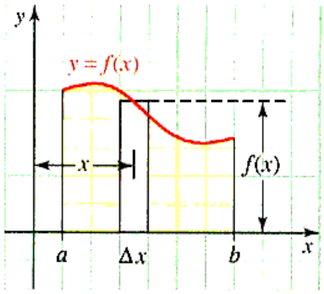



**Figure 6.21** Shell method

**Example 5 Volume using cylindrical shells**

Find the volume of the solid formed by revolving the region bounded by the $x$-axis and the graphs of $y = x^3 + x^2 + 1$, $x = 1$, and $x = 3$ about the $y$-axis.

***Solution*** The region to be rotated is shown in Figure 6.22, along with a typical vertical strip of height $f(x) = x^3 + x^2 + 1$ and width $\Delta x$.

The volume, by the method of shells, is

$$\begin{aligned} V &= 2\pi \int_1^3 x(x^3 + x^2 + 1)\, dx \\ &= 2\pi \int_1^3 (x^4 + x^3 + x)\, dx \\ &= 2\pi \left[ \frac{x^5}{5} + \frac{x^4}{4} + \frac{x^2}{2} \right]_1^3 \\ &= \frac{724}{5}\pi \end{aligned}$$

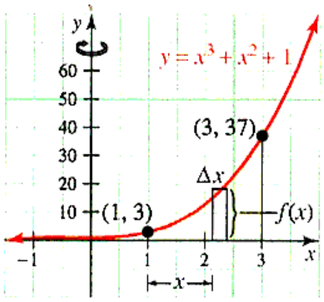



**Figure 6.22** Shell method about $y$-axis

When the axis of rotation is a line $x = L$ parallel to the $y$-axis, the distance of a typical vertical strip from the axis will no longer be $x$. Here is an example that illustrates how to compute volume of revolution in such a case.

**Example 6 Volume of revolution about a horizontal or vertical line**

Find the volume of the solid formed by revolving the region $D$ bounded by the curve $y = x^{-2}$ and the $x$-axis for $1 \le x \le 2$ about the line $x = -1$.

***Solution*** The region $D$ is shown in Figure 6.23, along with a typical vertical strip of height $f(x) = x^{-2}$.

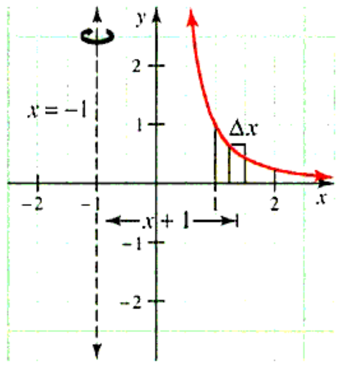



**Figure 6.23** Shell method about $x = -1$

If we were revolving this strip about the $y$-axis, the distance from the axis to the strip would be $L = x$, but since the axis of rotation is $x = -1$, the distance is $L = x - (-1) = x + 1$.

$$\begin{aligned} V &= 2\pi \int_1^2 (x+1)\left(\frac{1}{x^2}\right) dx \\ &= 2\pi \int_1^2 (x^{-1} + x^{-2})\, dx \\ &= 2\pi \left[\ln x - x^{-1}\right]_1^2 \\ &= 2\pi \ln 2 + \pi \end{aligned}$$

Sometimes you have a choice between using shells or washers to compute a volume of revolution. The decision as to which method to use often comes down to which leads to the easier integration. Which method would you choose in the following example?

**Example 7 Comparing the methods of shells and washers**

A cylindrical hole of radius $r$ is bored through the center of a solid sphere of radius $R$ $(r < R)$, as shown in Figure 6.24.

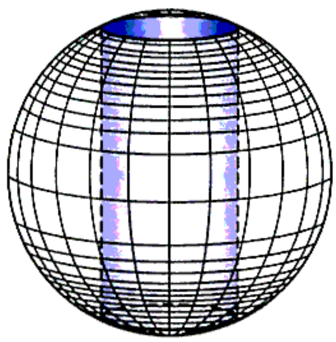


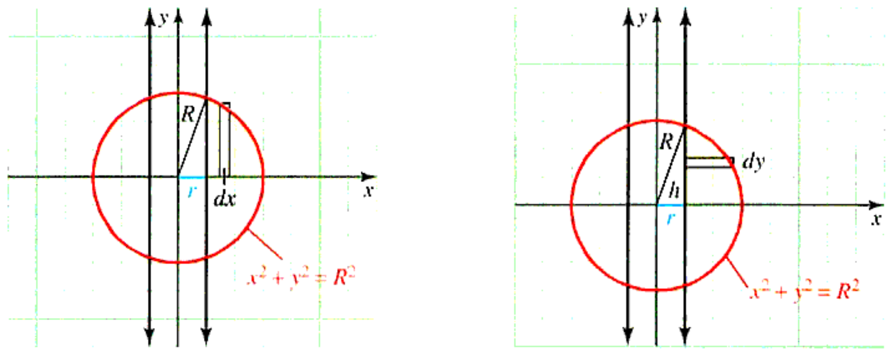
**Figure 6.24** Drill a hole in a sphere

Find the volume of the solid that remains by using: **a.** cylindrical shells **b.** washers

***Solution***

**a.** The required volume can be thought of as twice the volume generated when the shaded region of Figure 6.25a is revolved about the $y$-axis.

**b.** The required volume can be thought of as twice the volume generated when the shaded region of Figure 6.25b is revolved about the $y$-axis.




**a.** Strip *parallel* to the axis of rotation. Typical strip: height $y = \sqrt{R^2 - x^2}$ radius: $x$

**b.** Strip *perpendicular* to the axis of revolution. Typical washer: outer radius: $x = \sqrt{R^2 - y^2}$ inner radius: $r$ $h = \sqrt{R^2 - r^2}$

**Figure 6.25** Volume of a solid

**a.** cylindrical shells

$$\begin{aligned} V &= 2\int_r^R 2\pi xy\,dx \\ &= 4\pi \int_r^R x\sqrt{R^2 - x^2}\,dx \\ &= 4\pi \left[-\frac{1}{3}(R^2 - x^2)^{3/2}\right]_r^R \\ &= \frac{4\pi}{3}(R^2 - r^2)^{3/2} \end{aligned}$$

**b.** washers

$$\begin{aligned} V &= 2\int_0^h \pi\left[\left(\sqrt{R^2 - y^2}\right)^2 - r^2\right] dy \\ &= 2\pi \int_0^{\sqrt{R^2 - r^2}} [R^2 - y^2 - r^2]\,dy \\ &= 2\pi \left[(R^2 - r^2)y - \frac{y^3}{3}\right]_0^{\sqrt{R^2 - r^2}} \\ &= 2\pi \left[(R^2 - r^2)^{3/2} - \frac{1}{3}(R^2 - r^2)^{3/2}\right] \\ &= \frac{4\pi}{3}(R^2 - r^2)^{3/2} \end{aligned}$$

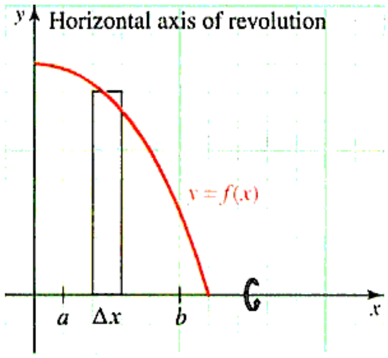
*Note that in the special case $r = 0$, a sphere, we get the familiar $V = \frac{4\pi}{3}R^3$.*

Table 6.1 compares and contrasts the disk, washer, and shell methods for computing volume.

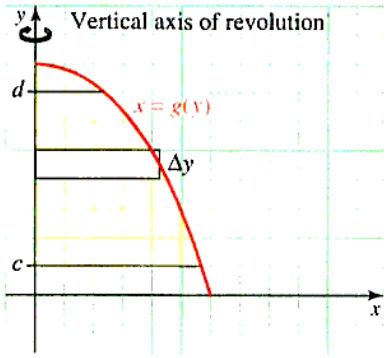
**Table 6.1** Volumes of revolution when the axis of revolution is either the $x$-axis or the $y$-axis

| | **Horizontal axis of revolution** | **Vertical axis of revolution** |
|---|---|---|

**Disk method**
A representative rectangle is **perpendicular** to the axis of revolution. The axis of revolution is a boundary of the region.

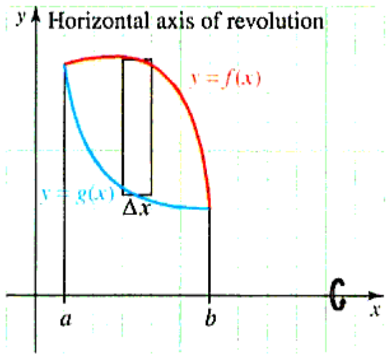



$$V = \pi \int_a^b \Big[\overbrace{f(x)}^{\textit{Length (height) of rectangle}}\Big]^2 \underbrace{dx}_{\textit{Width of rectangle is } \Delta x.}$$

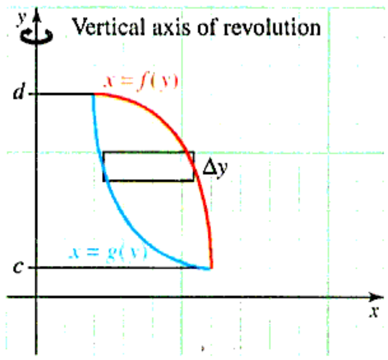



$$V = \pi \int_c^d \Big[\overbrace{g(y)}^{\textit{Length of rectangle}}\Big]^2 \underbrace{dy}_{\textit{Width of rectangle is } \Delta y.}$$

**Washer method**
A representative rectangle is **perpendicular** to the axis of revolution. The axis of revolution is *not* part of the boundary of the region.

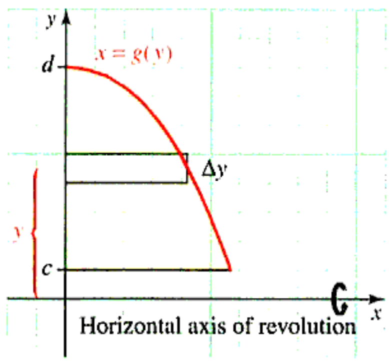



$$V = \pi \int_a^b \underbrace{\Big[\overbrace{f(x)}^{\textit{Top curve}}\Big]^2 - \Big[\overbrace{g(x)}^{\textit{Bottom curve}}\Big]^2}_{\textit{Length of rectangle}} \underbrace{dx}_{\textit{Width}}$$

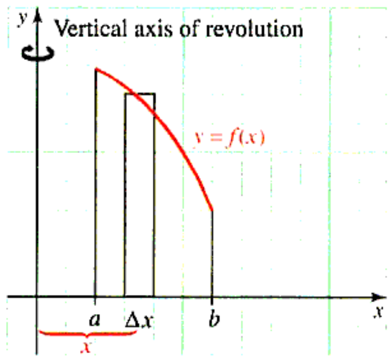



$$V = \pi \int_c^d \underbrace{\Big[\overbrace{f(y)}^{\textit{Right curve}}\Big]^2 - \Big[\overbrace{g(y)}^{\textit{Left curve}}\Big]^2}_{\textit{Length of rectangle}} \underbrace{dy}_{\textit{Width}}$$

**Shell method**
A representative rectangle is **parallel** to the axis of revolution.




$$V = 2\pi \int_c^d \underset{\substack{\uparrow \\ \textit{Distance to axis}}}{y} \Big[\overbrace{g(y)}^{\textit{Length of rectangle}}\Big] \underbrace{dy}_{\textit{Width}}$$




$$V = 2\pi \int_a^b \underset{\substack{\uparrow \\ \textit{Distance to axis}}}{x} \Big[\overbrace{f(x)}^{\textit{Length (height)}}\Big] \underbrace{dx}_{\textit{Width}}$$

## PROBLEM SET 6.2

**Level 1**

*In Problems 1-4, sketch the given region and then find the volume of the solid whose base is the given region and which has the property that each cross section perpendicular to the x-axis is a* ***square****.*

1. the triangular region bounded by the coordinate axes and the line $y = 3 - x$
2. the region bounded by the $x$-axis and the semicircle $y = \sqrt{16 - x^2}$
3. the region bounded by the line $y = x + 1$ and the parabola $y = x^2 - 2x + 3$
4. the region bounded above by $y = \sqrt{\sin x}$ and below by the $x$-axis on the interval $[0, \pi]$

*In Problems 5-8, sketch the region and then find the volume of the solid whose base is the given region and which has the property that each cross section perpendicular to the x-axis is an* ***equilateral triangle****.*

5. the region bounded by the circle $x^2 + y^2 = 4$
6. the region bounded by the curves $y = x^3$ and $y = x^2$
7. the region bounded above by $y = \sqrt{\cos x}$ and below by the $x$-axis on the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
8. the triangular region with vertices $A(1, 1)$, $B(3, 5)$, and $C(3, -2)$

*In Problems 9-12, sketch the region and then find the volume of the solid where the base is the given region and which has the property that each cross section perpendicular to the x-axis is a semicircle.*

9. the region bounded by the parabola $y = x^2$ and the line $2x + y - 3 = 0$
10. the region bounded above by $y = \cos x$, below by $y = \sin x$, and on the left by the $y$-axis
11. the region bounded above by the curve $y = \tan x$ and below by the $x$-axis, on the interval $\left[0, \frac{\pi}{4}\right]$
12. the region bounded by the $x$-axis and the curve $y = e^x$ between $x = 1$ and $x = 3$

*In Problems 13-18, find the volume of the solid formed when the region described is revolved about the x-axis using washers or disks.*

13. the region under the parabolic arc $y = \sqrt{x}$ on the interval $0 \le x \le 1$
14. the region under the curve $y = \sqrt[3]{x}$ on the interval $0 \le x \le 8$
15. the region bounded by the lines $y = x$, $y = 2x$, and $x = 1$
16. the region bounded by the lines $x = 0$, $x = 1$, $y = x + 1$, and $y = x + 2$
17. the region between $y = \sin x$ and $y = \cos x$ on $0 \le x \le \frac{\pi}{4}$
18. the region bounded by the curves $y = e^x$ and $y = e^{-x}$ on $[0, 2]$

*In Problems 19-24, use shells to find the volume of the solid formed by revolving the given region about the y-axis.*

19. the region bounded by the lines $y = 2x$, the $y$-axis, and $y = 1$
20. the region bounded by the parabolic arc $y = \sqrt{x}$, the $y$-axis, and the line $y = 1$
21. the region bounded by the parabola $y = 1 - x^2$, the $y$-axis, and the positive $x$-axis
22. the region bounded by the parabolas $y = x^2$, $y = 1 - x^2$, and the $y$-axis for $x \ge 0$
23. the region bounded by the curve $y = e^{-x^2}$, the $y$-axis, and the line $y = \frac{1}{2}$
24. the region inside the ellipse

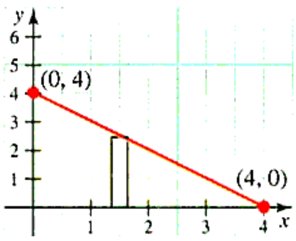
$$2(x - 3)^2 + 3(y - 2)^2 = 6$$

about the $y$-axis

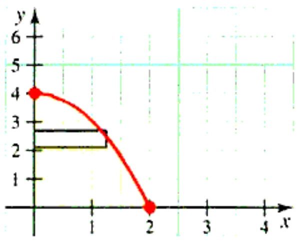
*In Problems 25-32, set up but do not evaluate an integral using the shaded strips for the volume generated when the given region is revolved about:*

**a.** *the x-axis* **b.** *the y-axis*
**c.** *the line $y = -1$* **d.** *the line $x = -2$*

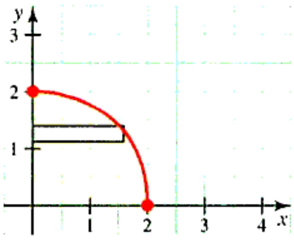
25. $y = 4 - x$, $0 \le x \le 4$



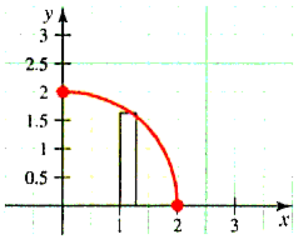

26. $y = 4 - x^2$, $0 \le x \le 2$



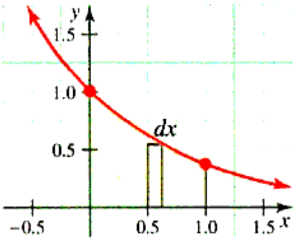
27. $y = \sqrt{4 - x^2}$, $0 \le x \le 2$



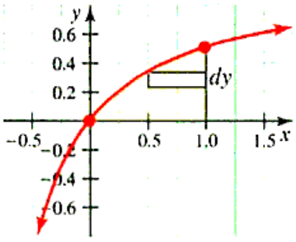
28. $y = \sqrt{4 - x^2}$, $0 \le x \le 2$



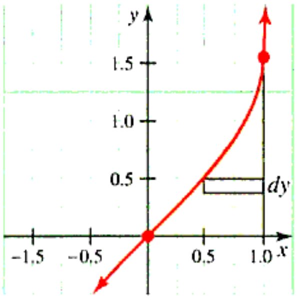
**29.** $y = e^{-x}$, $0 \le x \le 1$



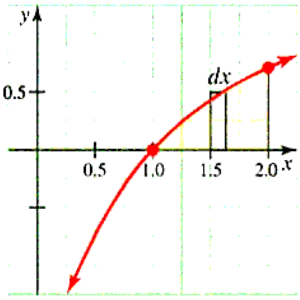

**30.** $y = \dfrac{x}{x+1}$, $0 \le x \le 1$




**31.** $y = \sin^{-1} x$, $0 \le x \le 1$




**32.** $y = \ln x$, $1 \le x \le 2$

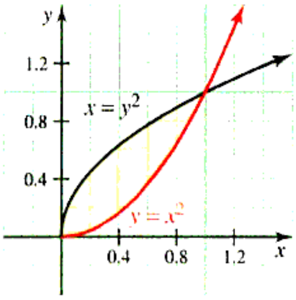



*In Problems* 33-38, *draw a representative strip and set up an integral for the volume of the solid formed by revolving the given region:*

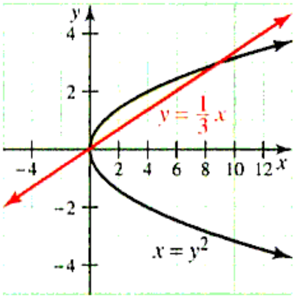
**a.** *about the x-axis* **b.** *about the y-axis*

*Set up the integral only*; *DO NOT EVALUATE.*

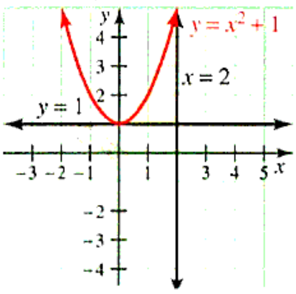
**33.** the region bounded by $x = y^2$ and $y = x^2$



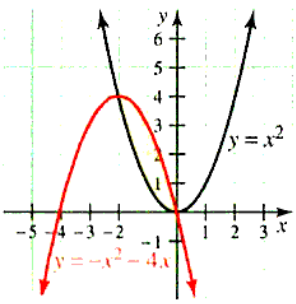

**34.** the region bounded by $y = \frac{1}{3}x$ and $x = y^2$



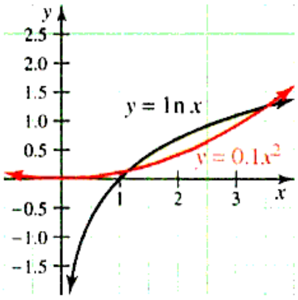

**35.** the region bounded by $y = 1$, $x = 2$, and $y = x^2 + 1$



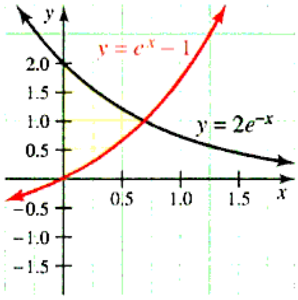

**36.** the region bounded by $y = x^2$ and $y = -x^2 - 4x$




**37.** the region bounded by $y = 0.1x^2$ and $y = \ln x$




**38.** the region bounded by $y = e^x - 1$, $y = 2e^{-x}$, and $x = 0$




## Level 2

*In each of Problems* 39-42, *find the volume by the*

**a.** *disk/washer method*
**b.** *shell method*

**39.** Revolve the region bounded by $y = x^2$ and $y = x^3$ about the $x$-axis.

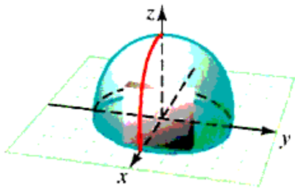
**40.** Revolve the region bounded by $y = x^2$ and $y = x^3$ about the $y$-axis.

**41.** Revolve the region bounded by $y = x$, $y = 2x$, and $y = 1$ about the $x$-axis.

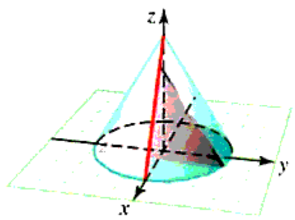
**42.** Revolve the region bounded by $y = x$, $y = 2x$, and $y = 1$ about the $y$-axis.

*In Problems* 43-46, *find the volume of the solid whose base is bounded by the circle* $x^2 + y^2 = 9$ *with the indicated cross sections taken perpendicular to the x-axis.*

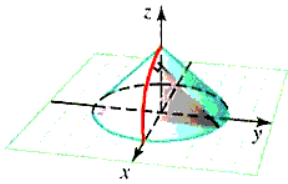
**43.** squares



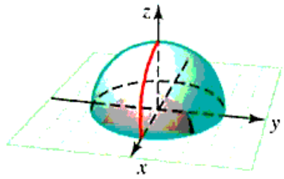
**44.** equilateral triangles



**45.** isosceles right triangles

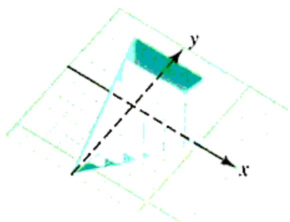



**46.** semicircles

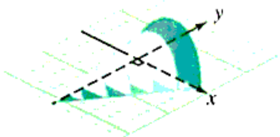



*In Problems 47-48, find the volume V of the solid with the given information regarding its cross section.*

**47.** The base of the solid is an equilateral triangle, each side of which has length 4. The cross sections perpendicular to a given altitude of the triangle are squares.

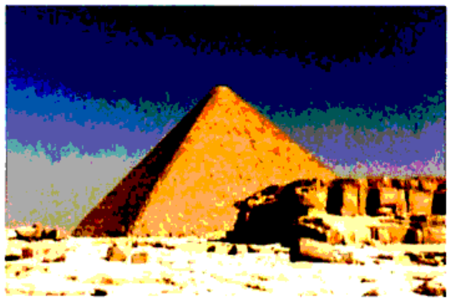



**48.** The base of the solid is an isosceles right triangle whose legs are each 4 units long. Each cross section perpendicular to a side is a semicircle.




**49.** The great pyramid of Cheops is approximately 480 ft tall and 750 ft square at the base. Find the volume of this pyramid by using the cross section method.





**Great pyramid of Cheops**

**50.** Find the volume of the solid generated when the region $y = x^{-1/2}$ on the interval $[1, 4]$ is revolved about:
**a.** the $x$-axis
**b.** the $y$-axis
**c.** the line $y = -2$

**51.** Let $R$ be the region in the first quadrant bounded by the line $y = kx$ and the line $x = k$, $k > 0$. Find the volume when $R$ is revolved about:
**a.** the $y$-axis
**b.** the line $x = 2k$

**52.** Derive the formula for the volume of a sphere of radius $r$ by revolving the region bounded by the semicircle $y = \sqrt{r^2 - x^2}$ about the $x$-axis.

**53.** The portion of the ellipse

$$\frac{x^2}{9} + \frac{y^2}{4} = 1$$

with $x \geq 0$ is rotated about the $y$-axis to form a solid $S$. A hole of radius 1 is drilled through the center of $S$, along the $y$-axis. Find the volume of the part of $S$ that remains.

**54.** Cross-sectional areas are measured at 1-foot intervals along the length of an irregularly shaped object, with the results listed in the following table ($x$ is in ft and $A$ is in ft$^2$):

| $x$ | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| $A$ | 1.12 | 1.09 | 1.05 | 1.03 | 0.99 | 1.01 |
| $x$ | 6 | 7 | 8 | 9 | 10 | |
| $A$ | 0.98 | 0.99 | 0.96 | 0.93 | 0.91 | |

Estimate the volume (correct to the nearest hundredth) by using the trapezoidal rule.

55. Cross-sectional areas are measured at 2 meter intervals along the length of an irregularly shaped object, with the results listed in the following table ($x$ is in meters and $A$ is in $m^2$):

| $x$ | 0 | 2 | 4 | 6 | 8 | 10 |
|---|---|---|---|---|---|---|
| $A$ | 1.12 | 1.09 | 1.05 | 1.03 | 0.99 | 1.01 |
| $x$ | 12 | 14 | 16 | 18 | 20 | |
| $A$ | 0.98 | 0.99 | 0.96 | 0.93 | 0.91 | |

Estimate the volume (correct to the nearest hundredth) by using Simpson's rule.

56. *Historical Quest*

**Johannes Kepler (1571-1630)**

*Johannes Kepler is usually remembered for his work in astronomy (see Section 10.3.) Kepler also made interesting mathematical discoveries. In fact, he has been described as "number-intoxicated." He was looking for mathematical harmonies in the physical universe. He is quoted in **The World of Mathematics**: "Nothing holds me; I will indulge my sacred fury; I will triumph over mankind by the honest confession that I have stolen the golden vases of the Egyptians to build up a tabernacle for my God far away from the confines of Egypt."**

In this *Quest* we look at Kepler's derivation for the volume of a torus (Figure 6.26), generated by revolving a circle of radius $a$ around a vertical axis at a distance $b$ from its center ($b \geq a$).

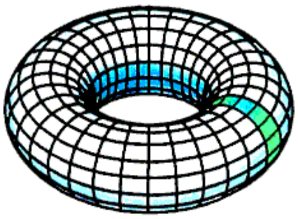


**Figure 6.26** Torus

He found $V = 2\pi^2 a^2 b$. He derived this formula by dissecting a torus into infinitely many thin vertical circular slices by considering planes perpendicular to the axis of revolution, as shown in Figure 6.27.

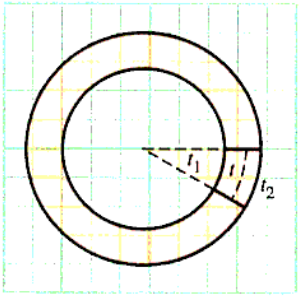



**Figure 6.27** Torus cross section

Note that each slice is thinner ($t_1$) on the inside (nearer the axis) and thicker ($t_2$) on the outside. Kepler assumed that the volume of each slice is $\pi a^2 t$ where $t = \frac{1}{2}(t_1 + t_2)$. Because $t$ is the average of its minimum and maximum thickness, it must be the thickness of the slice at its center. Use washers or shells to verify Kepler's formula for the volume of a torus.

## Level 3

57. A hemisphere of radius $r$ may be regarded as a solid whose base is the region bounded by the circle $x^2 + y^2 = r^2$ and with the property that each cross section perpendicular to the $x$-axis is a semicircle with a diameter in the base. Use this characterization and the method of cross sections to show that a sphere of radius $r$ has volume $V = \frac{4}{3}\pi r^3$.

58. A student, Frank Kornercutter, conjectures that a tetrahedron is nothing more than a solid figure with an equilateral triangular base and cross sections perpendicular to that base that are also equilateral triangles. Is Frank correct this time? Either prove that he is or show that his conjecture must be false.

59. A "spherical cap" is formed by truncating a sphere of radius $R$ at a point $h$ units from the center, as shown in Figure 6.28. Find the volume of this cap.

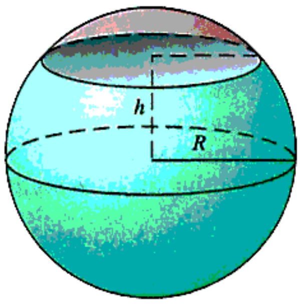



**Figure 6.28** Volume of a spherical cap

*Vol. I, James R. Newman (New York Simon & Schuster, 1956), p. 220.

**60.** The *frustum of a cone* is the solid region bounded by a cone and two parallel planes, as shown in Figure 6.29.

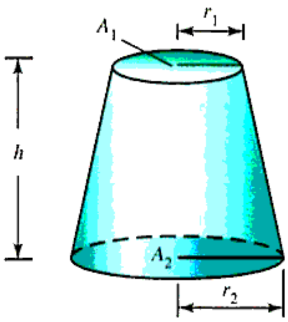



**Figure 6.29** Volume of the frustum of a cone

Suppose the planes are $h$ units apart and intersect the cone in plane regions of area $A_1$ and $A_2$, respectively. $R_1$ and $R_2$ respectively represent the radii of the circular bases.

Use integration to show that the frustum has volume

$$V = \frac{h}{3}\left(A_1 + \sqrt{A_1 A_2} + A_2\right)$$

or, equivalently,

$$V = \frac{\pi h}{3}\left(R_1^2 + R_1 R_2 + R_2^2\right)$$

## 6.3 Polar Forms and Area

**IN THIS SECTION:** *The polar coordinate system, polar graphs, summary of polar-form curves, intersection of polar-form curves, polar area*

Up to now, the only coordinate system we have used is a Cartesian coordinate system. In this section, we introduce the *polar coordinate system*, in which points are located by specifying their distance from a fixed point and their direction from a fixed line.

### The Polar Coordinate System

In the **polar coordinate system**, points are plotted in relation to a fixed point $O$, called the origin or **pole** and a fixed ray emanating from the origin called the **polar axis**. We then associate with each point $P$ in the plane an ordered pair of numbers $P(r, \theta)$, where $r$ is the distance from $O$ to $P$ and $\theta$ is the angle measured from the polar axis to the ray $OP$, as shown in Figure 6.30. The number $r$ is called the **radial coordinate** of $P$, and $\theta$ is the **polar angle**. The polar angle is regarded as positive if measured counterclockwise from the polar axis, and negative if measured clockwise. The origin $O$ has radial coordinate 0, and it is convenient to say that $O$ has polar coordinates $(0, \theta)$, for all angles $\theta$.

If the point $P$ has polar coordinates $(r, \theta)$, we say that the point $Q$ obtained by reflecting $P$ in the origin $O$ has coordinates $(-r, \theta)$. Thus, if you think of a pencil lying along the directed line segment $\overline{OP}$, with its midpoint at $O$ and tip at $P(r, \theta)$, then the eraser will be at $Q(-r, \theta)$, as illustrated in Figure 6.30. Note that $P(r, \theta)$ and $Q(-r, \theta)$ represent opposite points.

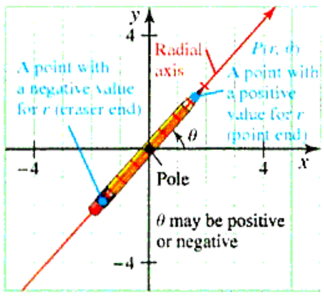



**Figure 6.30** Polar-form points

While each point in the plane is associated with exactly one pair of rectangular coordinates, a given point in polar coordinates has an infinite number of representations. For example, $(5, \frac{3\pi}{2})$, $\left(-5, \frac{\pi}{2}\right)$, and $\left(5, -\frac{\pi}{2}\right)$ all represent the same point in polar coordinates. The non-uniqueness of representation in the polar coordinate system causes some difficulties, but this can be handled by exercising a little care. We shall point out situations in our examples where the multiplicity of polar representation must be taken into account.

There is a simple trigonometric relationship linking polar and rectangular coordinates. Specifically, if we assume that the origin of the rectangular coordinate system is the pole and the positive $x$-axis is the polar axis, then the rectangular coordinates $(x, y)$ of each point are related to the polar coordinates $(r, \theta)$ of the same point by the formulas summarized in the following box.

**CHANGING COORDINATES** The procedure for changing from one coordinate system to another.

**Step 1** To change *from polar to rectangular* form use the formulas:

$$x = r\cos\theta \qquad y = r\sin\theta$$

**Step 2** To change from *rectangular to polar* form use the formulas

$$r = \sqrt{x^2+y^2} \qquad \tan\theta = \frac{y}{x}$$

if $x \neq 0$.

Note that the preceding formula gives $r$ as the distance from the pole to the point. This does **not** mean that ordered pairs in polar form need to have positive first components. For example $\left(1, \frac{3\pi}{2}\right)$ is the same point as $\left(-1, \frac{\pi}{2}\right)$.

You can remember these relationships by looking at Figure 6.31.

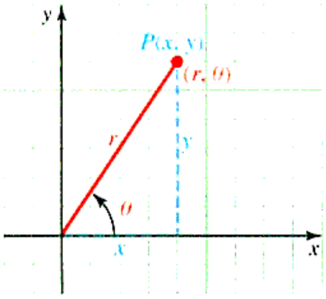


**Figure 6.31** Relationship between rectangular and polar coordinate systems

Also, note that $\tan\theta = \dfrac{y}{x}$ is the stated formula, but to find $\theta$, we first find the reference angle $\overline{\theta}$ and then place it in the proper quadrant by noting the signs of $x$ and $y$. Use $\overline{\theta} = \tan^{-1}\left|\dfrac{y}{x}\right|$, $x \neq 0$. If $x = 0$, $\overline{\theta} = \dfrac{\pi}{2}$.

**Example 1 Converting from polar to rectangular coordinates**

Change the polar coordinates $\left(-3, \frac{5\pi}{4}\right)$ to rectangular coordinates.

***Solution***
$$x = -3\cos\frac{5\pi}{4} = -3\left(-\frac{\sqrt{2}}{2}\right) = \frac{3\sqrt{2}}{2}$$
$$y = -3\sin\frac{5\pi}{4} = -3\left(-\frac{\sqrt{2}}{2}\right) = \frac{3\sqrt{2}}{2}$$

The rectangular coordinates are $\left(\frac{3\sqrt{2}}{2}, \frac{3\sqrt{2}}{2}\right)$.

**Example 2 Converting from rectangular to polar coordinates**

Write polar-form coordinates for the point with rectangular coordinates $\left(\frac{5\sqrt{3}}{2}, -\frac{5}{2}\right)$.

***Solution***
$$r = \sqrt{\left(\frac{5\sqrt{3}}{2}\right)^2 + \left(-\frac{5}{2}\right)^2} = \sqrt{\frac{75}{4}+\frac{25}{4}} = 5$$

Note that $\theta$ is in quadrant IV because $x$ is positive and $y$ is negative.

$$\overline{\theta} = \tan^{-1}\left|\frac{-\frac{5}{2}}{\frac{5\sqrt{3}}{2}}\right| = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6};\ \text{thus, } \theta = \frac{11\pi}{6}\ \text{(quadrant IV)}$$

Polar form coordinates are $\left(5, \frac{11\pi}{6}\right)$. A representation with a negative first component is $\left(-5, \frac{5\pi}{6}\right)$.

## Polar Graphs

The **graph** of an equation in polar coordinates is the set of all points $P$ whose polar coordinates $(r, \theta)$ satisfy the given equation. Circles, lines through the origin, and rays emanating from the origin have particularly simple equations in polar coordinates. *

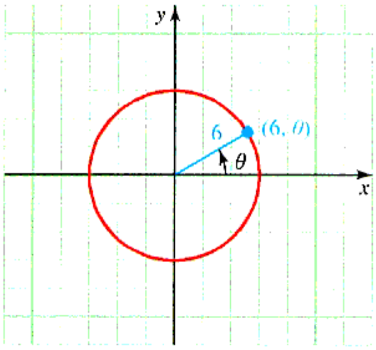
*Many books use what is called **polar graph paper**, but such paper is not really necessary. It also obscures the fact that polar curves and rectangular curves are both plotted as ordered pairs only with a different meaning attached to the ordered pairs. You can estimate the angles as necessary without polar graph paper.

**Example 3 Graphing circles, lines, and rays**

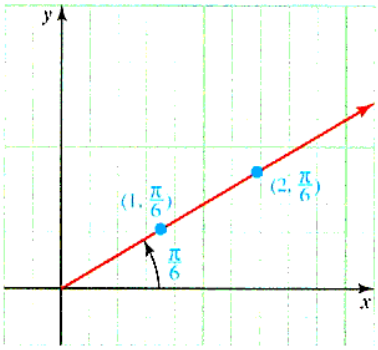
Graph: **a.** $r = 6$ **b.** $\theta = \dfrac{\pi}{6}$

***Solution***

**a.** The graph is the set of all points $(r, \theta)$ such that the first component is 6 for any angle $\theta$. This is a circle with radius 6 centered at the origin, as shown in Figure 6.32**a**.




**a.** The circle $r = 6$




**b.** The line $\theta = \dfrac{\pi}{6}$

**Figure 6.32** Two polar graphs

**b.** The graph is the set of all polar points $(r, \theta)$ with polar angle $\theta = \frac{\pi}{6}$. This is a line through the origin (pole) that makes an angle $\pi/6$ with the positive $x$-axis, as shown in Figure 6.32**b**. If we assume $r \geq 0$, then the graph is a ray (sometimes called a *half-line*).

**Example 4 Graph by converting to rectangular coordinates**

Graph $r = 4\cos\theta$ by first converting the polar form to rectangular form.

***Solution***

$$r = 4\cos\theta \quad \textit{Given}$$

$$r^2 = 4r\cos\theta \quad \textit{Multiply both sides by } r.$$

$$x^2 + y^2 = 4x \quad \textit{Change form (polar to rectangular).}$$

$$x^2 - 4x + y^2 = 0$$

$$(x-2)^2 + y^2 = 2^2 \quad \textit{Complete the square; add } 2^2 \textit{ to both sides.}$$

This is a circle with center at $(2, 0)$ and radius 2, as shown in Figure 6.33.

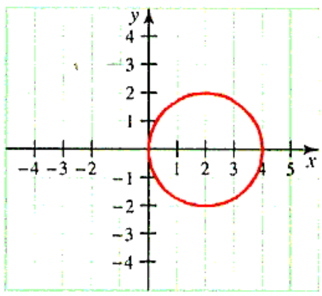



**Figure 6.33** Graph of $r = 4\cos\theta$

In our next example, we sketch a more complicated polar graph by plotting points in a polar coordinate plane. The graph is called a **cardioid** because of its heart-like shape.

**Example 5 A cardioid by plotting points**

Graph $r = 2(1 - \cos\theta)$.

***Solution*** Construct a table of values by choosing values for $\theta$ and approximating the corresponding values for $r$. Do not forget that even though we pick $\theta$ and find a value for $r$, we still represent the table values as ordered pairs of the form $(r, \theta)$.

| $\theta$ | $r$ |
|---|---|
| 0 | 0 |
| 1 | 0.9193954 |
| 2 | 2.832294 |
| 3 | 3.979985 |
| 4 | 3.307287 |
| 5 | 1.432676 |
| 6 | 0.079659 |

The points are connected as shown in Figure 6.34.

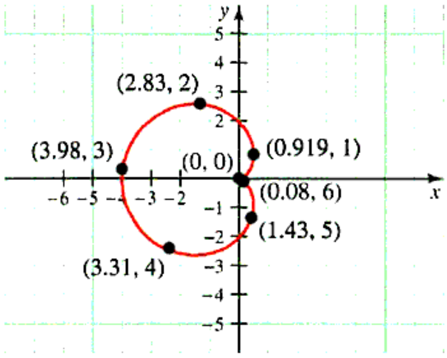
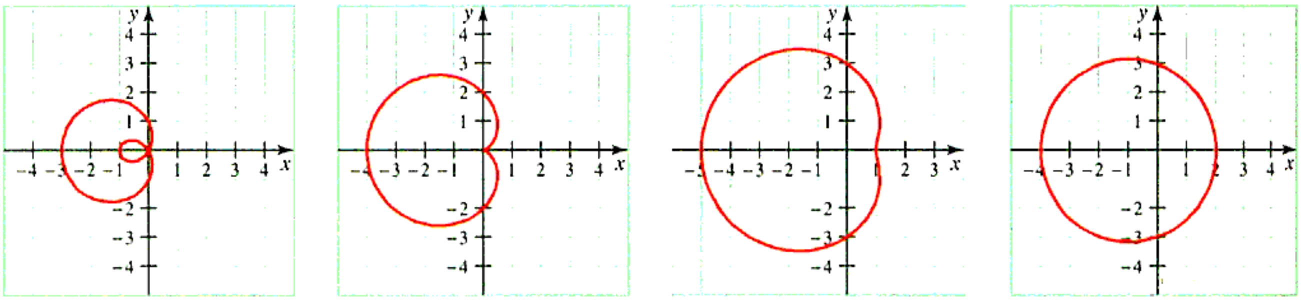



**Figure 6.34** Graph of $r = 2(1 - \cos\theta)$

The graph of any polar equation of the general form

$$r = b \pm a\cos\theta \quad \text{or} \quad r = b \pm a\sin\theta$$

is called a **limaçon** (derived from the Latin word *limax*, which means "slug"; in French, it is the word for "*snail*"). The special case where $a = b$ is the *cardioid*. Figure 6.35 shows four different kinds of limaçons that can occur. Note how the appearance of the graph depends on the ratio $b/a$.



**a.** $\frac{b}{a} < 1$; inner loop
Case I: ($r = 1 - 2\cos\theta$)

**b.** $\frac{b}{a} = 1$; cardioid
Case II: ($r = 2 - 2\cos\theta$)

**c.** $1 < \frac{b}{a} < 2$; dimple
Case III: ($r = 3 - 2\cos\theta$)

**d.** $\frac{b}{a} \geq 2$; convex
Case IV: ($r = 3 - \cos\theta$)

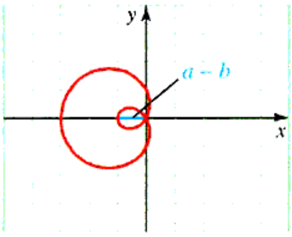
**Figure 6.35** Limaçons: $r = b \pm a\cos\theta$ or $r = b \pm a\sin\theta$

## Summary of Polar-Form Curves

There are several polar form curves which you can graph by plotting points or by using a graphing calculator (be sure to convert to polar mode). Table 6.2 gives you the names of some of the special types of polar-form curves you will encounter. There are many others, some of which are represented in the problems.

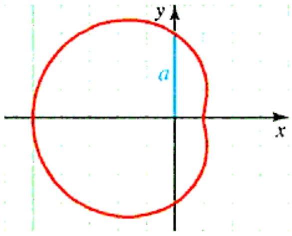
**Table 6.2** Directory of Polar-Form Curves

LIMAÇONS $r = b \pm a\cos\theta$ and $r = b \pm a\sin\theta$



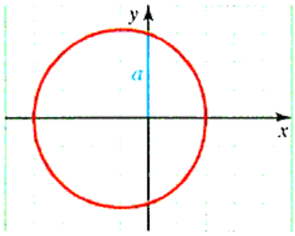

$r = b - a\cos\theta,\ \dfrac{b}{a} < 1$

standard form, inner loop



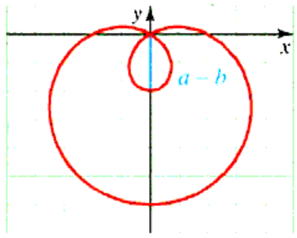

$r = b - a\cos\theta,\ 1 < \dfrac{b}{a} < 2$

standard form, dimple




$r = b - a\cos\theta,\ \dfrac{b}{a} \geq 2$

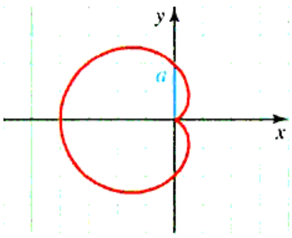
standard form, convex




$r = b - a\sin\theta,\ \dfrac{b}{a} < 1$

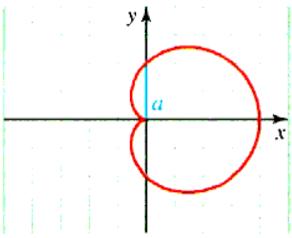
$\dfrac{\pi}{2}$ rotation; inner loop

CARDIOIDS $r = a(1 \pm \cos\theta)$ and $r = a(1 \pm \sin\theta)$ Limaçons in which $a = b$



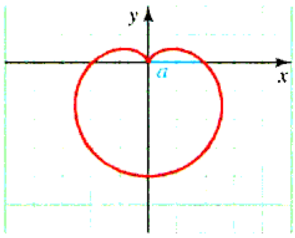

$r = a - a\cos\theta$

standard form



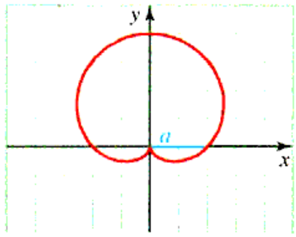

$r = a + a\cos\theta$

$\pi$ rotation




$r = a - a\sin\theta$

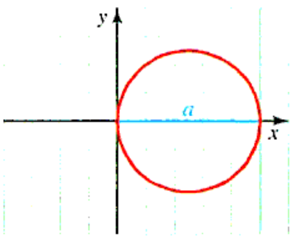
$\dfrac{\pi}{2}$ rotation




$r = a + a\sin\theta$

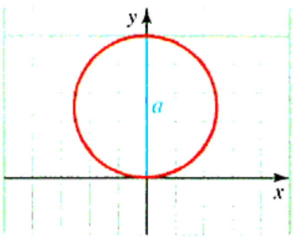
$\dfrac{3\pi}{2}$ rotation

ROSE CURVES
$r = a\cos n\theta$ and $r = a\sin n\theta$




$r = a\cos\theta$; circle

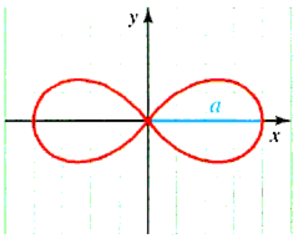
standard form; one petal




$r = a\sin\theta$; circle

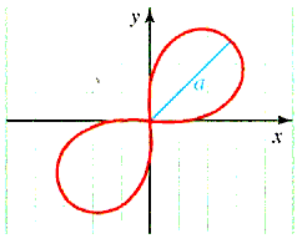
$\dfrac{\pi}{2}$ rotation; one petal

LEMNISCATES
$r^2 = a^2\cos 2\theta$ and $r^2 = a^2\sin 2\theta$



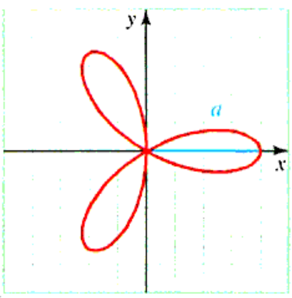

$r^2 = a^2\cos 2\theta$

standard form



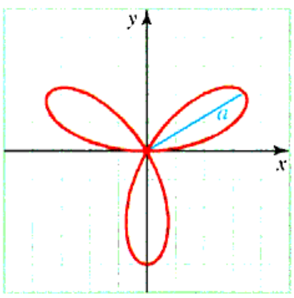

$r^2 = a^2\sin 2\theta$

$\dfrac{\pi}{4}$ rotation



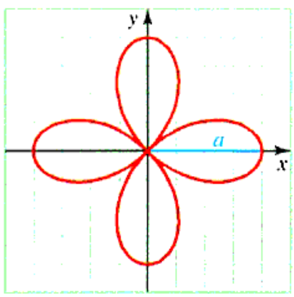

$r = a\cos 3\theta$

standard form; three petals



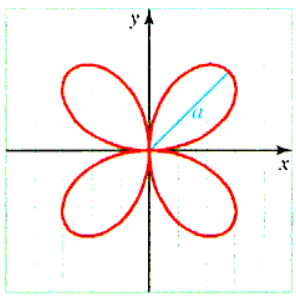

$r = a\sin 3\theta$

$\dfrac{\pi}{6}$ rotation; three petals




$r = a\cos 2\theta$

standard form; four petals




$r = a\sin 2\theta$

$\dfrac{\pi}{4}$ rotation; four petals

## Intersection of Polar-Form Curves

To find the points of intersection of graphs in rectangular form, you need only find the simultaneous solution of the equations that define those graphs. It is not even necessary to draw the graphs, because there is a one-to-one correspondence between ordered pairs satisfying an equation and points on its graph. However, in polar form, this one-to-one property is lost, so that without drawing the graphs you may fail to find all points of intersection. For example, $(-1, \pi)$ and $(1, 0)$ both represent the same point, in polar coordinates. Therefore, our method for finding the intersection of polar-form curves will include sketching the graphs.

**GRAPHICAL SOLUTION OF THE INTERSECTION OF POLAR FORM CURVES**

**Step 1** Find all simultaneous solutions of the given system of equations.
**Step 2** Determine whether the pole $r = 0$ lies on the two graphs.
**Step 3** Graph the curves to look for other points of intersection.

**Example 6 Intersection of polar form curves**

Find the points of intersection of the curves $r = \frac{3}{2} - \cos\theta$ and $\theta = \frac{2\pi}{3}$.

**Step 1** Solve the system by substitution:

$$r = \frac{3}{2} - \cos\frac{2\pi}{3} = \frac{3}{2} - \left(-\frac{1}{2}\right) = 2$$

The solution is $\left(2, \frac{2\pi}{3}\right)$.

**Step 2** If $r = 0$, the first equation has no solution because

$$0 = \frac{3}{2} - \cos\theta \qquad \text{or} \qquad \cos\theta = \frac{3}{2}$$

and a cosine cannot be larger than 1.

**Step 3** Now look at the graphs, as shown in Figure 6.36.

We see that $\left(-1, \frac{2\pi}{3}\right)$ may also be a point of intersection. It satisfies the equation $\theta = \frac{2\pi}{3}$, but what about $r = \frac{3}{2} - \cos\theta$?

$$\text{Check } \left(-1, \frac{2\pi}{3}\right) \quad -1 \stackrel{?}{=} \frac{3}{2} - \cos\frac{2\pi}{3}$$
$$= \frac{3}{2} - \left(-\frac{1}{2}\right)$$
$$= 2 \quad \text{Not satisfied}$$

However, if you check an alternative representation of $\left(-1, \frac{2\pi}{3}\right)$, we find:

$$\text{Check } \left(1, \frac{5\pi}{3}\right) \quad 1 \stackrel{?}{=} \frac{3}{2} - \cos\frac{5\pi}{3}$$
$$= \frac{3}{2} - \frac{1}{2}$$
$$= 1 \quad \text{Satisfied}$$

Be sure to check for points of intersection that you may have missed.

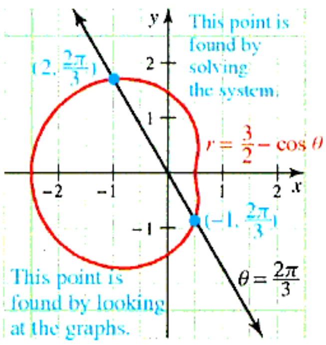



**Figure 6.36** Intersection of graphs

## Polar Area

To find the area of a region bounded by a polar graph, we use Riemann sums in much the same way as when we developed the integral formula for the area of a region described in rectangular form. However, instead of using rectangular areas as the basic units being summed, in polar form, we sum areas of *circular sectors*. A typical circular sector is shown in Figure 6.37. Recall the formula for the area of such a sector from trigonometry.

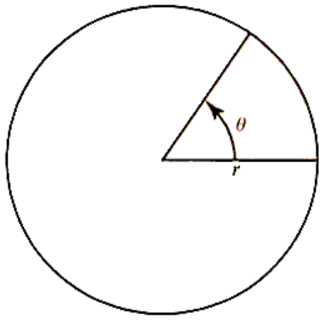



**Figure 6.37** Sector of a circle

**AREA OF A SECTOR** The area of a circular sector of radius $r$ is given by

$$A = \frac{1}{2}r^2\theta$$

where $\theta$ is the central angle of the sector measured in radians.

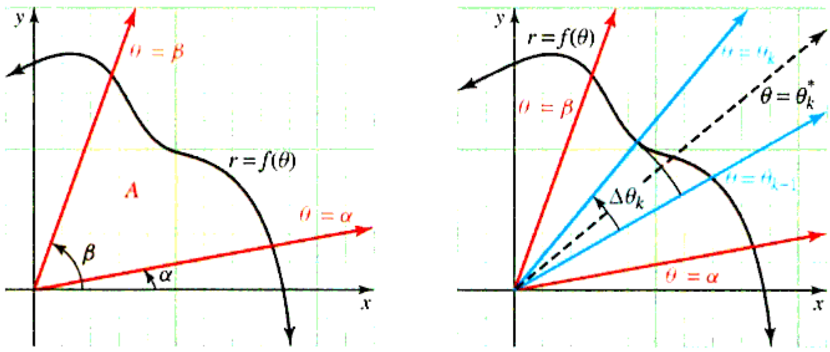
Using this formula for the area of a sector, we now state a theorem that gives us a formula for finding the area enclosed by a polar curve.

**Theorem 6.1 Area in polar coordinates**

Let $r = f(\theta)$ define a polar curve, where $f$ is continuous and $f(\theta) \geq 0$ on the closed interval $\alpha \leq \theta \leq \beta$, where $0 \leq \beta - \alpha \leq 2\pi$. Then the region bounded by the curve $r = f(\theta)$ and the rays $\theta = \alpha$ and $\theta = \beta$ has area

$$A = \frac{1}{2}\int_{\alpha}^{\beta} r^2 d\theta = \frac{1}{2}\int_{\alpha}^{\beta} [f(\theta)]^2\, d\theta$$

***Proof:*** The region is shown in Figure 6.38. To find the area bounded by the graphs of the polar functions, partition the region between $\theta = \alpha$ and $\theta = \beta$ by a collection of rays, say $\theta_0, \theta_1, \cdots, \theta_n$. Let $\alpha = \theta_0$ and $\beta = \theta_n$.




**a.** The region bounded by the polar curve $r = f(\theta)$ and the rays $\theta = a$ and $\theta = b$

**b.** The area $A$ can be estimated by adding the area of "small" (blue) circular sectors

**Figure 6.38** Area in polar form

Pick any ray $\theta = \theta_k^*$, with $\theta_{k-1} \leq \theta_k^* \leq \theta_k$. Then the area $\Delta A_k$ of the circular sector is approximately the same as the area of the region bounded by the graph of $f$ and the lines $\theta = \theta_{k-1}$ and $\theta = \theta_k$. Because this circular sector has radius $f(\theta_k^*)$ and central angle $\Delta\theta_k$, its area is

$$\Delta A_k \approx \frac{1}{2}(\text{radius})^2(\text{central angle}) = \frac{1}{2}\left[f(\theta_k^*)\right]^2 \Delta\theta_k$$

The sum $\sum_{k=1}^{n} \Delta A_k$ is an approximation to the total area $A$ bounded by the polar curve, and by taking the limit as $n \to \infty$, we obtain

$$\begin{aligned} A &= \lim_{n\to\infty} \sum_{k=1}^{n} \Delta A_k \\ &= \lim_{n\to\infty} \frac{1}{2} \sum_{k=1}^{n} \left[f(\theta_k^*)\right]^2 \Delta\theta_k \\ &= \frac{1}{2}\int_{\alpha}^{\beta} \left[f(\theta)\right]^2 d\theta \qquad \textit{Riemann sum} \end{aligned}$$

You will probably find that the most difficult part of the problem is deciding on the limits of integration. A sketch of the region will help with this.

**Example 7 Finding area of part of a cardioid**

Find the area of the top half ($0 \le \theta \le \pi$) of the cardioid $r = 1 + \cos\theta$.

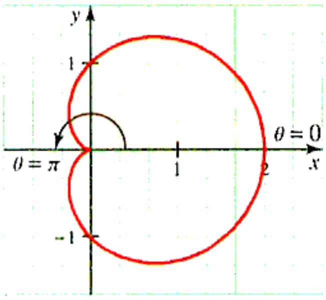



**Figure 6.39** Area of top half of a cardioid

***Solution*** The cardioid is shown in Figure 6.39.

Note that the top half of the graph lies between the rays $\theta = 0$ and $\theta = \pi$. Hence the required area is given by:

$$\begin{aligned} A &= \frac{1}{2}\int_0^{\pi} (1+\cos\theta)^2\, d\theta \\ &= \frac{1}{2}\int_0^{\pi} (1 + 2\cos\theta + \cos^2\theta)\, d\theta \\ &= \frac{1}{2}\int_0^{\pi} \left(1 + 2\cos\theta + \frac{1+\cos 2\theta}{2}\right) d\theta \qquad \textit{Trigonometric half-angle identity} \\ &= \frac{1}{2}\left[\theta + 2\sin\theta + \frac{\theta}{2} + \frac{\sin 2\theta}{4}\right]_0^{\pi} \\ &= \frac{1}{2}\left[\pi + 2(0) + \frac{\pi}{2} + \frac{0}{4} - 0\right] \\ &= \frac{3\pi}{4} \end{aligned}$$

**Example 8 Finding area enclosed by a four-leaved rose**

Find the area enclosed by the four-leaved rose $r = \cos 2\theta$.

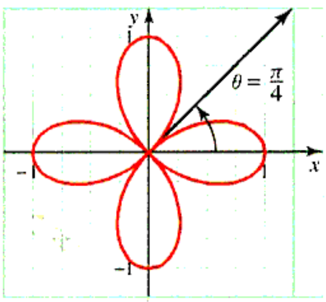



**Figure 6.40** Area enclosed by a four-leaved rose

***Solution*** The rose curve is shown in Figure 6.40.

We will find the area of the top half of the right loop (shaded portion) to make sure $f(\theta) \ge 0$. We see that this corresponds to angles with measures from $\theta = 0$ to $\theta = \frac{\pi}{4}$. If you have a calculator, the easiest way to see this is to plot the graph and then use the *trace* feature to find the $\theta$-values for the top half of the first leaf. By symmetry, the entire area enclosed by the four-leaved rose is 8 times the shaded region. Thus, the required area is given by

$$\begin{aligned} A &= 8\left[\frac{1}{2}\int_0^{\frac{\pi}{4}} \cos^2 2\theta\, d\theta\right] \\ &= 4\int_0^{\frac{\pi}{4}} \frac{1+\cos 4\theta}{2}\, d\theta \qquad \textit{Trigonometric half-angle identity} \end{aligned}$$

$$= 4\left[\frac{\theta}{2} + \frac{\sin 4\theta}{8}\right]_0^{\frac{\pi}{4}}$$
$$= 4\left[\frac{\pi}{8} + 0 - 0\right]$$
$$= \frac{\pi}{2}$$

**Example 9 Finding the area of a region between two polar curves**

Find the area of the region common to the circles $r = a\cos\theta$ and $r = a\sin\theta$.

***Solution*** The circles are shown in Figure 6.41.

Solve $a\cos\theta = a\sin\theta$ to find that the circles intersect at $\theta = \frac{\pi}{4}$ and at the pole. To set up the integrals properly, remember to think in terms of polar coordinates and not in terms of rectangular coordinates. Specifically, we must **scan radially**. This means that we need to find the area of intersection by adding the areas of the regions marked $R_1$ and $R_2$. Region $R_1$ is bounded by the circle $r = a\sin\theta$ and the rays $\theta = 0$, $\theta = \frac{\pi}{4}$. Region $R_2$ is bounded by the circle $r = a\cos\theta$ and $\theta = \frac{\pi}{4}$, $\theta = \frac{\pi}{2}$. Note the rays $\theta = 0$ and $\theta = \frac{\pi}{2}$ are not necessary as geometric boundaries for $R_1$ and $R_2$, but they are necessary to describe which part of each circle is being calculated.

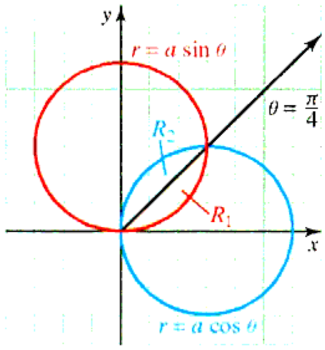


**Figure 6.41** Area bounded by two polar curves

$$A = \text{AREA OF } R_1 + \text{AREA OF } R_2$$
$$= \frac{1}{2}\int_0^{\frac{\pi}{4}} a^2\sin^2\theta\, d\theta + \frac{1}{2}\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} a^2\cos^2\theta\, d\theta$$
$$= \frac{a^2}{2}\left[\frac{\theta}{2} - \frac{\sin 2\theta}{4}\right]_0^{\frac{\pi}{4}} + \frac{a^2}{2}\left[\frac{\theta}{2} + \frac{\sin 2\theta}{4}\right]_{\frac{\pi}{4}}^{\frac{\pi}{2}} \qquad \textit{Trigonometric half-angle identities}$$
$$= \frac{a^2}{2}\left[\frac{\pi}{8} - \frac{1}{4} - 0\right] + \frac{a^2}{2}\left[\frac{\pi}{4} + 0 - \frac{\pi}{8} - \frac{1}{4}\right]$$
$$= \frac{a^2}{2}\left[\frac{\pi}{4} - \frac{1}{2}\right]$$
$$= \frac{1}{8}a^2(\pi - 2)$$

**Example 10 Finding the area between a circle and a limaçon**

Find the area between the circle $r = 5\cos\theta$ and the limaçon $r = 2 + \cos\theta$. Round your answer to the nearest hundredth of a square unit.

***Solution*** As usual, begin by drawing the graphs, as shown in Figure 6.42.

Note that both the limaçon and circle are symmetric with respect to the $x$-axis, so we can find the area in the first quadrant and multiply by 2.

Next, we need to find the points of intersection. We see that the curves do not intersect at the pole.

Now solve

$$5\cos\theta = 2 + \cos\theta$$
$$\cos\theta = \frac{1}{2}$$
$$\theta = \frac{\pi}{3} \qquad \textit{Solution in the first quadrant}$$

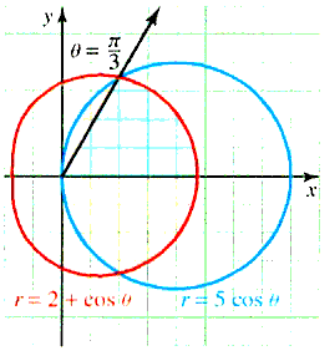


**Figure 6.42** Area bounded by a circle and a limaçon

To find the area, divide the region into two parts, along the ray $\theta = \frac{\pi}{3}$. The right part (blue region above the $x$-axis) is bounded by the limaçon $r = 2 + \cos\theta$ and the rays $\theta = 0$ and $\theta = \frac{\pi}{3}$. The left part (shaded region shown in pink above the $x$-axis) is bounded by the circle $r = 5\cos\theta$ and the rays $\theta = \frac{\pi}{3}$ and $\theta = \frac{\pi}{2}$. Using this preliminary information, we can now set up and evaluate a sum of integrals for the required area

$$\begin{aligned}
A &= 2[\text{AREA OF RIGHT PART} + \text{AREA OF LEFT PART}]\\
&= 2\left[\frac{1}{2}\int_0^{\frac{\pi}{3}} (2+\cos\theta)^2 d\theta + \frac{1}{2}\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} (5\cos\theta)^2 d\theta\right]\\
&= \int_0^{\frac{\pi}{3}} (4 + 4\cos\theta + \cos^2\theta)\,d\theta + \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} 25\cos^2\theta\,d\theta\\
&= \left[4\theta + 4\sin\theta + \frac{\theta}{2} + \frac{\sin 2\theta}{4}\right]_0^{\frac{\pi}{3}} + 25\left[\frac{\theta}{2} + \frac{\sin 2\theta}{4}\right]_{\frac{\pi}{3}}^{\frac{\pi}{2}}\\
&= \left[\frac{4\pi}{3} + \frac{4\sqrt{3}}{2} + \frac{\pi}{6} + \frac{\sqrt{3}}{8} - 0\right] + 25\left[\frac{\pi}{4} + 0 - \frac{\pi}{6} - \frac{\sqrt{3}}{8}\right]\\
&= \frac{43\pi}{12} - \sqrt{3}\\
&\approx 9.53
\end{aligned}$$

## PROBLEM SET 6.3

### Level 1

1. ■ What does this say? Discuss a procedure for finding the intersection of polar-form curves.
2. ■ What does this say? Discuss a procedure for finding the area enclosed by a polar curve.
3. Identify each of the curves as a cardioid, rose curve (state number of petals), lemniscate, limaçon, circle, line, or none of the above.

   **a.** $r^2 = 9\cos 2\theta$ **b.** $r = 2\sin\frac{\pi}{6}$
   **c.** $r = 3\sin 3\theta$ **d.** $r = 3\theta$
   **e.** $r = 2 - 2\cos\theta$ **f.** $\theta = \frac{\pi}{6}$
   **g.** $r^2 = \sin 2\theta$ **h.** $r - 2 = 4\cos\theta$

4. Identify each of the curves as a cardioid, rose curve (state number of petals), lemniscate, limaçon, circle, line, or none of the above.

   **a.** $r = 2\sin 2\theta$ **b.** $r^2 = 2\cos 2\theta$
   **c.** $r = 5\cos 60^\circ$ **d.** $r = 5\sin 8\theta$
   **e.** $r\theta = 3$ **f.** $r^2 = 9\cos(2\theta - \frac{\pi}{4})$
   **g.** $r = \sin 3\left(\theta + \frac{\pi}{6}\right)$ **h.** $\cos\theta = 1 - r$

5. Identify each of the curves as a cardioid, rose curve (state number of petals), lemniscate, limaçon, circle, line, or none of the above.

   **a.** $r = 2\cos 2\theta$ **b.** $r = 4\sin 30^\circ$
   **c.** $r + 2 = 3\sin\theta$ **d.** $r + 3 = 3\sin\theta$
   **e.** $\theta = 4$ **f.** $\theta = \tan\frac{\pi}{4}$
   **g.** $r = 3\cos 5\theta$ **h.** $r\cos\theta = 2$

*Graph the polar-form curves given in Problems 6-21.*

6. $r = 3,\ 0 \le \theta \le \frac{\pi}{2}$
7. $\theta = -\frac{\pi}{2},\ 0 \le r \le 3$
8. $r = 2\theta,\ \theta \ge 0$
9. $r = \theta + 1,\ 0 \le \theta \le \pi$
10. $r = 2\cos 2\theta$
11. $r = 5\sin 3\theta$
12. $r^2 = 16\cos 2\theta$
13. $r = 3\cos 3\left(\theta - \frac{\pi}{3}\right)$
14. $r = 5\cos 3\left(\theta - \frac{\pi}{4}\right)$
15. $r = \sin\left(2\theta + \frac{\pi}{3}\right)$
16. $r = 2 + \cos\theta$
17. $r^2 = 16\cos 2\left(\theta - \frac{\pi}{6}\right)$
18. $r = 1 + \sin\theta$
19. $r\cos\theta = 2$
20. $r = 1 + 3\cos\theta$
21. $r = -2\sin\theta$

*Find the points of intersection of the curves given in Problems 22-29.*

22. $\begin{cases} r = 4\cos\theta \\ r = 4\sin\theta \end{cases}$
23. $\begin{cases} r = 4\sin\theta \\ r = 2 \end{cases}$
24. $\begin{cases} r^2 = 9\cos 2\theta \\ r = 3 \end{cases}$
25. $\begin{cases} r = 3\theta \\ \theta = \frac{\pi}{3} \end{cases}$
26. $\begin{cases} r = 2(1 + \sin\theta) \\ r = 2(1 - \sin\theta) \end{cases}$
27. $\begin{cases} r^2 = \sin 2\theta \\ r = \sqrt{2}\sin\theta \end{cases}$
28. $\begin{cases} r = 2(1 - \cos\theta) \\ r = 4\sin\theta \end{cases}$
29. $\begin{cases} r = \dfrac{4}{1 - \cos\theta} \\ r = 2\cos\theta \end{cases}$

## Level 2

*Find the area of each polar region enclosed by* $f(\theta)$, $\theta = a, \theta = b$, *for* $a \le \theta \le b$ *in Problems* 30-37.

**30.** $f(\theta) = \sin\theta, 0 \le \theta \le \frac{\pi}{6}$

**31.** $f(\theta) = \cos\theta, 0 \le \theta \le \frac{\pi}{6}$

**32.** $f(\theta) = \sec\theta, -\frac{\pi}{4} \le \theta \le \frac{\pi}{4}$

**33.** $f(\theta) = \sqrt{\sin\theta}, \frac{\pi}{6} \le \theta \le \frac{\pi}{2}$

**34.** $f(\theta) = e^{\theta/2}, 0 \le \theta \le 2\pi$

**35.** $f(\theta) = \sin\theta + \cos\theta, 0 \le \theta \le \frac{\pi}{4}$

**36.** $f(\theta) = \frac{\theta}{\pi}, 0 \le \theta \le 2\pi$

**37.** $f(\theta) = \frac{\theta^2}{\pi}, 0 \le \theta \le 2\pi$

**38.** **Spirals** are interesting mathematical curves.



**Spiral staircase**

There are three special types of spirals:

**a.** A **spiral of Archimedes** has the form $r = a\theta$. Graph $r = 2\theta$.

**b.** A **hyperbolic spiral** has the form $r\theta = a$. Graph $r\theta = 2$.

**c.** A **logarithmic spiral** has the form $r = a^{k\theta}$. Graph $r = 2^{\theta}$.

**39.** The **strophoid** is a curve of the form $r = a\cos 2\theta \sec\theta$. Graph this curve for $a = 2$.

**40.** The **bifolium** has the form $r = a\sin\theta\cos^2\theta$. Graph this curve for $a = 1$.

**41.** The **folium of Descartes** has the form $r = \dfrac{3a\sin\theta\cos\theta}{\sin^3\theta + \cos^3\theta}$. Graph this curve for $a = 2$.

**42.** Find the area of one loop of the four-leaved rose $r = 2\sin 2\theta$.

**43.** Find the area enclosed by the three-leaved rose $r = a\sin 3\theta$.

**44.** Find the area of the region that is inside the circle $r = 4\cos\theta$ and outside the circle $r = 2$.

**45.** Find the area of the region that is inside the circle $r = a$ and outside the cardioid $r = a(1 - \cos\theta)$.

**46.** Find the area of the region that is inside the circle $r = \sin\theta$ and outside the cardioid $r = 1 - \cos\theta$.

**47.** Find the area of the region that is inside the circle $r = 6\cos\theta$ and outside the cardioid $r = 2(1 + \cos\theta)$.

**48.** Find the area of the portion of the lemniscate $r^2 = 8\cos 2\theta$ that lies in the region $r \ge 2$.

**49.** Find the area between the inner and outer loops of the limaçon $r = 2 - 4\sin\theta$.

**50.** Find the area to the right of the line $r\cos\theta = 1$ and inside the lemniscate $r^2 = 2\cos 2\theta$.

**51.** Find the area to the right of the line $r\cos\theta = 1$ and inside the lemniscate $r^2 = 2\sin 2\theta$.

**52.** Find the maximum value of the $y$-coordinate of points on the limaçon $r = 2 + 3\cos\theta$.

**53.** Find the maximum value of the $x$-coordinate of points on the cardioid $r = 3 + 3\sin\theta$.

## Level 3

**54.** **a.** Show that if the polar curve $r = f(\theta)$ is rotated about the pole through an angle $\alpha$, the equation for the new curve is $r = f(\theta - \alpha)$.

**b.** Use a rotation to sketch the curve $r = 2\sec\left(\theta - \frac{\pi}{3}\right)$.

**55.** The **ovals of Cassini** have the polar form

$$r^4 + b^4 - 2b^2r^2\cos 2\theta = k^4.$$

Graph the curve, where $b = 2, k = 3$.

**56.** Sketch the graph of

$$r = \frac{\theta}{\cos\theta}$$

for $0 \le \theta \le \frac{\pi}{2}$. In particular, show that the graph has a vertical asymptote at $x = \frac{\pi}{2}$.

**57.** If $f$ is a differentiable function of $\theta$, then show that the tangent line to the polar curve $r = f(\theta)$ at the point $P(r, \theta)$ has slope

$$m = \frac{f(\theta)\cos\theta + f'(\theta)\sin\theta}{-f(\theta)\sin\theta + f'(\theta)\cos\theta}$$

whenever the denominator is not zero.

*Hint*: $\dfrac{dy}{dx} = \dfrac{dy/d\theta}{dx/d\theta}$.

**58.** Because the formula for the slope in polar-form is complicated (see Problem 57), it is more convenient to measure the inclination of the tangent line to a polar curve in terms of the angle $\alpha$ extended from the radial line to the tangent line, as shown in Figure 6.43.

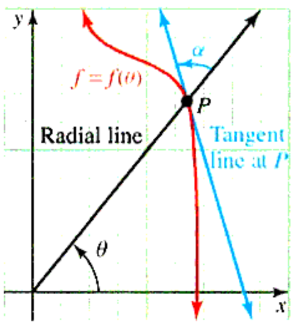



**Figure 6.43** Problem 58

Let $P$ be a point on the polar curve $r = f(\theta)$, and let $\alpha$ be the angle extending from the radial line to the tangent line, as shown in Figure 6.43. Assuming that $f'(\theta) \neq 0$, show that

$$\tan\alpha = \frac{f(\theta)}{f'(\theta)}$$

*Hint*: Use the formula

$$\tan\alpha = \tan(\phi - \theta) = \frac{\tan\phi - \tan\theta}{1 + \tan\phi\tan\theta}.$$

**59.** Use the formula in Problem 58 to find $\tan\alpha$ for each of the following:

**a.** the circle $r = a\cos\theta$
**b.** the cardioid $r = 2(1 - \cos\theta)$
**c.** the logarithmic spiral $r = 2e^{3\theta}$

**60.** **Journal Problem** (*School Science and Mathematics**) Graph the polar curve

$$r = 4 + 2\sin\frac{5\theta}{2}$$

and find the total area interior to this curve. Then find the area of the "star."

## 6.4 Arc Length and Surface Area

**IN THIS SECTION:** *The arc length of a curve, the area of a surface of revolution, polar arc length and surface area*

If you were asked to measure the length of a given curve by hand, you might proceed by first fitting a piece of string to the curve and then measuring its length with a ruler. In this section, we see how such measurements can be carried out mathematically using integration.

### The Arc Length of a Curve

If a function $f$ has a derivative that is continuous on some interval, then $f$ is said to be **continuously differentiable** on the interval. The portion of the graph of a continuously differentiable function $f$ that lies between $x = a$ and $x = b$ is called the **arc** of the graph on the interval $[a, b]$. To find, and define, the length of this arc, let $P$ be a partition of the interval $[a, b]$, with subdivision points $x_0, x_1, \cdots, x_n$, where $x_0 = a$ and $x_n = b$. Let $P_k$ denote the point $(x_k, y_k)$ on the graph, where $y_k = f(x_k)$. By joining the points $P_0, P_1, \cdots, P_n$, we obtain a polygonal path whose length approximates that of the arc. Figure 6.44 demonstrates the labeling for the case where $n = 6$.

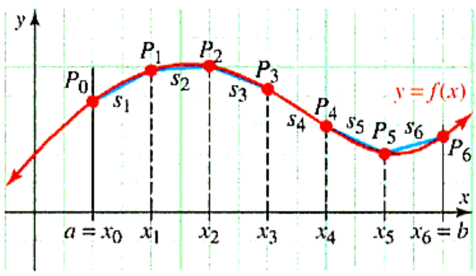



**Figure 6.44 Interactive** A polygonal path approximating an arc of a curve

The length of the polygonal path connecting the points $P_0, P_1, \cdots, P_n$ on the graph of $f$ is the sum

$$\sum_{k=1}^{n} \Delta s_k$$

**School Science and Mathematics*, Problem 3949 by V. C. Bailey, Vol. 83, 1983, p. 356.

where $\Delta s_k$ is the length of the segment joining $P_{k-1}$ to $P_k$. By applying the distance formula and rearranging terms with $\Delta x_k = x_k - x_{k-1}$ and $\Delta y_k = y_k - y_{k-1}$ we find that

$$\begin{aligned}\Delta s_k &= \sqrt{(x_k - x_{k-1})^2 + (y_k - y_{k-1})^2} \\ &= \sqrt{(\Delta x_k)^2 + (\Delta y_k)^2} \\ &= \sqrt{\frac{(\Delta x_k)^2 + (\Delta y_k)^2}{(\Delta x_k)^2}}\,\Delta x_k \\ &= \sqrt{1 + \left(\frac{\Delta y_k}{\Delta x_k}\right)^2}\,\Delta x_k\end{aligned}$$

It is reasonable to expect a connection between the ratio $\Delta y_k/\Delta x_k$ and the derivative $dy/dx = f'(x)$. Indeed, using the MVT, it can be shown (see Problem 55) that

$$\Delta s_k = \sqrt{1 + \left[f'(x_k^*)\right]^2}\,\Delta x_k$$

for some number $x_k^*$ between $x_{k-1}$ and $x_k$. Therefore, the arc length of the graph of $f$ on the interval $[a, b]$ may be estimated by the Riemann sum

$$\sum_{k=1}^{n} \sqrt{1 + \left[f'(x_k^*)\right]^2}\,\Delta x_k$$

This estimate may be improved by increasing the number of subdivision points in the partition $P$ of the interval $[a, b]$ in such a way that the subinterval lengths tend to zero. Thus, it is reasonable to define the actual arc length to be the limit

$$\lim_{\|P\|\to 0} \sum_{k=1}^{n} \sqrt{1 + \left[f'(x_k^*)\right]^2}\,\Delta x_k$$

Notice that if $f'$ is continuous on the interval $[a, b]$, then so is $\sqrt{1 + \left[f'(x)\right]^2}$. Thus, this limit exists and is the integral of $\sqrt{1 + \left[f'(x)\right]^2}$ with respect to $x$ on the interval $[a, b]$. These observations lead us to the following definition.*

**ARC LENGTH** Let $f$ be a function whose derivative $f'$ is continuous on the interval $[a, b]$ and differentiable on $(a, b)$. Then the **arc length**, $s$, of the graph of $y = f(x)$ between $x = a$ and $x = b$ is given by the integral

$$s = \int_a^b \sqrt{1 + \left[f'(x)\right]^2}\,dx$$

Similarly, for the graph of $x = g(y)$, where $g'$ is continuous on the interval $[c, d]$, the arc length from $y = c$ to $y = d$ is

$$s = \int_c^d \sqrt{1 + \left[g'(y)\right]^2}\,dy$$

*Here is an easy way to remember the arc length formula:*

$$\begin{aligned} ds &= \sqrt{(dx)^2 + (dy)^2} \\ &= \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx \\ &= \sqrt{\left(\frac{dx}{dy}\right)^2 + 1}\,dy.\end{aligned}$$

*We have used integration to obtain a meaningful definition of arc length for a function $f(x)$ with a continuous derivative $f'(x)$. It is possible to define arc length for certain curves $y = f(x)$ when $f(x)$ does not have a continuous derivative, but we will not pursue this more general topic.

**Example 1 Arc length of a curve**

Find the arc length (rounded to two decimal places) of the curve $y = x^{3/2}$ on the interval $[0, 4]$.

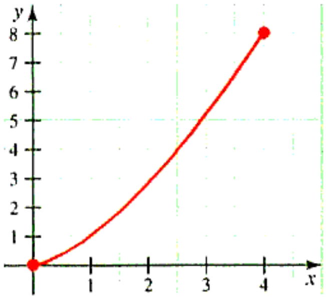


**Figure 6.45** What is the length of this curve?

***Solution*** The graph is shown in Figure 6.45.

Let $f(x) = x^{3/2}$; therefore, $f'(x) = \frac{3}{2}x^{1/2}$.

$$\begin{aligned} s &= \int_0^4 \sqrt{1 + \left[\frac{3}{2}x^{1/2}\right]^2}\,dx \\ &= \int_0^4 \sqrt{1 + \frac{9}{4}x}\;dx \\ &= \left[\frac{4}{9}\cdot\frac{2}{3}\left(1 + \frac{9}{4}x\right)^{3/2}\right]_0^4 \\ &= \frac{8}{27}\left[10^{3/2} - 1^{3/2}\right] \\ &\approx 9.0734 \end{aligned}$$

**Example 2 Arc length of a curve $x = g(y)$**

Find the arc length of the curve $x = \frac{1}{3}y^3 + \frac{1}{4}y^{-1}$ from $y = 1$ to $y = 3$.

***Solution*** Because $g(y) = \frac{1}{3}y^3 + \frac{1}{4}y^{-1}$, we have $g'(y) = y^2 - \dfrac{1}{4}y^{-2} = \dfrac{4y^4 - 1}{4y^2}$, which is continuous throughout the interval from $y = 1$ to $y = 3$. Therefore, the arc length is

$$\begin{aligned} s &= \int_1^3 \sqrt{1 + [g'(y)]^2}\,dy \\ &= \int_1^3 \sqrt{1 + \left(\frac{4y^4 - 1}{4y^2}\right)^2}\,dy \\ &= \int_1^3 \sqrt{1 + \frac{16y^8 - 8y^4 + 1}{16y^4}}\,dy \\ &= \int_1^3 \sqrt{\frac{(4y^4 + 1)^2}{(4y^2)^2}}\,dy \\ &= \int_1^3 \frac{4y^4 + 1}{4y^2}\,dy \\ &= \int_1^3 \left(y^2 + \frac{1}{4}y^{-2}\right)dy \\ &= \left[\frac{1}{3}y^3 + \frac{1}{4}\cdot\frac{y^{-1}}{-1}\right]_1^3 \\ &= \frac{53}{6} \end{aligned}$$

**Example 3 Estimating arc length using numerical integration**

Find the length of the curve defined by $y = \sin x$ on $[0, 2\pi]$.

***Solution*** The region is shown in Figure 6.46.

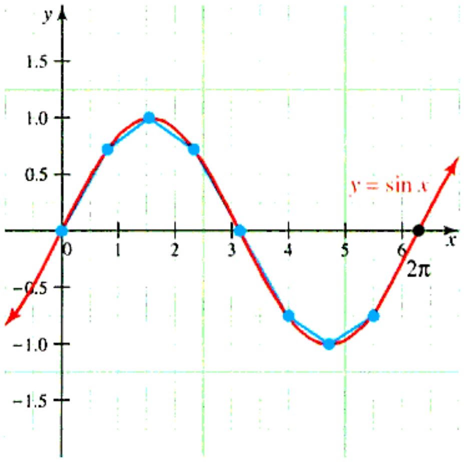



**Figure 6.46 Interactive** Finding the length of an arc (8 subdivisions shown)

Because $y' = \cos x$, we have, from the arc length formula,

$$s = \int_0^{2\pi} \sqrt{1 + \cos^2 x}\, dx$$

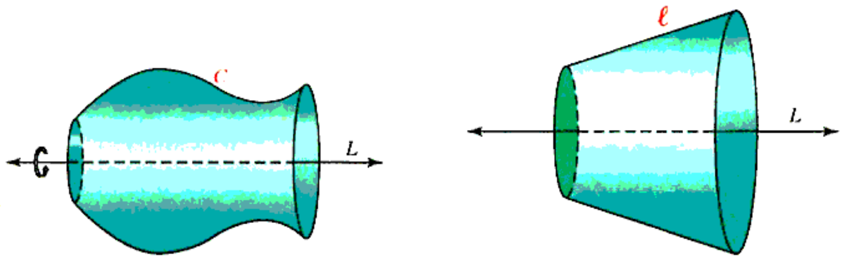
We do not have techniques to allow us to evaluate this integral, so we turn to numerical integration. If we consider four rectangles, we find, by various methods:

| Method ($n = 4$) | Approximation |
|---|---|
| Rectangles: | |
| Left endpoints | 7.584476 |
| Right endpoints | 7.584476 |
| Midpoints | 7.695299 |
| Trapezoidal rule | 7.584476 |
| Simpson's rule | 7.150712 |

In practice, you would choose just *one* of the methods whose approximate values are given. You might also note that, for this example, Simpson's rule performs worse than the trapezoidal, midpoint, or even the left- and right-endpoint methods. For more accurate results, you might wish to use a computer program. The actual arc length (rounded to six places) computed as an area is 7.640396.

## The Area of a Surface of Revolution

When the arc of a curve is revolved about a line $L$ it generates a surface called a **surface of revolution**, as shown in Figure 6.47**a**. In particular, if the generating arc is a line segment, the frustum of a cone is generated, as shown in Figure 6.47**b**.



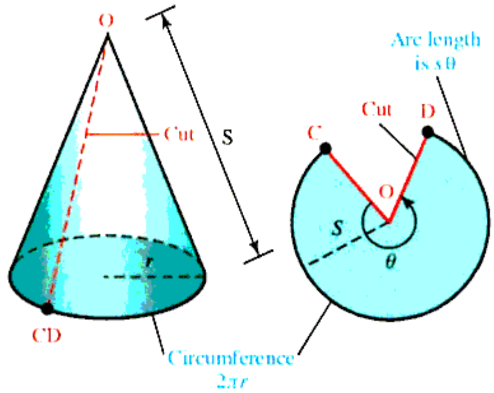

**a.** The surface generated by revolving the curve $C$ about a line $L$

**b.** The line segment of length $\ell$ generates the frustum of a cone

**Figure 6.47** Surfaces of revolution

We will obtain a formula for the lateral area of the frustum of a cone, which will then be used as part of an approximation scheme for the area of a general surface of revolution. Accordingly, first note that a cone with slant height $s$ and base radius $r$ has lateral surface area is $A = \pi rs$. To see this, cut the cone in Figure 6.48**a** along the dashed line and flatten it out as shown in Figure 6.48**b**. The angle $\theta$ is determined by noting that the circumference of the circular base of the cone is $2\pi r$, which is the same as the arc length $s\theta$ after the cone is flattened. Thus, $2\pi r = s\theta$, and $\theta = \dfrac{2\pi r}{s}$. The lateral surface area of the cone is the area of sector $COD$; namely,

$$A = \frac{1}{2}s^2\theta = \frac{1}{2}s^2\left(\frac{2\pi r}{s}\right) = \pi rs$$




**a.** A cone with base radius $r$ and slant height $s$ is cut along the line from $CD$ to $O$

**b.** The flattened cone is a sector with central angle $\theta$ in a circle of radius $s$

**Figure 6.48** A cone with base radius $r$ and slant height s has area $A = \pi rs$

Finally, the frustum of the cone shown in Figure 6.49 has slant height $\ell$, top radius $r_1$, and bottom radius $r_2$, and can be formed by removing a small cone of base radius $r_1$ and slant height $\ell_1$ from the larger cone of base radius $r_2$ and slant height $\ell_2 = \ell_1 + \ell$.

Thus, the lateral surface area $F$ of the frustum can be found by subtracting the area of the smaller cone from that of the larger:

$$\begin{aligned} F &= \pi r_2\ell_2 - \pi r_1\ell_1 \\ &= \pi r_2(\ell_1 + \ell) - \pi r_1\ell_1 \\ &= \pi[(r_2 - r_1)\ell_1 + r_2\ell] \end{aligned}$$

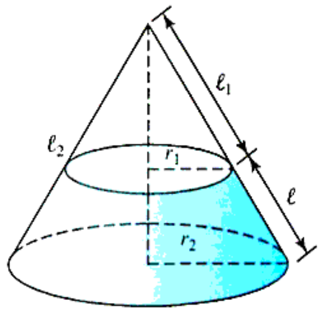



**Figure 6.49** Frustum of a cone

Using similar triangles, we see that

$$\frac{\ell_1}{r_1} = \frac{\ell_1 + \ell}{r_2} \qquad \text{or} \qquad r_1\ell = (r_2 - r_1)\ell_1$$

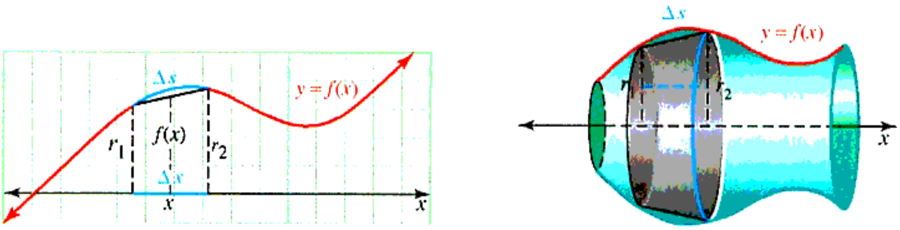
so that

$$\begin{aligned} F &= \pi[(r_2 - r_1)\ell_1 + r_2\ell] \\ &= \pi[r_1\ell + r_2\ell] && \textit{By substitution} \\ &= 2\pi\left(\frac{r_1 + r_2}{2}\right)\ell \end{aligned}$$

*You might remember the formula* $F = \pi(r_1 + r_2)\ell$.

where $\frac{1}{2}(r_1 + r_2)$ is the average of the radii of the frustum.

Next, assuming that the function $f(x)$ is continuous with $f(x) \geq 0$ on the interval $[a, b]$, we will derive a formula for the area of the surface generated by revolving the curve $y = f(x)$ about the $x$-axis. Rather than argue formally with Riemann sums, we will proceed heuristically. Figure 6.50**a** shows an approximating line segment with length $\Delta s$, whose endpoints are, respectively, distances $r_1$ and $r_2$ above the $x$-axis, and Figure 6.50**b** shows the surface of revolution.




**a.** The approximating line segment of length $\Delta s$ is at an average height $f(x)$ above the $x$-axis

**b.** Revolving the segment about the $x$-axis generates a frustum (gray) used to approximate surface area

**Figure 6.50** Approximating the area of a surface

Let $f(x)$ be the average height of the segment above the $x$-axis, we generate the frustum of a cone, with surface area

$$\Delta S = 2\pi\left(\frac{r_1 + r_2}{2}\right)\Delta s = 2\pi f(x)\sqrt{1 + [f'(x)]^2}\,\Delta x$$

where $\Delta x$ is the projection of the line segment on the $x$-axis. The area of the entire surface of revolution $S$ can then be obtained by integrating this expression over the interval $[a, b]$. These considerations lead us to the following definition.

**SURFACE AREA** Suppose $f'$ is continuous on the interval $[a, b]$. Then the surface generated by revolving about the $x$-axis the arc of the curve $y = f(x)$ on $[a, b]$ has **surface area**

$$S = 2\pi\int_a^b f(x)\sqrt{1 + [f'(x)]^2}\,dx$$

*An easy-to-remember form for surface area is $S = 2\pi\int y\,ds$ since $ds = \sqrt{1 + [f'(x)]^2}\,dx$.*

You may find it instructive to derive this formula by the more rigorous approach outlined in Problem 56. We close with two examples that illustrate the use of the surface area formula.

**Example 4 Area of a surface of revolution**

Find the area of the surface generated by revolving about the $x$-axis the arc of the curve $y = x^3$ on $[0, 1]$.

***Solution*** The graph is shown in Figure 6.51.
Because $f(x) = x^3$, we have $f'(x) = 3x^2$, which is certainly continuous on the interval $[0, 1]$.

$$S = 2\pi\int_0^1 x^3\sqrt{1 + (3x^2)^2}\,dx$$

$$= 2\pi\int_0^1 x^3\sqrt{1 + 9x^4}\,dx$$

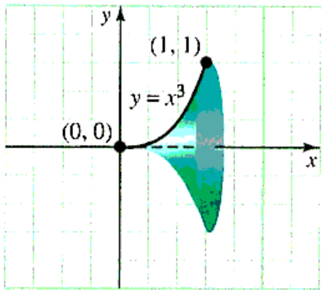



**Figure 6.51** Graph of surface of revolution

$$= 2\pi \int_1^{10} u^{1/2}\left(\frac{du}{36}\right) \qquad \textit{Let } u = 1 + 9x^4;\ du = 36x^3\,dx$$
*If $x = 0$, then $u = 1$; if $x = 1$, then $u = 10$.*

$$= \frac{\pi}{18}\left[\frac{2}{3}u^{3/2}\right]_1^{10}$$

$$= \frac{\pi}{27}\left(10\sqrt{10} - 1\right)$$

This is a surface area of about 3.563 square units.

**Example 5 Derive the formula for the surface area of a sphere**

Find a formula for the surface area of a sphere of radius $r$.

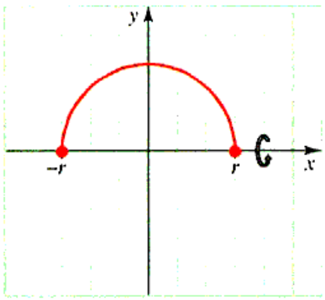


**Figure 6.52** Graph of semicircle $y = \sqrt{r^2 - x^2}$

***Solution*** We can generate the surface of the sphere by revolving the semicircle $y = \sqrt{r^2 - x^2}$ about the $x$-axis (see Figure 6.52)

We find that

$$y' = -x(r^2 - x^2)^{-1/2} = \frac{-x}{\sqrt{r^2 - x^2}}$$

Because the semicircle intersects the $x$-axis at $x = r$ and $x = -r$, the interval of integration is $[-r, r]$. Finally, by applying the formula for the surface area, we find

$$S = 2\pi \int_{-r}^{r} \sqrt{r^2 - x^2}\sqrt{1 + \left(\frac{-x}{\sqrt{r^2 - x^2}}\right)^2}\,dx$$

$$= 2\pi \int_{-r}^{r} \sqrt{(r^2 - x^2)\left(\frac{r^2 - x^2 + x^2}{r^2 - x^2}\right)}\,dx$$

$$= 2\pi \int_{-r}^{r} r\,dx$$

$$= 2\pi r(r + r)$$

$$= 4\pi r^2$$

To generalize the formula for surface area to apply to any vertical or horizontal axis of revolution, suppose that an axis of revolution is $R(x)$ units from a typical element of arc on the graph of $y = f(x)$, as shown in Figure 6.53.

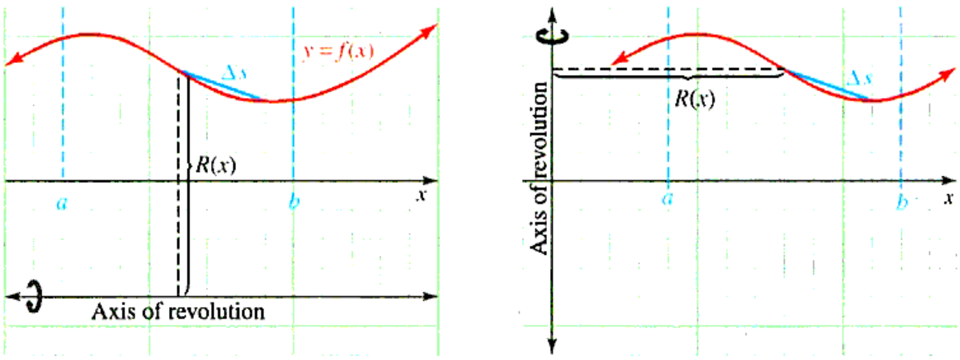


**Figure 6.53** Surface of revolution about a general horizontal or vertical axis

Then $2\pi R(x)$ is the circumference of a circle of radius $R(x)$, and it can be shown that an element of surface area is

$$\Delta S = 2\pi R(x)\Delta s = 2\pi R(x)\sqrt{1 + [f'(x)]^2}\,\Delta x$$

Thus, the surface of revolution on the interval $[a, b]$ has area

$$S = 2\pi \int_a^b R(x)\sqrt{1 + [f'(x)]^2}\,dx$$

In particular, if the graph of $y = f(x)$ is revolved about the $y$-axis, an element of the arc is $R(x) = x$ units from the $y$-axis, and the resulting surface has area

$$S = 2\pi \int_a^b x\sqrt{1 + [f'(x)]^2}\,dx$$

## Polar Arc Length and Surface Area

Sometimes we wish to find arc length and surface area in polar form. Figure 6.54 shows the portion of a polar curve subtended by a central angle $\Delta\theta$. If $\Delta\theta$ is small, the corresponding arc length $\Delta s$ may be thought of as the hypotenuse of a right triangle with sides $\Delta r$ and $r\Delta\theta$, the length of the circular arc $AB$ with radius $r$.

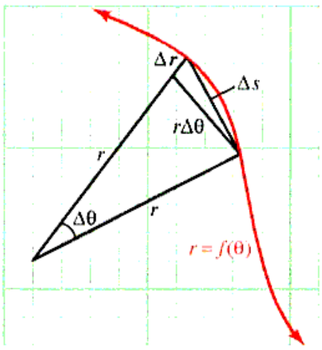


**Figure 6.54** Polar arc length

We have

$$\Delta s = \sqrt{(r\Delta\theta)^2 + (\Delta r)^2} \qquad \textit{Pythagorean theorem}$$

$$= \sqrt{r^2 + \left(\frac{\Delta r}{\Delta\theta}\right)^2}\,\Delta\theta$$

By formalizing these observations, we are led to compute polar arc length by the following formula.

**ARC LENGTH IN POLAR COORDINATES** The length of a polar curve $r = f(\theta)$ for $\alpha \le \theta \le \beta$ is given by the integral

$$S = \int_\alpha^\beta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\,d\theta$$

**Example 6 Computing polar arc length**

Find the length of the circle $r = 2\sin\theta$.

***Solution*** This circle is shown in Figure 6.55, and since this is a unit circle, we know the circumference to be $2\pi$.

We will verify this conclusion using the arc length formula. Note that $0 \le \theta \le \pi$.

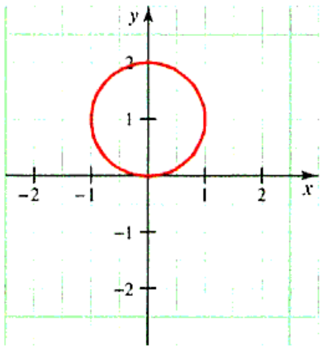


**Figure 6.55** Circle $r = 2\sin\theta$

$$\begin{aligned}
s &= \int_0^\pi \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\,d\theta && \textit{Arc length formula where } \alpha = 0 \textit{ and } \beta = \pi \\
&= \int_0^\pi \sqrt{(2\sin\theta)^2 + (2\cos\theta)^2}\,d\theta && r = 2\sin\theta;\ \frac{dr}{d\theta} = 2\cos\theta \\
&= \int_0^\pi 2\,d\theta && \sin^2\theta + \cos^2\theta = 1 \\
&= 2\pi
\end{aligned}$$

If a polar curve $r = f(\theta)$ between $\theta = \alpha$ and $\theta = \beta$ is revolved about the $x$-axis, it generates a surface of area

$$A = \int_\alpha^\beta 2\pi y\, ds$$

$$= 2\pi \int_\alpha^\beta (r \sin\theta) \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$$

We conclude this section with an example showing the use of this formula.

**Example 7 Computing surface area in polar coordinates**

Find the area of the surface generated by revolving about the $x$-axis the top half of the cardioid $r = 1 + \cos\theta$.

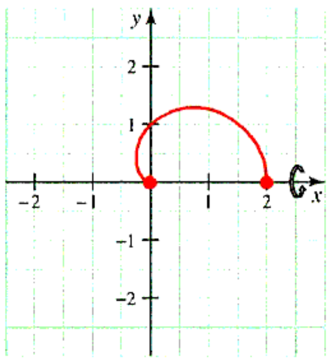



**Figure 6.56** Top half of cardioid $r = 1 + \cos\theta$ $(0 \le \theta \le \pi)$

***Solution*** The cardioid is shown in Figure 6.56.

Since $r = 1 + \cos\theta$ and $r' = -\sin\theta$, the surface area is given by

$$A = 2\pi \int_0^\pi (r\sin\theta)\sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$$

$$= 2\pi \int_0^\pi (1+\cos\theta)\sin\theta \sqrt{(1+\cos\theta)^2 + (-\sin\theta)^2}\, d\theta$$

$$= 2\pi \int_0^\pi (1+\cos\theta)\sin\theta \sqrt{1 + 2\cos\theta + \cos^2\theta + \sin^2\theta}\, d\theta$$

$$= 2\pi \int_0^\pi \sqrt{2}\,(1+\cos\theta)^{3/2} \sin\theta\, d\theta \qquad \textit{Since } \sin^2\theta + \cos^2\theta = 1$$

$$= 2\sqrt{2}\pi \left[-\frac{2}{5}(1+\cos\theta)^{5/2}\right]_0^\pi \qquad \textit{Substitute } u = 1 + \cos\theta.$$

$$= \frac{32\pi}{5}$$

## PROBLEM SET 6.4

**Level 1**

*Find the length of the arc of the curve $y = f(x)$ on the intervals given in Problems* 1-18.

1. $f(x) = 3x + 2$ on $[-1, 2]$
2. $f(x) = 5 - 4x$ on $[-2, 0]$
3. $f(x) = 1 - 2x$ on $[1, 3]$
4. $f(x) = 2 - 3x$ on $[0, 1]$
5. $f(x) = \frac{2}{3}x^{3/2} + 1$ on $[0, 4]$
6. $f(x) = \frac{2}{3}x^{3/2} + 2$ on $[1, 4]$
7. $f(x) = \frac{1}{3}(2 + x^2)^{3/2}$ on $[0, 3]$
8. $f(x) = \frac{1}{3}(2 + x^2)^{3/2}$ on $[1, 4]$
9. $f(x) = \frac{1}{12}x^5 + \frac{1}{5}x^{-3}$ on $[1, 2]$
10. $f(x) = \frac{1}{3}x^3 + \frac{1}{4}x^{-1}$ on $[1, 4]$
11. $f(x) = \frac{1}{8}x^2 - \ln x$ on $[1, 2]$
12. $f(x) = \frac{1}{4}x^4 + \frac{1}{8}x^{-2}$ on $[1, 2]$
13. $f(x) = \sqrt{e^{2x} - 1} - \sec^{-1}(e^x)$ on $[0, \ln 2]$
14. $f(x) = \sqrt{e^{2x} - 1} - \sec^{-1}(e^x)$ on $[0, \ln 3]$
15. $f(x) = x^3 + \frac{1}{12}x^{-1}$ on $[1, 2]$
16. $f(x) = \frac{1}{6}x^3 + \frac{1}{2}x^{-1}$ on $[1, 2]$
17. $f(x) = x^{3/2} - \frac{1}{3}x^{1/2}$ on $[1, 9]$
18. $f(x) = x^{3/2} - \frac{1}{3}x^{1/2}$ on $[4, 16]$
19. Find the length of the curve defined by $9x^2 = 4y^3$ between the points $(0, 0)$ and $\left(2\sqrt{3}, 3\right)$.
20. Find the length of the curve defined by $(y + 1)^2 = 4x^3$ between the points $(0, -1)$ and $(1, 1)$.
21. Find the length of the curve defined by

$$\int_1^x \sqrt{t^2 - 1}\, dt$$

on $[1, 2]$.

**22.** Find the length of the curve defined by

$$\int_1^x \sqrt{t^2+1}\,dt$$

on $[0, 1]$.

*Find the surface area generated when the graph of each function given in Problems 23-28 on the prescribed interval is revolved about the x-axis.*

**23.** $f(x) = 2x + 1$ on $[0, 2]$
**24.** $f(x) = 2x - 1$ on $[1, 2]$
**25.** $f(x) = \sqrt{x}$ on $[2, 6]$
**26.** $f(x) = \sqrt{1 - x^2}$ on $[0, 1/2]$
**27.** $f(x) = \frac{1}{3}x^3 + \frac{1}{4}x^{-1}$ on $[1, 2]$
**28.** $f(x) = \frac{1}{4}x^4 + \frac{1}{8}x^{-2}$ on $[1, 2]$

*Find the length of the polar curves given in Problems 29-36.*

**29.** $r = \cos\theta$
**30.** $r = \sin\theta + \cos\theta$
**31.** $r = e^{3\theta}, 0 \le \theta \le \frac{\pi}{2}$
**32.** $r = e^{1-\theta}, 0 \le \theta \le 1$
**33.** $r = \theta^2, 0 \le \theta \le 1$
**34.** $r = \theta^2, 1 \le \theta \le 2$
**35.** $r = \cos^2 \dfrac{\theta}{2}$
**36.** $r = \sin^2 \dfrac{\theta}{2}$

*In Problems 37-40, find the surface area generated when the given polar curve is revolved about the x-axis.*

**37.** $r = 5, 0 \le \theta \le \frac{\pi}{3}$
**38.** $r = 1 - \cos\theta, 0 \le \theta \le \pi$
**39.** $r = \csc\theta, \frac{\pi}{4} \le \theta \le \frac{\pi}{3}$
**40.** $r = \cos^2 \frac{\theta}{2}, 0 \le \theta \le \pi$

## Level 2

*In Problems 41 and 42, use numerical integration to find the arc length.*

**41.** $f(x) = \sin x$ on $[0, \pi]$
**42.** $f(x) = \tan x$ on $[0, 1]$
**43.** Find the area of the surface generated when the arc of the curve

$$y = \frac{1}{3}x^3 + (4x)^{-1}$$

between $x = 1$ and $x = 3$ is revolved about

**a.** the $x$-axis **b.** the $y$-axis

**44.** Find the area of the surface generated when the arc of the curve

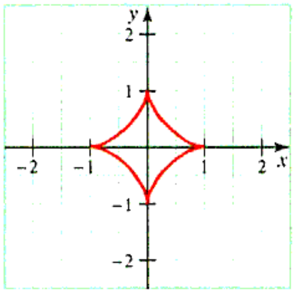
$$x = \frac{3}{5}y^{5/3} - \frac{3}{4}y^{1/3}$$

between $y = 0$ and $y = 1$ is revolved about

**a.** the $y$-axis **b.** the $x$-axis
**c.** the line $y = -1$

*Find the surface area generated when the graph of each function given in Problems 45-48 on the prescribed interval is revolved about the y-axis. Give your answers to the nearest hundredth.*

**45.** $f(x) = \frac{1}{3}(12 - x)$ on $[0, 3]$
**46.** $f(x) = \frac{2}{3}x^{3/2}$ on $[0, 3]$
**47.** $f(x) = \frac{1}{3}\sqrt{x}(3 - x)$ on $[1, 3]$
**48.** $f(x) = 2\sqrt{4 - x}$ on $[1, 3]$
**49.** The graph of the equation $x^{2/3} + y^{2/3} = 1$ is an **astroid**.




Find the length of this particular astroid by finding the length of the first quadrant portion,

$$y = (1 - x^{2/3})^{3/2} \text{ on } [0, 1]$$

By symmetry, the length of the entire curve is four times the length of the arc in the first quadrant.

**50.** Find the area of the surface generated by revolving the portion of the astroid with $y \ge 0$ (see Problem 49)

$$x^{2/3} + y^{2/3} = 1 \text{ on } [-1, 1]$$

about the $x$-axis.

**51.** The following table gives the slope $f'(x_k)$ for points $x_k$ located at 0.3 unit intervals on a certain curve defined by $y = f(x)$. Use the trapezoidal rule to estimate the arc length of $y = f(x)$ over the interval $[0, 3]$.

| $x_k$ | $f'(x_k)$ | $x_k$ | $f'(x_k)$ |
|---|---|---|---|
| 0.0 | 3.7 | 1.8 | 4.6 |
| 0.3 | 3.9 | 2.1 | 4.9 |
| 0.6 | 4.1 | 2.4 | 5.2 |
| 0.9 | 4.1 | 2.7 | 5.5 |
| 1.2 | 4.2 | 3.0 | 6.0 |
| 1.5 | 4.4 | | |

**52.** Use the table in Problem 51 to estimate the surface area generated when $y = f(x)$ is revolved about the $y$-axis over the interval $[0, 3]$.

**53.** Show that a cone of radius $r$ and height $h$ has lateral surface area

$$S = \pi r\sqrt{r^2 + h^2}$$

*Hint*: Revolve part of the line $hy = rx$ about the $x$-axis.

**Level 3**

**54.** Show that when the arc of the graph of $f(x)$ between $x = a$ and $x = b$ is revolved about the $y$-axis, the surface generated has area

$$S = 2\pi \int_a^b x\sqrt{1 + [f'(x)]^2}\,dx$$

**55.** If $f'(x)$ exists throughout the interval $[x_{k-1}, x_k]$, show that there exists a number $x_k^*$ in this interval for which

$$\sqrt{(\Delta x_k)^2 + (\Delta y_k)^2} = \sqrt{1 + [f'(x_k^*)]^2}\Delta x$$

where $\Delta x_k = x_k - x_{k-1}$ and $\Delta y_k = f(x_k) - f(x_{k-1})$. *Hint*: Use the mean value theorem.

**56.** Verify the surface area formula by completing the following steps (see Figure 6.57).

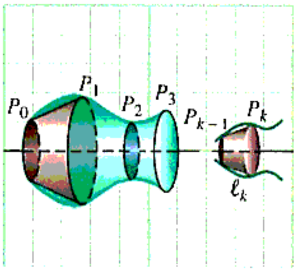



**Figure 6.57** Surface area formula

**a.** Let $P = \{x_0, x_1, \cdots, x_n\}$ be a partition of the interval $[a, b]$, and for $k = 0, 1, \cdots, n$, let $P_k$ denote the point $(x_k, y_k)$ where $y_k = f(x_k)$. Show that when the line segment between $P_{k-1}$ and $P_k$ is revolved about the $x$-axis, it generates a frustum of a cone with surface area

$$\pi(y_{k-1} + y_k)\ell_k$$

where $\ell_k$ is the distance between $P_{k-1}$ and $P_k$.

**b.** Notice that $\frac{1}{2}(y_{k-1} + y_k)$ is a number between $y_{k-1}$ and $y_k$. Use the intermediate value theorem (Section 2.3) to show that

$$y_{k-1} + y_k = 2f(c_k)$$

for some number $c_k$ between $x_{k-1}$ and $x_k$.

**c.** Explain why the surface generated when the arc of the graph of $f$ between $x = a$ and $x = b$ is revolved about the $x$-axis has area

$$\lim_{\|P\|\to 0}\sum_{k=1}^{n} 2\pi f(c_k)\sqrt{1 + [f'(x_k^*)]^2}\Delta x_k$$

**d.** Notice that as $\|P\| \to 0$, the numbers $c_k$ and $x_k^*$ are "squeezed" together. Use this observation to show that the surface area is given by the integral

$$S = 2\pi \int_a^b f(x)\sqrt{1 + [f'(x)]^2}\,dx$$

**57.** Let $n$ be a positive integer and $C, D$ be constants such that

$$n(n-1) = \frac{1}{16CD}$$

Find the length of the curve

$$y = Cx^{2n} + Dx^{2(1-n)}$$

from $x = a$ to $x = b$.

**58.** The center of a circle of radius $r$ is located at $(R, 0)$, where $R > r$. When the circle is revolved about the $y$-axis, it generates a *torus* (a doughnut-shaped solid). What is the surface area of the torus?

**59.** Suppose it is known that the arc length of the curve $y = f(t)$ on the interval $0 \le t \le x$ is

$$L(x) = \ln(\sec x + \tan x)$$

for every $x$ on $0 \le x \le 1$. If the curve $y = f(x)$ passes through the origin, what is $f(x)$?

**60.** Find the length of

$$y = \frac{x^3}{24} + \frac{2}{x}$$

from $x = 2$ to $x = 3$. Find a general formula for the length of the curve

$$y = Cx^n + Dx^{2-n}$$

from $x = a$ to $x = b$ for

$$n(n-2) = \frac{1}{4CD}$$

## 6.5 PHYSICAL APPLICATIONS: WORK, LIQUID FORCE, AND CENTROIDS

**IN THIS SECTION:** *Work, modeling fluid pressure and force, modeling the centroid of a plane region, volume theorem of Pappus*

Integration plays an important role in many different areas of physics. **Mechanics** is the branch of physics that deals with the effects of forces on objects. In this section, we show how integration can be used to compute work and the force exerted by a liquid.

### Work

In physics, "force" is an influence that tends to cause motion in a body. When a constant force of magnitude $F$ is applied to an object through a distance $d$, it performs **work** measured by the product $W = Fd$.

> **WORK DONE BY A CONSTANT FORCE** If a body moves a distance $d$ in the direction of an applied constant force $F$, the **work** $W$ done is
>
> $$W = Fd$$

*If there is no movement, there is no work!*

For example, the work done in lifting a 90 lb bag of concrete 3 ft is $W = Fd = (90 \text{ lb})(3 \text{ ft}) = 270$ ft-lb. Notice that this definition does not conform to everyday use of the word *work*. If you work at lifting the concrete all day, but you are not able to move the sack of concrete, then no work has been done. Table 6.3 shows some common units of work and force.

**Table 6.3** Common units of work and force

| Mass | Distance | Force | Work |
|---|---|---|---|
| kg | m | newton (N) | joule |
| g | cm | dyne (dyn) | erg |
| slug | ft | pound | ft-lb |

To find the work done in moving an object against a variable force, calculus is required. Suppose $F$ is a continuous function and $F(x)$ is a variable force that acts on an object moving along the $x$-axis from $x = a$ to $x = b$. We shall heuristically define the work done by this force. Partition the interval $[a, b]$ into $n$ subintervals, and let $\Delta x_k$ be the length of the $k$th subinterval $I_k$. If $\Delta x_k$ is sufficiently small, we can expect the force to be essentially constant on $I_k$, equal, for instance, to $F(x_k^*)$, where $x_k^*$ is a point chosen arbitrarily from the interval $I_k$. It is reasonable to expect that the work $\Delta W_k$ required to move the object on the interval $I_k$ is approximately $\Delta W_k = F(x_k^*)\Delta x_k$, and the total work is estimated by the sum

$$\sum_{k=1}^{n} F(x_k^*)\Delta x_k$$

as indicated in Figure 6.58.

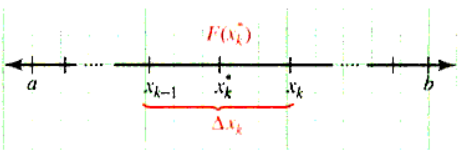



**Figure 6.58** Work performed by a variable force

By taking the limit as the norm of the partition approaches 0, we find the total work may be modeled by

$$W = \lim_{\|P\|\to 0} \sum_{k=1}^{n} F(x_k^*)\Delta x_k = \int_a^b F(x)\,dx$$

**WORK DONE BY A VARIABLE FORCE** The work done by the variable force $F(x)$ in moving an object along the $x$-axis from $x = a$ to $x = b$ is given by

$$W = \int_a^b F(x)\,dx$$

**Example 1 Work done by a variable force**

An object located $x$ ft from a fixed starting position is moved along a straight road by a force of $F(x) = (3x^2 + 5)$ lb. What work is done by the force to move the object
**a.** through the first 4 ft? **b.** from 1 ft to 4 ft?

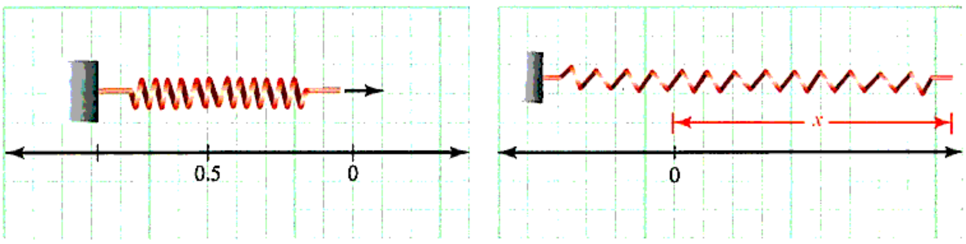
***Solution***

**a.** $W = \int_0^4 (3x^2 + 5)\,dx = (x^3 + 5x)\Big|_0^4 = 84$ ft-lb

**b.** $W = \int_1^4 (3x^2 + 5)\,dx = (x^3 + 5x)\Big|_1^4 = 78$ ft-lb

**Hooke's law**, named for the English physicist Robert Hooke (1635-1703), states that *when a spring is pulled $x$ units past its equilibrium (rest) position, there is a restoring force $F(x) = kx$ that pulls the spring back toward equilibrium.** (See Figure 6.59.) The constant $k$ in this formula is called the **spring constant.**

*Hooke's law:* A force $F(x) = kx$ acts to restore the spring to its equilibrium position.




A spring at rest (equilibrium position) A spring stretched $x$ units past equilibrium

**Figure 6.59** Hooke's law

*Actually, Hooke's law applies only in ideal circumstances, and a spring for which the law applies is sometimes called an ideal (or linear) spring.

**Example 2 Modeling work using Hooke's formula**

The natural length of a certain spring is 10 cm. If it requires 2 ergs of work to stretch the spring to a total length of 18 cm, how much work will be performed in stretching the spring to a total length of 20 cm? (See Figure 6.60.)

***Solution*** Assume that the point of equilibrium is at 0 on a number line, and let $x$ be the length the free end of the spring is extended past equilibrium. Because the stretching force of the spring is $F(x) = kx$, the work done in stretching the spring $b$ cm beyond equilibrium is

$$W = \int_0^b F(x) = \int_0^b kx\,dx = \frac{1}{2}kb^2$$

We are given that $W = 2$ when $b = 8$ (since 18 cm is 8 cm beyond equilibrium), so

$$2 = \frac{1}{2}k(8)^2 \quad \text{implying} \quad k = \frac{1}{16}$$

Thus, the work done in stretching the spring $b$ cm beyond equilibrium is

$$W = \frac{1}{2}\left(\frac{1}{16}\right)b^2 = \frac{1}{32}b^2 \text{ ergs}$$

In particular, when the total length of the spring is 20 cm, it is extended $b = 10$ cm, and the required work is

$$W = \frac{1}{32}(10)^2 = \frac{25}{8} = 3.125 \text{ ergs}$$

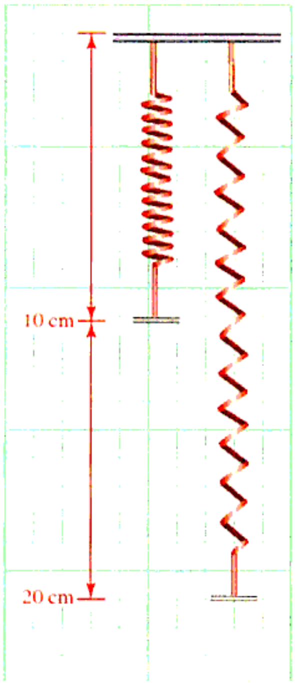
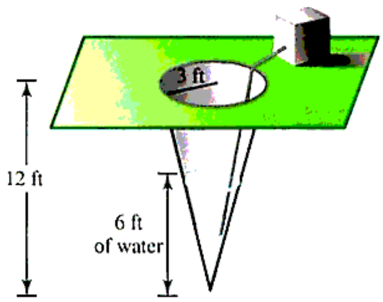


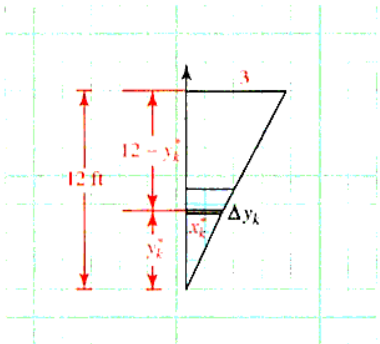

**Figure 6.60** Stretching a spring

**Example 3 Modeling the work performed in pumping water out of a tank**

A tank in the shape of a right circular cone of height 12 ft and radius 3 ft is inserted into the ground with its vertex pointing down and its top at ground level, as shown in Figure 6.61**a**. If the tank is filled with water (weight density $\rho g = 62.4$ lb/ft$^3$)* to a depth of 6 ft, how much work is performed in pumping all the water in the tank to ground level? What changes if the water is pumped to a height of 3 ft above ground level?




**a.** A conical water tank




**b.** A disk of water is $12 - y_k^*$ units from the top

**Figure 6.61** Work in pumping water

*In non-English-speaking countries, where the metric system is commonly used, density is generally considered to be mass density $\rho$. In English-speaking countries where the British and US system of measurement is commonly used, density is generally considered to be weight density $\delta = \rho g$, where $g$ is the acceleration due to gravity.

***Solution*** Set up a coordinate system with the origin at the vertex of the cone and the $y$-axis as the axis of symmetry (Figure 6.61**b**). Partition the interval $0 \le y \le 6$ (remember, the water only comes up 6 ft). Choose a representative point $y_k^*$ in the $k$th subinterval, and construct a thin disklike slab of water $y_k^*$ units above the vertex of the cone. Our plan is to think of the water in the tank as a collection of these "slabs" of water piled on top of each other. We shall find the work done to raise a typical water disk and then compute the total work by using integration to add the contributions of all such slabs.

Note that the force required to lift the slab is equal to the weight of the slab, which equals its volume multiplied by the weight per cubic foot of water. Let $x_k^*$ be the radius of the $k$th slab. By similar triangles (Figure 6.61**b**), we have

$$\frac{x_k^*}{y_k^*} = \frac{3}{12} \quad \text{which implies} \quad x_k^* = \frac{y_k^*}{4}$$

Hence, the volume of the cylindrical slab is

$$\Delta V = \pi (x_k^*)^2 \Delta y_k = \pi \underbrace{\left(\frac{y_k^*}{4}\right)^2}_{\text{Radius}} \overbrace{\Delta y_k}^{\text{Thickness}}$$

so that the weight of the $k$th slab is modeled by

$$62.4\pi \left(\frac{y_k^*}{4}\right)^2 \Delta y_k \text{ lb}$$

From Figure 6.61**b**, we see that the slab of water must be raised $12 - y_k^*$ feet, which means that the work $\Delta W$ required to raise this $k$th slab is

$$\Delta W = \underbrace{62.4\pi \left(\frac{y_k^*}{4}\right)^2 \Delta y_k}_{\text{Weight (force)}} \underbrace{(12 - y_k^*)}_{\text{Distance}} = 62.4\pi \left(\frac{y_k^*}{4}\right)^2 (12 - y_k^*)\Delta y_k$$

Finally, to compute the total work, we add the work required to lift each thin slab and take the limit of the sum as the norm of the partition approaches 0.

$$\begin{aligned} W &= \lim_{\|P\| \to 0} 62.4\pi \sum_{k=1}^{n} \left(\frac{y_k^*}{4}\right)^2 (12 - y_k^*)\Delta y_k \\ &= 62.4\pi \int_0^6 \left(\frac{y}{4}\right)^2 (12 - y)dy \\ &= \frac{62.4}{16}\pi \int_0^6 (12y^2 - y^3)dy \\ &= 3.9\pi \left[4y^3 - \frac{1}{4}y^4\right]_0^6 \\ &= 2,106\pi \end{aligned}$$

This is about 6,616 ft-lb.

If the water is pumped to a height of 3 ft above ground level, all that changes is the distance moved by the $k$th slab of water. It becomes $12 + 3 - y_k^* = 15 - y_k^*$, and the work is given by

$$W = 62.4\pi \int_0^6 \left(\frac{y}{4}\right)^2 (15 - y)\, dy \approx 9,263 \text{ ft-lb}$$

## Modeling Fluid Pressure and Force

Anyone who has dived into water has probably noticed that the pressure (that is, the force per unit area) due to the water's weight increases with depth. Careful observations show that the water pressure at any given point is directly proportional to the depth at that point. The same principle applies to other fluids.

In physics, **Pascal's principle** (named for Blaise Pascal, 1623-1662, a French mathematician and scientist) states that fluid pressure is the same in all directions (see Figure 6.62).

This means that the pressure must be the same at all points on a surface submerged horizontally, and in this case, the fluid force on the surface is given by the formula in the following box.

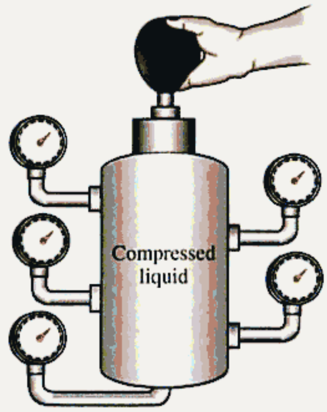



**Figure 6.62** Pascal's principle

**FLUID FORCE** If a surface of area $A$ is submerged horizontally at a depth $h$ in a fluid, the weight of the fluid exerts a force of

$$F = (\text{PRESSURE})(\text{AREA}) = \delta hA = \rho ghA$$

on the surface, where $\delta$ is the weight density, $\rho$ is the mass density, and $g$ is the acceleration due to gravity. This is called **fluid force** or **hydrostatic force**.

Table 6.4 shows the fluid density for some common fluids.

**Table 6.4** Weight density, $\delta = \rho g$ (lb/ft$^3$)

| Fluid | Density |
|---|---|
| water | 62.4 |
| seawater | 64.0 |
| gasoline | 42.0 |
| kerosene | 51.2 |
| SAE 20 oil | 57.0 |
| milk | 64.5 |
| mercury | 849.0 |

It turns out that this fluid force does not depend on the shape of the container or its size. For instance, each of the containers in Figure 6.63 has the same pressure on its base because the fluid depth $h$ and the base area $A$ are the same in each case.

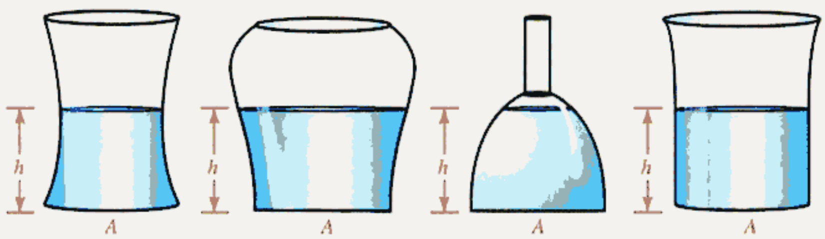
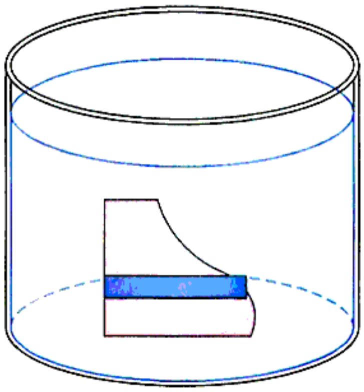
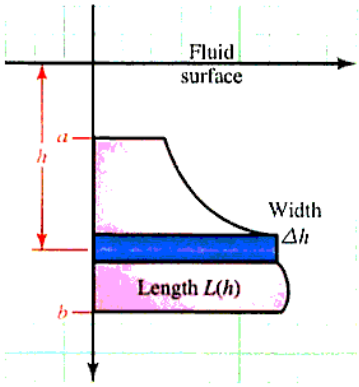



**Figure 6.63** The fluid force $F = \delta h\, A$ does not depend on the shape or size of the container

When the surface is submerged vertically or at an angle, however, this simple formula does not apply because different parts of the surface are at different depths. We will use integration to compute the fluid force in such cases. Consider a plate that is submerged vertically in a fluid of density $\delta$ as shown in Figure 6.64**a**. We set up a coordinate system with the horizontal axis on the surface of the fluid and the positive $h$-axis (depth) pointing down. Thus, greater depths correspond to larger values of $h$ (see Figure 6.64**b**). For simplicity, we assume the plate is oriented so its top and bottom are located $a$ units and $b$ units below the surface, respectively.



**a.** A plate submerged vertically in a fluid of density $\delta$




**b.** A typical slab has area $A = L(h)\Delta h$

**Figure 6.64** Hydrostatic force

Our strategy will be to think of the plate as a pile of subplates or slabs, each so thin that we may regard it as being at a constant depth below the surface. A typical slab (a horizontal strip) is shown in Figure 6.64**b**. Note that it has thickness $\Delta h$ and length $L = L(h)$, so that its area is $\Delta A = L\Delta h$. Because the slab is at a constant depth $h$ below the surface, the fluid force on its surface is

$$\Delta F = (\text{PRESSURE})(\text{AREA}) = \delta h \Delta A = \delta h L(h)\Delta h$$

and by integrating as $h$ varies from $a$ to $b$, we can model the total force on the plate.

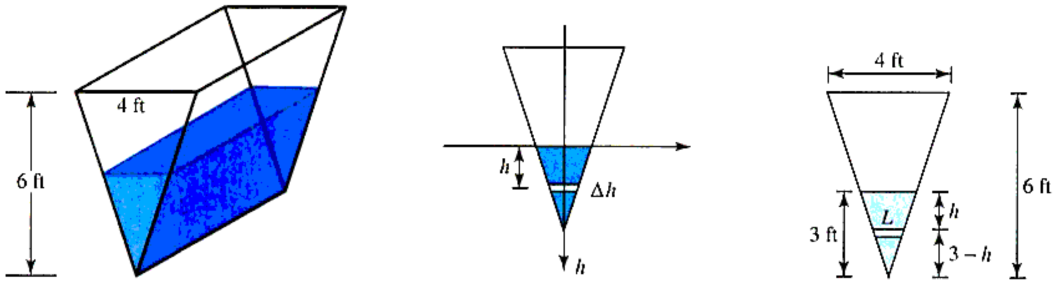
**FLUID (HYDROSTATIC) FORCE** Suppose a flat surface (a plate) is submerged vertically in a fluid of weight density $\delta = \rho g$ (lb/ft$^3$) and that the submerged portion of the plate extends from $h = a$ to $h = b$ on the vertical axis. Then the total force $F$ exerted by the fluid is given by

$$F = \int_a^b \delta h L(h)\, dh$$

where $h$ is the depth and $L(h)$ is the corresponding length of a typical horizontal approximating strip.

**Example 4 Fluid force on a vertical surface**

The cross sections of a certain trough are inverted isosceles triangles with height 6 ft and base 4 ft, as shown in Figure 6.65**a**. Suppose the trough contains water to a depth of 3 ft. Find the total fluid force on one end.




**a.** A trough with half-filled triangular cross sections

**b.** Side view of trough

**c.** Use similar triangles to find $\frac{L}{4} = \frac{3-h}{6}$

**Figure 6.65** Fluid force

***Solution*** First, we set up a coordinate system in which the horizontal axis lies at the fluid surface and the positive vertical axis (the $h$-axis) points down (see Figure 6.65**b**). Next, we find expressions for the length and depth of a typical thin slab (horizontal strip) in terms of the variables. We assume the slab has thickness $\Delta h$ and length $L$. Then by similar triangles (see Figure 6.65**c**) we have

$$\frac{L}{4} = \frac{3-h}{6} \qquad \text{so that} \qquad L = \frac{2}{3}(3-h)$$

We see that the approximating slab has area $\Delta A = L\Delta h = \frac{2}{3}(3-h)\Delta h$. Finally, we multiply the product of the fluid's weight density at a given depth ($\rho h$) and the area of the approximating slab ($\Delta A$) and integrate on the interval of depths occupied by the vertical plate.

$$F = \int_0^3 \underbrace{\frac{2}{3}\delta h(3-h)\,dh}_{\Delta F \,=\, \textit{force on a typical slab}}$$

*Note: (weight density)(depth)(area of strip)*

$$\delta \qquad h \qquad \tfrac{2}{3}(3-h)\Delta h$$

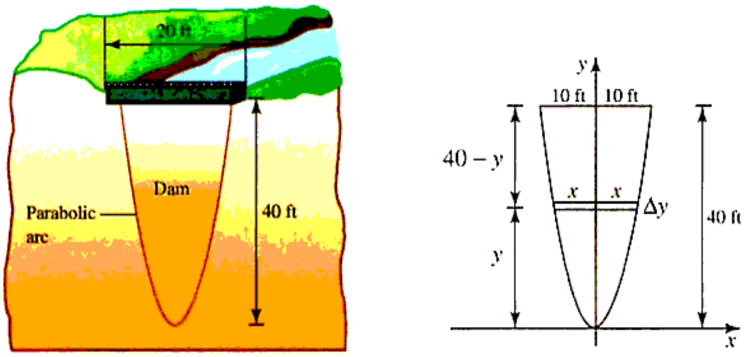
$$= \frac{2}{3}(62.4)\int_0^3 (3h - h^2)\,dh \qquad \rho = 62.4\ lb/ft^3 \textit{ for water}$$

$$= 41.6\left(\frac{3}{2}h^2 - \frac{1}{3}h^3\right)\Bigg|_0^3$$

$$= 187.2 \text{ lb}$$

**Example 5 Modeling the force on one face of a dam**

A reservoir is filled with water to the top of a dam. If the dam is in the shape of a parabola 40 ft high and 20 ft wide at the top, as shown in Figure 6.66**a**, what is the total fluid force on the face of the dam?




**a.** Cross section of a dam

**b.** A typical horizontal strip is $40 - y$ ft below the water

**Figure 6.66** Fluid force

***Solution*** Instead of putting the $x$-axis on the surface of the water (at the top of the dam), we place the origin at the vertex of the parabola and the positive $y$-axis along the parabola's axis of symmetry (Figure 6.66**b**). The advantage of this choice of axes is that the parabola can be represented by an equation of the form $y = cx^2$. Because we know that $y = 40$ when $x = 10$, it follows that

$$40 = c(10)^2 \quad \text{so that} \quad c = \frac{2}{5} \text{ and thus } y = \frac{2}{5}x^2$$

The typical horizontal strip on the face of the dam is located $y$ feet above the $x$-axis, which means it is $h = 40 - y$ feet below the surface of the water. The strip is $\Delta y$ feet wide and $L = 2x$ feet long. Write $L$ as a function of $y$:

$$y = \frac{2}{5}x^2 \quad \text{so that} \quad x = \sqrt{\frac{5}{2}y}$$

and

$$L(y) = 2x = 2\left(\sqrt{\frac{5}{2}y}\right) = \sqrt{10y}$$

Therefore,

$$\Delta A = L\Delta y = \sqrt{10y}\,\Delta y$$

and the total force may be modeled by the integral

$$F = \int_0^{40} \underbrace{\delta}_{\textit{Weight density of water} = 62.4} \overbrace{h}^{\textit{Depth below water}: h = 40 - y} \underbrace{L(y)\,dy}_{\textit{Area of strip}}$$

$$= \int_0^{40} 62.4(40 - y)\sqrt{10y}\,dy$$

$$= 62.4\sqrt{10}\int_0^{40} (40y^{1/2} - y^{3/2})\,dy$$

$$= 62.4\sqrt{10}\left[\frac{80}{3}y^{3/2} - \frac{2}{5}y^{5/2}\right]_0^{40}$$

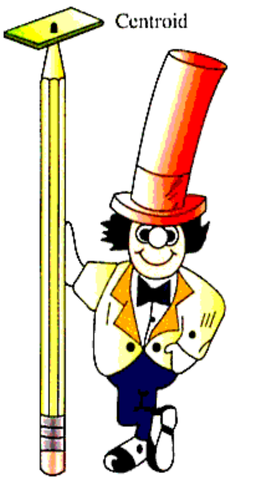
$$= 62.4\sqrt{10}\,y^{3/2}\left(\frac{80}{3} - \frac{2}{5}y\right)\Big|_0^{40}$$

$$= 532{,}480 \text{ lb}$$

## Modeling the Centroid of a Plane Region

In mechanics, it is often important to determine the point where an irregularly shaped plate will balance. The **moment of a force** measures its tendency to produce rotation in an object and depends on the magnitude of the force and the point on the object where it is applied. Since the time of Archimedes (287-212 B.C.), it has been known that the balance point of an object occurs where all its moments cancel out (so there is no rotation).

The **mass** of an object is a measure of its **inertia**; that is, its propensity to maintain a state of rest or uniform motion. A thin plate whose material is distributed uniformly, so that its density $\rho$ (mass per unit area) is constant, is called a **homogeneous lamina**. The balance point of such a lamina is called its **centroid** and may be thought of as its geometrical center. We shall see how centroids can be computed by integration in this section, and shall examine the topic even further in Section 12.6.




*Theoretically, the lamina should balance on a point placed at its centroid.*

Consider a homogeneous lamina that covers a region $R$ bounded by the curves $y = f(x)$ and $y = g(x)$ on the interval $[a, b]$, and consider a thin, vertical approximating strip within $R$, as shown in Figure 6.67.

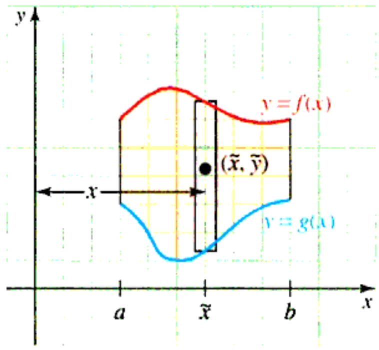



**Figure 6.67** Homogeneous lamina with a vertical approximating strip

The mass of the strip is given by

$$\Delta m = \underset{\text{Density}}{\rho}\ \underbrace{[f(x) - g(x)]\Delta x}_{\text{Area of strip}}$$

and the total mass of the lamina may be found by integration:

$$m = \rho\int_a^b [f(x) - g(x)]\,dx$$

Note that $m = \text{AREA}$ if $\rho = 1$.

The centroid (geometrical center) of the approximating strip is $(\tilde{x}, \tilde{y})$, where $\tilde{x} = x$ and $\tilde{y} = \frac{1}{2}[f(x) + g(x)]$. The **moment of the approximating strip about the $y$-axis** is defined to be the product

$$\Delta M_y = \underset{\text{Distance of the strip from the y-axis}}{\tilde{x}} \cdot \underbrace{\Delta m}_{\text{Mass of strip}} = \tilde{x}\{\rho[f(x) - g(x)]\Delta x\}$$

This product provides a measure of the tendency of the strip to rotate about the $y$-axis. Similarly, the **moment of the strip about the $x$-axis** is defined to be the product

$$\Delta M_x = \underset{\substack{\uparrow \\ \text{Distance of the strip from the y-axis}}}{\tilde{y}} \cdot \underbrace{\Delta m}_{\text{mass of strip}} = \underbrace{\frac{1}{2}[f(x)+g(x)]}_{\text{Average distance of the strip from the x-axis}}\{\rho[f(x)-g(x)]\Delta x\}$$

$$= \frac{1}{2}\rho\{[f(x)]^2 - [g(x)]^2\}\Delta x$$

This product provides a measure of the tendency of the strip to rotate about the $x$-axis.

Integrating on $[a, b]$, we find the moments of the entire lamina $R$ about the $y$-axis and $x$-axis may be modeled by

$$M_y = \rho \int_a^b x[f(x) - g(x)]\,dx \qquad \text{and} \qquad M_x = \frac{1}{2}\rho \int_a^b \{[f(x)]^2 - [g(x)]^2\}\,dx$$

If the entire mass $m$ of the lamina $R$ were located at the point $(\overline{x}, \overline{y})$, then its moments about the $x$-axis and $y$-axis would be $m\overline{y}$ and $m\overline{x}$, respectively. Thus, we have $m\overline{x} = M_y$ and $m\overline{y} = M_x$, so that

$$\overline{x} = \frac{M_y}{m} = \frac{\rho \int_a^b x[f(x) - g(x)]\,dx}{\rho \int_a^b [f(x) - g(x)]\,dx} \qquad \text{and}$$

$$\overline{y} = \frac{M_x}{m} = \frac{\frac{1}{2}\rho \int_a^b \{[f(x)]^2 - [g(x)]^2\}\,dx}{\rho \int_a^b [f(x) - g(x)]\,dx}$$

We can now summarize these results.

Let $f$ and $g$ be continuous and satisfy $f(x) \geq g(x)$ on the interval $[a, b]$, and consider a thin plate (lamina) of uniform density $\rho$ that covers the region $R$ between the graphs of $y = f(x)$ and $y = g(x)$ on the interval $[a, b]$. Then

**MASS** The **mass** of $R$ is: $m = \rho \displaystyle\int_a^b [f(x) - g(x)]\,dx$

**CENTROID** The **centroid** of $R$ is the point $(\overline{x}, \overline{y})$ such that

$$\overline{x} = \frac{M_y}{m} = \frac{\int_a^b x[f(x) - g(x)]\,dx}{\int_a^b [f(x) - g(x)]\,dx}$$

and

$$\overline{y} = \frac{M_x}{m} = \frac{\frac{1}{2}\int_a^b \{[f(x)]^2 - [g(x)]^2\}\,dx}{\int_a^b [f(x) - g(x)]\,dx}$$

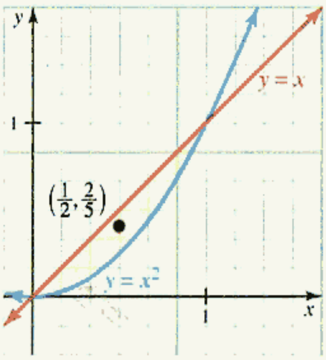



**Figure 6.68** The centroid of a planar region

**Example 6 Centroid of a thin plate**

A homogeneous lamina $R$ has constant density $\rho = 1$ and is bounded by the parabola $y = x^2$ and the line $y = x$. Find the mass and the centroid of $R$.

***Solution*** We see that the line and the parabola intersect at the origin and at the point $(1, 1)$, as shown in Figure 6.68.

Because $\rho = 1$, the mass is the same as the area of the region $R$. That is,

$$m = A = \int_0^1 (x - x^2)\,dx = \left(\frac{x^2}{2} - \frac{x^3}{3}\right)\Big|_0^1 = \frac{1}{6}$$

and we find that the region has moments $M_y$ and $M_x$ about the $y$-axis and $x$-axis, respectively, where

$$M_y = \int_0^1 x(x - x^2)\,dx = \left(\frac{x^3}{3} - \frac{x^4}{4}\right)\Big|_0^1 = \frac{1}{12}$$

and

$$\begin{aligned} M_x &= \frac{1}{2}\int_0^1 (x^2 - x^4)\,dx \\ &= \frac{1}{2}\left(\frac{x^3}{3} - \frac{x^5}{5}\right)\Big|_0^1 \\ &= \frac{1}{2}\left(\frac{1}{3} - \frac{1}{5}\right) \\ &= \frac{1}{15} \end{aligned}$$

Thus, the centroid of the region $R$ has coordinates

$$\overline{x} = \frac{M_y}{m} = \frac{\frac{1}{12}}{\frac{1}{6}} = \frac{1}{2} \quad \text{and} \quad y = \frac{M_x}{m} = \frac{\frac{1}{15}}{\frac{1}{6}} = \frac{2}{5}$$

## Volume Theorem of Pappus

There is a remarkable result attributed to Pappus of Alexandria (ca. 300 A.D.), whom many regard as the last great Greek geometer, which gives a connection between centroids and volumes of revolution.

**Theorem 6.2 Volume theorem of Pappus**

The solid generated by revolving a region $R$ about a line outside its boundary (but in the same plane) has volume $V = As$, where $A$ is the area of $R$ and $s$ is the distance traveled by the centroid of $R$.

***Proof:*** First, choose a coordinate system in which the $y$-axis coincides with the axis of revolution. Figure 6.69 shows a typical region $R$ together with a vertical approximating rectangle. We shall assume that this rectangle has area $\Delta A$ and that it is located $x$ units from the $y$-axis.

Notice that when the rectangle is revolved about the $y$-axis, it generates a shell of volume $\Delta V = 2\pi x \Delta A$. Thus, by partitioning the region $R$ into a number of rectangles and taking the limit of the sum of the volumes of all the related approximating shells as the norm of the partition approaches 0, we find that

$$\begin{aligned} V &= \lim_{\|P\|\to 0} \sum_{k=1}^{n} 2\pi x_k \Delta A_k \\ &= \int 2\pi x\; dA \\ &= 2\pi \int x\; dA \end{aligned}$$

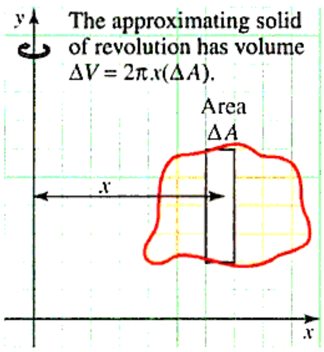


**Figure 6.69** The volume theorem of Pappus

$$= 2\pi\overline{x}\int dA \qquad \textit{Because } \overline{x} = \frac{\int x\,dA}{\int dA}$$

$$= 2\pi\overline{x}A \qquad \textit{Because } \int dA = A$$

$$= As \qquad \textit{Where } s = 2\pi\overline{x} \textit{ is the distance traveled by the centroid.}$$

*The circumference of a circle of radius $\bar{x}$.*

We close this section with an example that illustrates the use of the volume theorem of Pappus.

**Example 7 Volume of a torus using the volume theorem of Pappus**

When a circle of radius $r$ is revolved about a line in the plane of the circle located $R$ units from its center ($R > r$), the solid figure so generated is called a *torus*, as shown in Figure 6.70. Show that the torus has volume $V = 2\pi^2 r^2 R$.

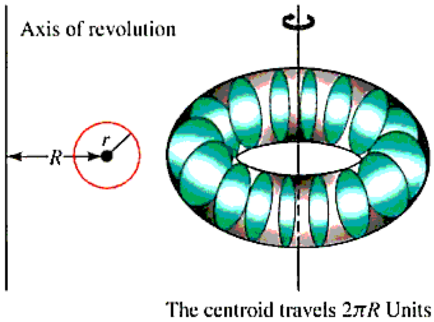



**Figure 6.70** The volume of a torus

***Solution*** The circle has area $A = \pi r^2$ and its center (which is its centroid) travels $2\pi R$ units. Comparing this example to the volume theorem of Pappus, we see $s = 2\pi R$. Therefore, the torus has volume

$$V = (2\pi R)A = 2\pi R(\pi r^2) = 2\pi^2 r^2 R$$

Incidentally, the volume found in Example 7 using the theorem of Pappus was worked in Problem 56 of Section 6.2 by using washers.

## PROBLEM SET 6.5

**Level 1**

1. ■ What does this say? Discuss work and how to find it. (*Hint:* The answer is *not* to apply at the unemployment office!)
2. ■ What does this say? Discuss fluid force and how to find it.
3. ■ What does this say? What is meant by a centroid? Outline a procedure for finding both the mass and the centroid.
4. ■ What does this say? State and discuss the volume theorem of Pappus.
5. What is the work done in lifting an 850-lb billiard table 15 ft?
6. What is the work done in lifting a 50-lb bag of salt 5 ft?
7. A bucket weighing 75 lb when filled and 10 lb when empty is pulled up the side of a 100-ft building. How much more work is done in pulling up the full bucket than the empty bucket?
8. Suppose it takes 4 ergs of work to stretch a spring 10 cm beyond its natural length. How much more work is needed to stretch it 4 cm further?
9. A 5-lb force will stretch a spring 9 in. (= 0.75 ft) beyond its natural length. How much work is required to stretch it 1 ft beyond its natural length?
10. A spring whose natural length is 10 cm exerts a force of 30 newtons when stretched to a length of 15 cm. (*Note*: 15 cm − 10 cm = 5 cm = 0.05 m.) Find the spring constant, and then determine the work done in

stretching the spring 7 cm ($= 0.07$ m) beyond its natural length.

**11.** A 30-lb ball hangs at the bottom of a cable that is 50 ft long and weighs 20 lb. The entire length of cable hangs over a cliff. Find the work done to raise the cable and get the ball to the top of the cliff.

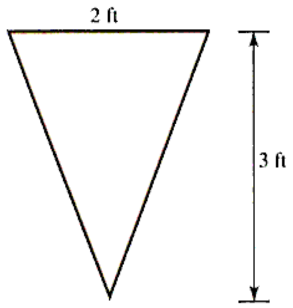
**12.** A 30-ft rope weighing 0.4 lb/ft hangs over the edge of a building 100 ft high. How much work is done in pulling the rope to the top of the building? Assume that the top of the rope is flush with the top of the building and the lower end of the rope is swinging freely.

**13.** An object moving along the $x$-axis is acted upon by a force $F(x) = |\sin x|$. Find the total work done by the force in moving the object from $x = 0$ cm to $x = 2\pi$ cm. *Note*: Distance is in cm, so force is in dynes.

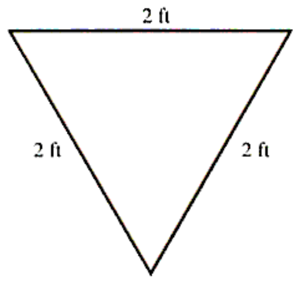
**14.** An object moving along the $x$-axis is acted upon by a force $F(x) = x^4 + 2x^2$. Find the total work done by the force in moving the object from $x = 1$ cm to $x = 2$ cm. *Note*: Distance is in cm, so force is in dynes.

*In Problems 15-20 the given figure is the vertical cross section of a tank containing the indicated fluid. Find the fluid force against the end of the tank. The weight densities are given in Table 6.4 on page 473.*

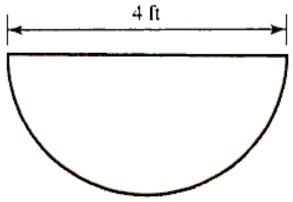
**15.** seawater



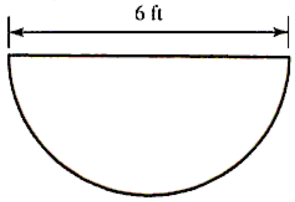

**16.** water



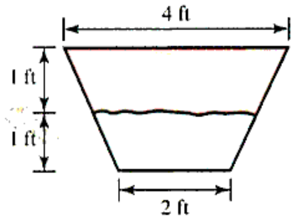

**17.** SAE 20 oil



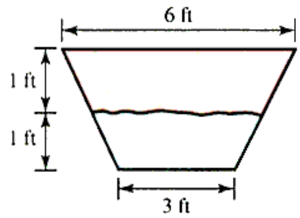

**18.** gasoline




**19.** kerosene

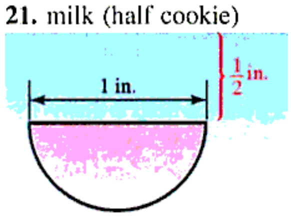



**20.** mercury

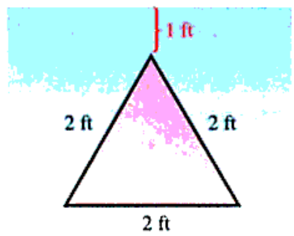



*In Problems 21-26, set up, but do not evaluate, an integral for the fluid force against the indicated vertical plate.*

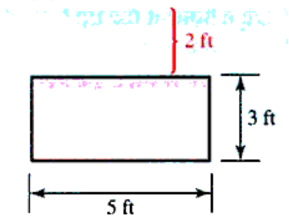
**21.** milk (half cookie)



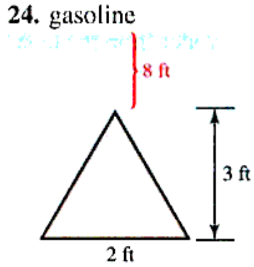

**22.** water



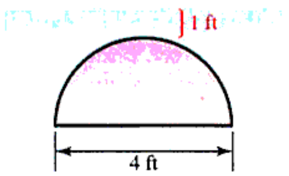

**23.** SAE 20 oil



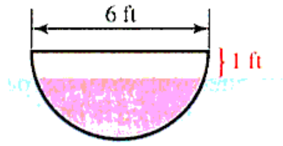

**24.** gasoline




**25.** kerosene




**26.** mercury




*Find the centroid ($\rho = 1$) of each planar region in Problems 27-32.*

**27.** the region bounded by the parabola $y = x^2 - 9$ and the $x$-axis

**28.** the region in the first quadrant bounded by the curves $y = x^3$ and $y = \sqrt[3]{x}$

**29.** the region bounded by the curve $y = x^{-1}$, the $x$-axis, and the lines $x = 1$ and $x = 2$

**30.** the region bounded by the parabola $y = 4 - x^2$ and the line $y = x + 2$

**31.** the region bounded by $y = \dfrac{1}{\sqrt{x}}$ and the $x$-axis on $[1, 4]$

**32.** the region bounded by the curve $y = x^{-1}$ and the line $2x + 2y = 5$

*Use the theorem of Pappus to compute the volume of the solids generated by revolving the regions given in Problems 33-36 about the prescribed axis.*

**33.** the triangular region with vertices $(-3, 0)$, $(0, 5)$, and $(2, 0)$, about the line $y = -1$

**34.** the region bounded by the curve $y = \sqrt{x}$, the $x$-axis, and the line $x = 4$ about the line $x = -1$

**35.** the semicircular region $x = \sqrt{4 - y^2}$, about the line $x = -2$

**36.** the semicircular region $y = \sqrt{1 - x^2}$, about the line $y = -1$

## Level 2

**37.** A tank in the shape of an inverted right circular cone of height 6 ft and top radius 3 ft is half full.

**a.** How much work is performed in pumping all the water over the top edge of the tank?

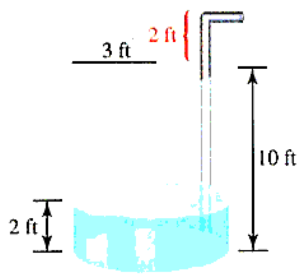
**b.** How much work is performed in draining all the water from the tank through a hole at the tip?

**38.** The centroid of any triangle lies at the intersection of the medians, two-thirds the distance from each vertex to the midpoint of the opposite side. Locate the centroid of the triangle with vertices $(0, 0)$, $(7, 3)$, and $(7, -2)$ using this method, and then check your result by calculus.

**39.** A hemispherical bowl with radius 10 ft is filled with water to a level of 6 ft. Find the work done to the nearest ft-lb to pump all the water to the top of the bowl.

**40.** A tank is formed by rotating the curve $y = x^2$ about the $y$-axis for $0 \leq x \leq 4$, where $x$ is in feet. If the tank is filled with water to a depth of 12 feet, how much work is performed in pumping all the water to a height of 3 ft above the top of the tank?

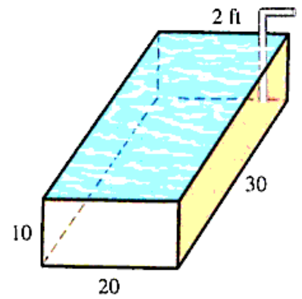
**41.** A cylindrical tank of radius 3 ft and height 10 ft is filled to a depth of 2 ft with a liquid of weight density $\delta = 40$ lb/ft$^3$. Find the work done in pumping all the liquid to a height of 2 ft above the top of the tank.




**42.** A holding tank has the shape of a rectangular parallelepiped 20 ft by 30 ft by 10 ft.

**a.** How much work is done in pumping all the water to the top of the tank?

**b.** How much work is done in pumping all the water out of the tank to a height of 2 ft above the top of the tank?




**43.** A swimming pool 20 ft by 15 ft by 10 ft deep is filled with water. A cube 1 ft on a side lies on the bottom of the pool. Find the total fluid force on the five exposed faces of the cube.

**44.** Suppose a log of radius 1 ft lies at the bottom of the pool in Problem 43. What is the total fluid force on one end of the log?

**45.** An object weighing 800 lb on the surface of the earth is propelled to a height of 200 mi above the earth. How much work is done against gravity? *Hint*: Assume the radius of the earth is 4,000 mi, and use Newton's law $F = -k/x^2$, for the force on an object $x$ miles from the center of the earth.

**46.** An oil can in the shape of a rectangular parallelepiped is filled with kerosene (weight density 51.2 lb/ft$^3$). What is the total force on a side of the can if it is 6 in high and has a square base 4 in. on a side?

**47.** **Modeling Problem** According to *Coulomb's law* in physics, two similarly charged particles repel each other with a force inversely proportional to the square of the distance between them. Suppose the force is 12 dynes when they are 5 cm apart.

**a.** How much work is done in moving one particle from a distance of 10 cm to a distance of 8 cm from the other?

**b.** Set up and analyze a model to determine the amount of work performed in moving one particle from an "infinite" distance to a distance of 8 cm from the other. State any assumptions that you must make.

**48.** A dam is in the shape of an isosceles trapezoid 200 ft at the top, 100 ft at the bottom, and 75 ft high. When water is up to the top of the dam, what is the force on its face?

**49.** **Modeling Problem** Figure 6.71 shows a swimming pool, part of whose bottom is an inclined plane. Set up and analyze a model for determining the fluid force on the bottom when the pool is filled to the top with water.

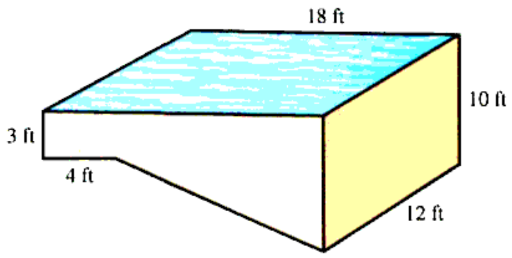



**Figure 6.71** Swimming pool dimensions

**50. Modeling Problem** Figure 6.72 shows a dam whose face against the water is an inclined rectangle.

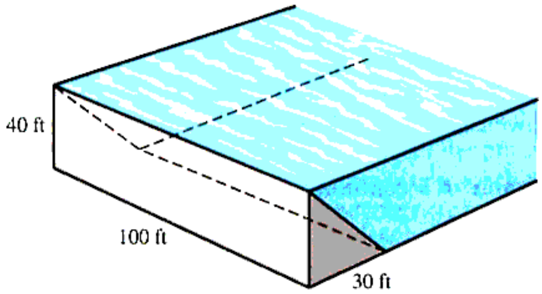



**Figure 6.72** Dam with face at an inclined rectangle

**a.** Set up and analyze a model for determining the fluid force against the face of the dam when the water is level with the top.

**b.** What would the fluid force be if the face of the dam were a vertical rectangle (not inclined)?

*A region R with uniform density is said to have* ***moments of inertia*** *$I_x$ about the x-axis and $I_y$ about the y-axis, where*

$$I_x = \int y^2 dA \quad and \quad I_y = \int x^2 dA$$

*The dA represents the area of an approximating strip. In Problems 51-52, find $I_x$ and $I_y$ for the given region.*

**51.** the region bounded by the parabola $y = \sqrt{x}$, the $x$-axis, and the line $x = 4$

**52.** the region bounded by the curve $y = \sqrt[3]{1 - x^3}$ and the coordinate axes

**53. Modeling Problem** If a region has area $A$ and moment of inertia $I_y$ about the $y$-axis, then it is said to have a **radius of gyration**

$$\rho_y = \sqrt{\frac{I_y}{A}}$$

with respect to the $y$-axis. Set up and analyze a model to determine the radius of gyration for the triangular region bounded by the line $y = 4 - x$ and the positive coordinate axes.

**54.** An irregular plate is measured at 0.2-ft intervals from top to bottom, and the corresponding widths noted. The plate is then submerged vertically in water with its top 1 ft below the surface. Use Simpson's rule to estimate the total force on the face of the plate.

| Depth, $x_k$ | Width, $L_k$ |
|---|---|
| 1.0 | 0.0 |
| 1.2 | 3.3 |
| 1.4 | 3.6 |
| 1.6 | 3.7 |
| 1.8 | 3.7 |
| 2.0 | 3.6 |
| 2.2 | 3.4 |
| 2.4 | 2.9 |
| 2.6 | 0.0 |

**Level 3**

**55.** A right circular cone is generated when the region bounded by the line $y = x$ and the vertical lines $x = 0$ and $x = r$ is revolved about the $x$-axis. Use Pappus' theorem to show that the volume of this cone is $V = \frac{1}{3}\pi r^3$.

**56.** Show that the centroid of a rectangle is the point of intersection of its diagonals.

**57.** Find the volume of the solid figure generated when a square of side $L$ is revolved about a line that is outside the square, parallel to two of its sides, and located $s$ units from the closer side.

**58.** Let $R$ be the triangular region bounded by the line $y = x$, the $x$-axis, and the vertical line $x = r$. When $R$ is rotated about the $x$-axis, it generates a cone of volume $V = \frac{1}{3}\pi r^3$. Use Pappus' theorem to determine $\overline{y}$, the $y$-coordinate of the centroid of $R$. Then use similar reasoning to find $\overline{x}$.

**59.** Let $R$ be a region in the plane, and suppose that $R$ can be partitioned into two subregions $R_1$ and $R_2$. Let $(\overline{x}_1, \overline{y}_1)$ and $(\overline{x}_2, \overline{y}_2)$ be the centroids of $R_1$ and $R_2$, respectively, and let $A_1$ and $A_2$ be the areas of the subregions. Show that $(\overline{x}, \overline{y})$ is the centroid of $R$, where

$$\overline{x} = \frac{A_1 x_1 + A_2 x_2}{A_1 + A_2} \quad \text{and} \quad \overline{y} = \frac{A_1 y_1 + A_2 y_2}{A_1 + A_2}$$

**60.** Find the volume of the solid figure generated by revolving an equilateral triangle of side $L$ about one of its sides.

## 6.6 Applications to Business, Economics, and Life Sciences

**IN THIS SECTION:** *Future and present value of a flow of income; cumulative change: net earnings; consumer's and producer's surplus; survival and renewal, the flow of blood through an artery*

In Section 6.5, we saw how integration can be applied to the physical sciences. The goal of this section is to explore applications to a variety of other areas, especially business, economics, and the life sciences.

### Future and Present Value of a Flow of Income

Recall from Chapter 2 that when $P_0$ dollars are invested at the continuous compounding annual rate $r$, then after $t$ years, the account is worth $A = P_0 e^{rt}$ dollars. The amount $A$ is called the *future value* of the initial value $P_0$ dollars, but what is the future value if the money were not deposited just once, in a lump sum $P_0$, but continuously over the entire time period? The difficulty is that at any given time, the "old" money deposited earlier in the time period has been accumulating interest longer than "new" money deposited more recently. To compute future value, in this case, our strategy will be to approximate the continuous income flow by a sequence of discrete deposits called an **annuity**. Then, the amount of the approximating annuity is a sum whose limit is a definite integral that gives the future value of the income flow. The general procedure is illustrated in Example 1.

**Example 1 Computing future value of a continuous income flow**

Money is transferred into an account continuously at the rate of 1,500 dollars/yr. If the account earns interest at the rate of 7% compounded continuously, what is the future value of the account 5 years from now?

***Solution*** To approximate the future value of the income flow, divide the 5-yr time period $0 \le t \le 5$ into $n$ intervals, each of length $\Delta_n t = (5-0)/n$, and for $k = 1, 2, \cdots, n$, let $t_{k-1}$ denote the beginning of the $k$th subinterval $[t_{k-1}, t_k]$. The amount of money deposited during the $k$th subinterval is $1{,}500\Delta_n t$. Therefore, if as an approximation, we assume that all this money is deposited at time $t_{k-1}$, then since $5 - t_{k-1}$ years remain in the 5-yr investment period, it follows that this money would grow to $(1{,}500\Delta_n t)e^{0.07(5-t_{k-1})}$ dollars; that is,

$$\begin{bmatrix}\text{FUTURE VALUE OF MONEY DEPOSITED}\\ \text{DURING THE } k\text{TH SUBINTERAVAL}\end{bmatrix} = 1{,}500e^{0.07(5-t_{k-1})}\Delta_n t$$

This approximation is illustrated in Figure 6.73.

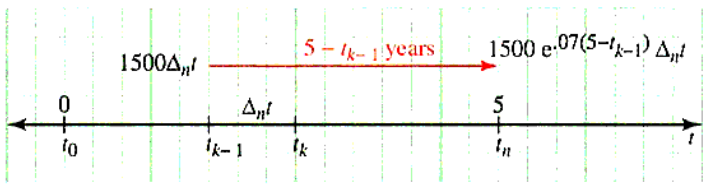



**Figure 6.73** Approximate future value of money deposited during the $k$th subinterval

The future value of the entire income stream can be approximated by the sum of the future values of the money deposited during all $n$ time subintervals. Hence, we have

$$[\text{FUTURE VALUE OF INCOME FLOW}] \approx \sum_{k=1}^{n} 1{,}500e^{0.07(5-t_{k-1})}\Delta_n t$$

As $n$ increases without bound, the length of each subinterval tends toward zero, and there is less and less error in our assumption that all the money in the $k$th subinterval is deposited at time $t = t_{k-1}$. Therefore, it is reasonable to compute the total future value of the income flow by taking the limit of the approximating sum as $n \to \infty$. We recognize this sum as a Riemann sum, and the limit as the definite integral. That is,

$$\begin{aligned}
[\text{FUTURE VALUE OF INCOME FLOW}] &= \lim_{n\to\infty} \sum_{k=1}^{n} 1{,}500e^{0.07(5-t_{k-1})}\Delta_n t \\
&= \int_0^5 1{,}500e^{0.07(5-t)}\,dt \\
&= 1{,}500e^{0.35}\int_0^5 e^{-0.07t}\,dt \\
&= 1{,}500e^{0.35}\left[\frac{e^{-0.07t}}{-0.07}\right]_0^5 \\
&= 8{,}980.02
\end{aligned}$$

Recall from Chapter 2, that the *present value*, $PV$, of $A$ dollars invested at the annual interest rate $r$ compounded continuously for a term $T$ years is the amount of money $PV = Ae^{-rt}$ that must be deposited today at the prevailing interest rate to generate $A$ dollars at the end of the term. Likewise, the present value of an income stream $f(t)$ generated continuously at a certain rate over a specified term is the amount of money that would have to be deposited at the beginning of the term at the prevailing interest rate to generate the income stream over the entire term. As with future value, the strategy is to approximate the continuous income stream by a sequence of discrete payments; that is, by an annuity. The present value of the approximating annuity is a sum, and by taking the limit of that sum, we obtain an integral

$$\int_0^T f(t)e^{-rt}\,dt$$

that represents the present value of the entire income flow. The details of this approximation scheme are outlined in the problem set. Here is a summary of the formulas to be used for computing future and present value of an income flow.

**FUTURE VALUE AND PRESENT VALUE** Let $f(t)$ be the amount of money deposited at time $t$ over the time period $[0, T]$ in an account that earns interest at the annual rate $r$ compounded continuously. Then the **future value** of the income flow over the time period is given by

$$FV = \int_0^T f(t)e^{r(T-t)}\,dt$$

and the **present value** of the same income flow over the time period is

$$PV = \int_0^T f(t)e^{-rt}\,dt$$

## Cumulative Change: Net Earnings

In certain applications, we are given the rate of change $Q'(x)$ of a quantity $Q(x)$ and are required to find the net change $Q(b) - Q(a)$ in $Q(x)$ as $x$ varies from $x = a$ to $x = b$. But because $Q(x)$ is an antiderivative of $Q'(x)$, the fundamental theorem of calculus tells us that the net change is given by the definite integral

$$Q(b) - Q(a) = \int_a^b Q'(x)\,dx$$

Since the definite integral on the right "adds" all the instantaneous changes in $Q(x)$ over the time period $[a, b]$, the integral can also be thought of as **cumulative change**. For example, if $N(t)$ is the population of a certain species that is growing at the rate $N'(t)$ at time $t$, then the cumulative change in population over the time period $[t_1, t_2]$ is given by the integral

$$N(t_2) - N(t_1) = \int_{t_1}^{t_2} N'(t)\,dt$$

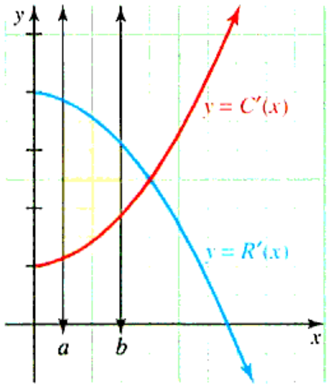



**Figure 6.74** Net earnings as an area

A business application involves the **net earnings** of an industrial process. Suppose $C(x)$ represents the total cost of producing $x$ units and $R(x)$ represents the revenue generated by the sale of those units. Then $E(x) = R(x) - C(x)$ represents earnings, and if the marginal cost $C'(x)$ and marginal revenue $R'(x)$ are known, then the net earnings for the production range $a \leq x \leq b$ are given by the definite integral

$$\begin{aligned} E(b) - E(a) &= \int_a^b E'(x)\,dx \\ &= \int_a^b [R'(x) - C'(x)]\,dx \end{aligned}$$

Net earnings can be interpreted geometrically as an area between the marginal revenue and marginal cost curves as shown in Figure 6.74. These ideas are illustrated in Example 2.

**Example 2 Computing net earnings**

For a certain industrial process, the marginal cost and marginal revenue (in thousands of dollars) associated with producing $x$ units are

$$C'(x) = 0.1x^2 + 4x + 10 \quad \text{and} \quad R'(x) = 70 - x$$

respectively. What are the net earnings of the process as $x$ ranges from $x = 0$ to $x = x_m$, where $x_m$ is the level of production at which marginal cost equals marginal revenue?

***Solution*** First, note that marginal cost equals marginal revenue when

$$\begin{aligned} 0.1x^2 + 4x + 10 &= 70 - x \\ 0.1x^2 + 5x - 60 &= 0 \\ x^2 + 50x - 600 &= 0 \\ (x + 60)(x - 10) &= 0 \\ x &= -60,\ 10 \end{aligned}$$

Reject the negative value, giving the solution $x_m = 10$, so that the net earnings as $x$ ranges from $x = 0$ to $x = 10$ are given by

$$\begin{aligned} E(10) - E(0) &= \int_0^{10} [R'(x) - C'(x)]\,dx \\ &= \int_0^{10} [70 - x - (0.1x^2 + 4x + 10)]\,dx \end{aligned}$$

$$= \int_0^{10} [-0.1x^2 - 5x + 60]\,dx$$

$$= \frac{950}{3}$$

The net earnings (rounded to the nearest dollar) are $316,667, and are shown as the area between the marginal cost and marginal revenue curves in Figure 6.75.

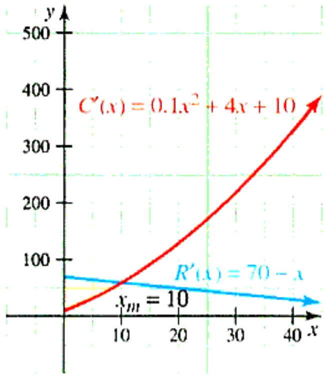


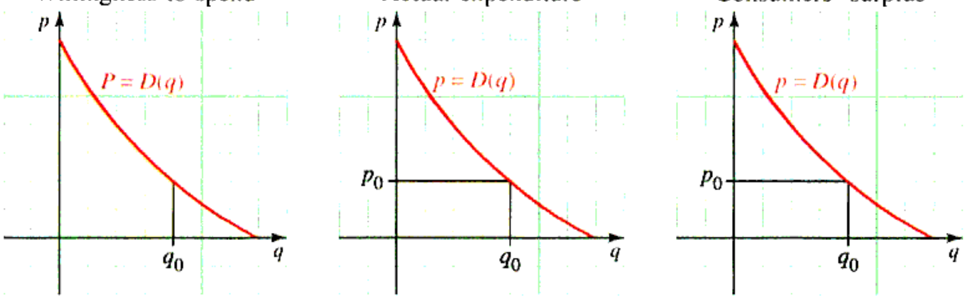

**Figure 6.75** Net earnings for the marginal cost and marginal revenue

## Consumer's and Producer's Surplus

In a competitive economy, the total amount that consumers actually spend on a commodity is usually less than the total amount they would have been willing to spend. The difference between the two amounts can be thought of as a savings realized by consumers and is known in economics as the **consumer's surplus**.

To get a better feel for the concept of consumer's surplus, consider an example of a couple who are willing to spend $500 for their first flat-screen HDTV, $300 for a second set, and only $50 for a third set. If the market price is $300, then the couple would buy only two sets and would spend a total of $2 \times \$300 = \$600$. This is less than the $\$500 + \$300 = \$800$ that the couple would have been willing to spend to get the two sets. The savings of $\$800 - \$600 = \$200$ is the couple's consumer's surplus. Consumer's surplus has a simple geometric interpretation, which is illustrated in Figure 6.76.

Willingness to spend − Actual expenditure = Consumers' surplus




$$\int_0^{q_0} D(q)dq \quad - \quad \int_0^{q_0} p_0\,dq \quad = \quad \int_0^{q_0} [D(q) - p_0]\,dq$$

**Figure 6.76** Geometric interpretation of consumer's surplus

Note that $p$ and $q$ denote the market price and the corresponding demand, respectively. The first figure shows the region under the demand curve $p = D(q)$ from $q = 0$ to $q = q_0$, and the area of this region represents the total amount that consumers are willing to spend to get $q_0$ units of the commodity. The rectangle in the middle has an area of $p_0 q_0$ and represents the actual consumer expenditure for $q_0$ units at $p_0$ dollars per unit. The difference between these two areas (figure at the right) represents the consumer's surplus. That is, consumer's surplus is the area of the region between the demand curve $p = D(q)$ and the horizontal line $p = p_0$ so that

$$\int_0^{q_0} [D(q) - p_0]\,dq = \int_0^{q_0} D(q)\,dq - \int_0^{q_0} p_0\,dq = \int_0^{q_0} D(q)\,dq - p_0 q_0$$

**CONSUMER'S SURPLUS** If $q_0$ units of a commodity are sold at a price of $p_0$ dollars per unit, and if $p = D(q)$ is the consumer's demand function for the commodity, then

$$\text{CONSUMER'S SURPLUS} = \begin{bmatrix}\text{TOTAL AMOUNT} \\ \text{CONSUMERS ARE} \\ \text{WILLING TO SPEND} \\ \text{FOR } q_0 \text{ UNITS}\end{bmatrix} - \begin{bmatrix}\text{ACTUAL CONSUMER} \\ \text{EXPENDITURE FOR} \\ q_0 \text{ UNITS}\end{bmatrix}$$

$$CS = \int_0^{q_0} D(q)\,dq - p_0 q_0$$

Producer's surplus is the other side of the coin from consumer's surplus. In particular, a supply function $p = S(q)$ gives the price per unit that producers are willing to accept in order to supply $q$ units to the marketplace. However, any producer who is willing to accept less than $p_0 = S(q_0)$ dollars for $q_0$ units gains from the fact that the price is actually $p_0$. Then **producer's surplus** is the difference between what producers would be willing to accept for supplying $q$ units and the price they actually receive. Reasoning as we did with consumer's surplus, we obtain the following formula for producer's surplus.

**PRODUCER'S SURPLUS** If $q_0$ units of a commodity are sold at a price of $p_0$ dollars per unit, and if $p = S(q)$ is the producer's supply function for the commodity, then

$$\text{PRODUCER'S SURPLUS} = \begin{bmatrix}\text{ACTUAL CONSUMER} \\ \text{EXPENDITURE} \\ \text{FOR } q_0 \text{ UNITS}\end{bmatrix} - \begin{bmatrix}\text{TOTAL AMOUNT} \\ \text{PRODUCER WILLING TO} \\ \text{ACCEPT WHEN } q_0 \\ \text{UNITS ARE SUPPLIED}\end{bmatrix}$$

$$PS = p_0 q_0 - \int_0^{q_0} S(q)\,dq$$

### Example 3 Computing consumer's and producer's surplus

A tire manufacturer estimates that $q$ (thousand) radial tires will be purchased (that is, demanded) by wholesalers when the price is

$$p = D(q) = -0.1q^2 + 90$$

dollars/tire, and the same number of tires will be supplied when the price is

$$p = S(q) = 0.2q^2 + q + 50$$

dollars/tire.

**a.** Find the equilibrium price (that is, where supply equals demand) and the quantity supplied and demanded at that price.

**b.** Determine the consumer's and producer's surplus at the equilibrium price.

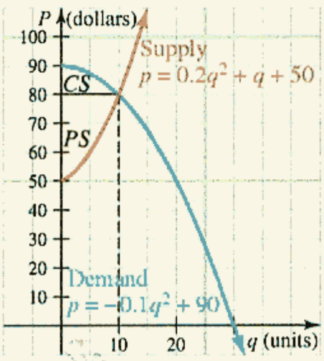



**Figure 6.77** Consumer's and producer's surplus at equilibrium for the given supply and demand functions

***Solution***

**a.** The supply and demand curves are shown in Figure 6.77.

Supply equals demand when

$$
\begin{aligned}
0.2q^2 + q + 50 &= -0.1q^2 + 90 \\
0.3q^2 + q - 40 &= 0 \\
3q^2 + 10q - 400 &= 0 \\
(3q + 40)(q - 10) &= 0 \\
q &= 10 \qquad \left(\text{reject } q = -\frac{40}{3}\right)
\end{aligned}
$$

and $p = S(10) = D(10) = 80$. Thus, equilibrium occurs at a price of \$80/tire, when 10,000 tires are supplied and demanded.

**b.** Using $p_0 = 80$ and $q_0 = 10$, we find that the consumer's surplus is

$$
\begin{aligned}
\int_0^{10} (-0.1q^2 + 90)dq - (80)(10) &= \left[-0.1\left(\frac{q^3}{3}\right) + 90q\right]_0^{10} - (80)(10) \\
&= \frac{2{,}600}{3} - (80)(10) \\
&= \frac{200}{3}
\end{aligned}
$$

The producer's surplus is

$$
\begin{aligned}
(80)(10) - \int_0^{10} (0.2q^2 + q + 50)\,dq &= (80)(10) - \left[0.2\left(\frac{q^3}{3}\right) + \left(\frac{q^2}{2}\right) + 50q\right]_0^{10} \\
&= (80)(10) - \frac{1{,}850}{3} \\
&= \frac{550}{3}
\end{aligned}
$$

Since $q$ is in thousands, the consumer's surplus is about \$66,667, and producer's surplus is about \$183,333. The consumer's surplus and the producer's surplus are the regions labeled $CS$ and $PS$, respectively, in Figure 6.77.

## Survival and Renewal

There is an important category of problems in which a survival function gives the fraction of individuals in a group of the population that can be expected to remain in the group and a renewal function gives the rate at which new members can be expected to join the group. Such **survival and renewal** problems arise in many fields, including sociology, ecology, demography, and even finance, where the "population" is the number of dollars in an investment account and "survival and renewal" refers to results of an investment strategy. The following example illustrates the general procedure for modeling with survival and renewal functions.

### Example 4 Modeling enrollment using survival and renewal

A new county mental health clinic has just opened. Statistics from similar facilities suggest that the fraction of patients who will still be receiving treatment at the clinic $t$ months after their initial visit is given by the function $f(t) = e^{-t/20}$. The clinic initially accepts 300 people for treatment and plans to accept new patients at the rate of 10 per month. Approximately how many people will be receiving treatment at the clinic 15 months from now?

***Solution*** Since $f(15)$ is the fraction of patients whose treatment continues at least 15 months, it follows that of the current 300 patients, only $300\,f(15)$ will still be receiving treatment 15 months from now.

To approximate the number of *new* patients who will be receiving treatment 15 months form now, divide the 15-month time interval $0 \le t \le 15$ into $n$ equal subintervals of length $\Delta t$ months and let $t_{j-1}$ denote the beginning of the $j$th subinterval. Since new patients are accepted at the rate of 10/mo, the number of new patients accepted during the $j$th subinterval is $10\Delta t$. Fifteen months from now, approximately $15 - t_{j-1}$ months will have elapsed since these $10\Delta t$ new patients had their initial visits, and so approximately $(10\Delta t)f(15 - t_{j-1})$ of them will still be receiving treatment at that time (see Figure 6.78).

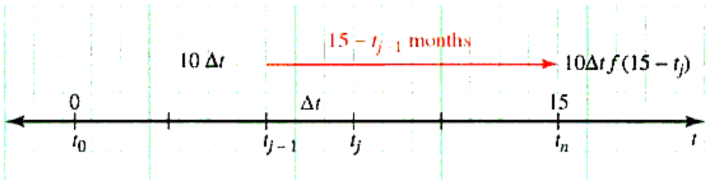



**Figure 6.78** New members arriving during the $j$th subinterval

It follows that the total number of new patients still receiving treatment 15 months from now can be approximated by the sum

$$\sum_{j=1}^{n} 10\,f(15 - t_{j-1})\Delta t$$

Adding this to the number of current patients who will still be receiving treatment in 15 months, we obtain

$$P \approx 300\,f(15) + \sum_{j=1}^{n} 10\,f(15 - t_{j-1})\Delta t$$

where $P$ is the total number of patients who will be receiving treatment 15 months from now. As $n$ increases without bound, the approximation improves and approaches the true value of $P$. That is,

$$\begin{aligned}
P &= 300\,f(15) + \lim_{n\to\infty} \sum_{j=1}^{n} 10\,f(15 - t_{j-1})\Delta t \\
&= 300\,f(15) + \int_0^{15} 10\,f(15 - t)\,dt \\
&= 300e^{-3/4} + 10e^{-3/4}\int_0^{15} e^{t/20}dt && f(t) = e^{-t/20};\ so\, f(15) = e^{-3/4} \\
& && and\, f(15 - t) = e^{-(15-t)/20} = e^{-3/4}e^{t/20} \\
&= 300e^{-3/4} + 200e^{-3/4}e^{t/20}\Big|_0^{15} \\
&= 300e^{-3/4} + 200(1 - e^{-3/4}) \\
&\approx 247.24
\end{aligned}$$

That is, 15 months from now, the clinic will be treating approximately 247 patients.

## The Flow of Blood Through an Artery

In biology, Poiseuille's law of flow* asserts that the speed of blood in an artery is a function of the distance of the blood from the artery's central axis. That is, the speed (in cm/s) of blood that is $r$ centimeters from the central axis of the artery may be modeled by

$$S(r) = k(R^2 - r^2)$$

where $R$ is the radius of the artery and $k$ is a constant. In the following example, you will see how to use this information to compute the volume rate (in cm$^3$/s) at which blood flows through the artery.

**Example 5 Modeling blood flow**

Find an expression for the volume rate (in cm$^3$/s) at which blood flows through an artery of radius $R$ if the speed of the blood $r$ cm from the central axis is $S(r) = k(R^2 - r^2)$, where $k$ is a constant.

***Solution*** To find the approximate volume of blood that flows through a cross section of the artery each second, divide the interval $0 \le r \le R$ into $n$ equal subintervals of width $\Delta r$ cm and let $r_j$ denote the beginning of the $j$th subinterval. These subintervals determine $n$ concentric rings, as illustrated in Figure 6.79.

If $\Delta r$ is small, the area of the $j$th ring is approximately the area of a rectangle whose length is the circumference of the (inner) boundary of the ring and whose width is $\Delta r$. That is,

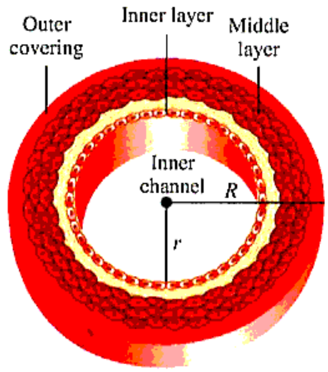



**Figure 6.79 Interactive**
Subdividing a cross section of an artery into concentric rings

$$\text{AREA OF THE } j\text{TH RING} \approx 2\pi r_j \Delta r$$

If we multiply the area of the $j$th ring (cm$^2$) by the speed (cm/s) of the blood flowing through this ring, we obtain the volume rate (cm$^3$/s) at which blood flows through the $j$th ring. Since the speed of blood flowing through the $j$th ring is approximately $S(r_j)$ cm/s, it follows that

$$\begin{aligned}
\begin{matrix}\text{VOLUME RATE OF FLOW}\\ \text{THROUGH THE } j\text{TH RING}\end{matrix} &= (\text{AREA OF } j\text{TH RING})\begin{pmatrix}\text{SPEED OF BLOOD}\\ \text{THROUGH THE}\\ j\text{TH RING}\end{pmatrix}\\
&= (2\pi r_j \Delta r)S(r_j)\\
&= 2\pi r_j \Delta r)[k(R^2 - r_j^2)]\\
&= 2\pi k(R^2 r_j - r_j^3)\Delta r
\end{aligned}$$

The volume rate of flow of blood through the entire cross section is the sum of $n$ such terms, one for each of the $n$ concentric rings. That is,

$$\begin{aligned}
\text{VOLUME RATE OF FLOW} &\approx \sum_{j=1}^{n} 2\pi k(R^2 r_j - r_j^3)\Delta r\\
&= \lim_{n\to\infty} \sum_{j=1}^{n} 2\pi k(R^2 r_j - r_j^3)\Delta r
\end{aligned}$$

*As $n$ increases without bound, this approximation approaches the true value of the volume rate of flow.*

*The law of flow is named for the same French physician, Jean Louis Poiseuille (1799-1869), responsible for the resistance to flow law cited in Section 4.7.

$$= \int_0^R 2\pi k(R^2 r - r^3)dr$$

$$= 2\pi k\left(\frac{R^2}{2}r^2 - \frac{1}{4}r^4\right)\Big|_0^R$$

$$= \frac{\pi k R^4}{2} \qquad \text{This is in cm}^3/\text{s.}$$

## PROBLEM SET 6.6

### Level 1

1. ■ What does this say? What is cumulative change for a quantity $Q(x)$?
2. ■ What does this say? What are the future and present values of an income flow?
3. ■ What does this say? What are consumer's and producer's surplus?
4. ■ What does this say? What do we mean by survival and renewal?

*Find the consumer's surplus for the given demand function defined by $D(q)$ at the point that corresponds to the sales level $q$ as given in Problems 5-8.*

5. $D(q) = 150 - 6q$
   a. $q_0 = 5$ b. $q_0 = 12$
6. $D(q) = 3.5 - 0.5q$
   a. $q_0 = 1$ b. $q_0 = 1.5$
7. $D(q) = 2.5 - 1.5q$
   a. $q_0 = 1$ b. $q_0 = 0$
8. $D(q) = 100 - 8q$
   a. $q_0 = 4$ b. $q_0 = 10$

*In Problems 9-12, $p = D(q)$ is the price (in dollars/unit) at which $q$ units of a particular commodity will be demanded by the market (that is, all $q$ units will be sold at this price), and $q_0$ is a specified level of production. In each case, find the price $p_0 = D(q_0)$ at which $q_0$ units will be demanded and compute the corresponding consumer's surplus. Sketch the demand curve $y = D(q)$ and shade the region whose area represents the consumer's surplus.*

9. $D(q) = 150 - 2q - 3q^2$; $q_0 = 6$ units
10. $D(q) = 2(64 - q^2)$; $q_0 = 3$ units
11. $D(q) = 75e^{-0.04q}$; $q_0 = 3$ units
12. $D(q) = 40e^{-0.05q}$; $q_0 = 5$ units

*In Problems 13-16, $p = S(q)$ is the price (in dollars/unit) at which $q$ units of a particular commodity will be supplied to the market by producers, and $q_0$ is a specified level of production. In each case, find the price $p_0 = S(q_0)$ at which $q_0$ units will be supplied and compute the corresponding producer's surplus. Sketch the supply curve $y = S(q)$ and shade the region whose area represents the producer's surplus.*

13. $S(q) = 0.5q + 15$; $q_0 = 5$ units
14. $S(q) = 0.3q^2 + 30$; $q_0 = 4$ units
15. $S(q) = 17 + 11e^{0.01q}$; $q_0 = 7$ units
16. $S(q) = 10 + 15e^{0.03q}$; $q_0 = 3$ units

### Level 2

*In Problems 17-20, find the consumer's surplus at the point of market equilibrium [the level of production where supply $S(q)$ equals demand $D(q)$].*

| | Demand function | Supply function |
|---|---|---|
| 17. | $D(q) = 27 - q^2$ | $S(q) = \frac{1}{4}q^2 + \frac{1}{2}q + 5$ |
| 18. | $D(q) = 14 - q^2$ | $S(q) = 2q^2 + 2$ |
| 19. | $D(q) = 25 - q^2$ | $S(q) = 5q^2 + 1$ |
| 20. | $D(q) = 32 - 2q^2$ | $S(q) = \frac{1}{3}q^2 + 2q + 5$ |

21. It is estimated that $t$ weeks from now, contributions in response to a fund-raising campaign will be coming in at the rate of

$$R'(t) = 5{,}000e^{-0.2t}$$

dollars/wk, while campaign expenses are expected to accumulate at the constant rate of \$676/wk.
   a. For how many weeks does the rate of revenue exceed the rate of cost?
   b. What net earnings will be generated by the campaign during the period of time determined in part **a**.
   c. Interpret the net earnings in part **b** as an area between two curves.
22. A manufacturer has determined that when $q$ units of a particular commodity are produced, the price at which all the units can be sold is $p = D(q)$ dollars per unit, where $D$ is the demand function given by

$$D(q) = \frac{300}{(0.1q+1)^2}$$

**a.** How many units can the manufacturer expect to sell if the commodity is priced at $p_0 = \$12$/unit?
**b.** Find the consumer's surplus that corresponds to the level of production $q_0$ found in part **a.**

**23.** Money is transferred continuously into an account at the constant rate of \$3,000/yr. The account earns interest at the annual rate of 8% compounded continuously.
**a.** What is the future value of the income flow in 3 years?
**b.** What is the present value of the income flow over a three-year term?

**24.** The management of a small firm expects to generate income at the rate of

$$f(t) = 2{,}000e^{0.03t}$$

dollars/yr. The prevailing rate of interest is 7% compounded continuously.
**a.** What is the future value for the firm over a five-year term?
**b.** What is the present value of the firm over the same five-year term?

**25.** Suppose an industrial machine that is $x$ years old generates revenue

$$R'(x) = 6{,}025 - 10x^2$$

dollars and costs

$$C'(x) = 4{,}000 + 15x^2$$

dollars to operate and maintain.
**a.** For how many years is it profitable to use this machine? [Recall that profit is $P(x) = R(x) - C(x)$.]
**b.** What are the net earnings generated by the machine during its period of profitability? Interpret this amount as the area between two curves.

**26.** After $t$ hours on the job, one factory worker is producing at the rate of

$$Q_1'(t) = 60 - 2(t-1)^2$$

per hour, and a second is producing

$$Q_2'(t) = 50 - 5t$$

units per hour. If both arrive on the job at 8:00 A.M., how many more units (to the nearest unit) will the first worker have produced by noon than the second worker? Interpret your answer as the area between two curves.

**27.** Suppose that $x$ years from now, one investment plan will be generating profit at the rate of

$$P_1'(x) = 100 + x^2$$

dollars per year, whereas a second plan will be generating profit at the rate of

$$P_2'(x) = 190 + 2x$$

dollars per year. Neither plan generates a profit in the beginning (when $x = 0$).
**a.** For how many years will the second plan be more profitable? That is, for how many years will $P_2$ be greater than $P_1$?
**b.** How much excess profit will you earn if you invest in the second plan instead of the first for the period of time in part **a**? Interpret the excess profit as the area between the two profit curves.

**28.** Parts for a piece of heavy machinery are sold in units of 1,000. The demand for the parts (in dollars) is given by $p(x) = 110 - x$. The total cost is given by $C(x) = x^3 - 25x^2 + 2x + 30$ (dollars).
**a.** For what value of $x$ is the profit

$$P(x) = xp(x) - C(x)$$

maximized?
**b.** Find the consumer's surplus with respect to the price that corresponds to maximum profit.

**29.** Repeat Problem 28 with

$$p(x) = 124 - 2x$$

and

$$C(x) = 2x^3 - 59x^2 + 4x + 76$$

**30.** Suppose when $q$ units of a commodity are produced, the demand is $p(q) = 45 - q^2$ dollars per unit, and the marginal cost is

$$\frac{dC}{dq} = 6 + \frac{1}{4}q^2$$

Assume there is no overhead (so $C(0) = 0$).
**a.** Find the total revenue and the marginal revenue.
**b.** Find the value of $q$ (to the nearest unit) that maximizes profit.
**c.** Find the consumer's surplus at the value of $q$ where profit is maximized. (Use the exact value of $q$.)

**31.** Repeat Problem 30 with

$$p(q) = \frac{1}{4}(10-q)^2 \text{ and } \frac{dC}{dq} = \frac{3}{4}q^2 + 5$$

**32.** A company plans to hire additional personnel. Suppose it is estimated that as $x$ new people are hired, it will cost $C(x) = 0.2x$ thousand dollars and that these $x$ people will bring in $R(x) = \sqrt{3x}$ thousand dollars in additional revenue. How many new people would be hired to maximize the profit? How much net revenue would the company gain by hiring these people?

**33.** Suppose that the demand function for a certain commodity is

$$D(q) = 20 - 4q^2$$

and that the marginal cost is

$$C'(q) = 2q + 6$$

where $q$ is the number of units produced. Find the consumer's surplus at the sales level $q_0$ where profit is maximized.

**34.** The resale value of a certain industrial machine decreases over a ten-year period at a rate that changes with time. When the machine is $x$ years old, its value is decreasing at the rate of $220(x - 10)$ dollars/yr. By how much does the machine depreciate during the second year?

**35.** The promoters of a county fair estimate that $t$ hours after the gates open at 9:00 A.M. visitors will be entering the fair at the rate of

$$-4(t + 2)^3 + 54(t + 2)^2$$

people/hr. How many people will enter the fair between 10:00 A.M. and noon?

**36.** At a certain factory, the marginal cost is $6(q - 5)^2$ dollars/unit when the level of production is $q$ units. By how much will the total manufacturing cost increase if the level of production is raised from 10 to 13 units?

**37.** A certain oil well that yields 400 barrels of crude oil a month will run dry in two years. The price of crude oil is currently \$108/barrel and is expected to rise at a constant rate of \$0.03 per barrel per month. If the oil is sold as soon as it is extracted from the ground, what will be the total future revenue from the well?



**38.** It is estimated that $t$ days from now a farmer's crop (in bushels) will be increasing at the rate of

$$0.3t^2 + 0.6t + 1$$

bu/day. By how much will the value of the crop increase during the next five days if the market price remains fixed at \$3/bu?

**39.** It is estimated that the demand for a manufacturer's product is increasing exponentially at the rate of 2%/yr. If the current demand is 5,000 units/yr and if the price remains fixed at \$400/unit, how much revenue will the manufacturer receive from the sale of the product over the next two years?

**40.** After $t$ hours on the job, a factory worker can produce $100(2 + \sin t)$ units/hr. How many units does a worker who arrives on the job at 8:00 A.M. produce between 10:00 A.M. and noon?

**41.** Suppose that $t$ years from now, one investment plan will be generating profit at the rate of

$$P_1'(t) = 100 + t^2$$

hundred dollars per year, while a second investment will be generating profit at the rate of $P_2'(t) = 220 + 2t$ hundred dollars per year.

**a.** For how many years does the rate of profitability of the second investment exceed that of the first?

**b.** Compute the net excess profit assuming that you invest in the second plan for the time period determined in part **a**.

**c.** Sketch the rate of profitability curves $y = P_1'(t)$ and $y = P_2'(t)$ and shade the region whose area represents the net excess profit computed in part **b**.

**42.** Answer the questions in Problem 41 for two investments with respective rates of profitability $P_1'(t) = 130 + t^2$ hundred dollars per year and $P_2'(t) = 306 + 5t$ hundred dollars per year.

**43.** Answer the questions in Problem 41 for two investments with respective rates of profitability

$$P_1'(t) = 60e^{0.12t}$$

thousand dollars per year and

$$P_2'(t) = 160e^{0.08t}$$

thousand dollars per year.

**44.** Answer the questions in Problem 41 for two investments with respective rates of profitability

$$P_1'(t) = 90e^{0.1t}$$

thousand dollars per year and

$$P_2'(t) = 140e^{0.07t}$$

thousand dollars per year.

**45.** Suppose that when it is $t$ years old, a particular industrial machine generates revenue at the rate

$$R'(t) = 6{,}025 - 8t^2$$

dollars/yr and that operating and servicing costs accumulate at the rate of

$$C'(t) = 4{,}681 + 13t^2$$

dollars/yr.

**a.** How many years pass before the machine ceases to be profitable?

**b.** Compute the net earnings generated by the machine over the time period determined in part **a**.

**c.** Sketch the revenue rate curve $y = R'(t)$ and the cost rate curve $y = C'(t)$ and shade the region whose area represents the net earnings computed in part **b**.

**46.** Answer the questions in Problem 45 for a machine that generates revenue at the rate

$$R'(t) = 7{,}250 - 18t^2$$

dollars/yr and for which costs accumulate at the rate

$$C'(t) = 3{,}620 + 12t^2$$

dollars/yr.

**47.** After $t$ hours on the job, one factory worker is producing

$$Q_1'(t) = 60 - 2(t-1)^2$$

units/hr, while a second worker is producing

$$Q_2'(t) = 50 - 5t$$

units/hr.

**a.** If both arrive on the job at 8:00 A.M., how many more units will the first worker have produced by noon than the second worker?

**b.** Interpret the answer in part **a** as the area between two curves.

**48.** A national consumers' association has compiled statistics suggesting that the fraction of its members who are still active $t$ months after joining is given by the function $f(t) = e^{-0.2t}$. A new local chapter has 200 charter members and expects to attract new members at the rate of 10/mo. How many members can the chapter expect to have at the end of 8 months?

**49.** The population density $r$ miles from the center of a certain city is

$$D(r) = 25{,}000e^{-0.05r^2}$$

people/mi$^2$. How many people live between 1 and 2 miles from the city center?

**50.** The population density $r$ miles from the center of a certain city is

$$D(r) = 5{,}000e^{-0.1r^2}$$

people/mi$^2$. How many people live within 3 miles from the city center?

**51.** Calculate the volume rate (in cm$^3$/s) at which blood flows through an artery of radius 0.1 cm if the speed of the blood $r$ cm from the central axis is $8(1 - 100r^2)$ cm/s.

**52. Modeling Problem** Blood flows through an artery of radius $R$. At a distance $r$ cm from the central axis of the artery, the speed of the blood is given by

$$S(r) = k(R^2 - r^2)$$

Show that the average speed of the blood is one-half the maximum speed.

**53.** A study indicates that $x$ months from now the population of a certain town will be increasing at the rate of $5 + 3x^{2/3}$ people/mo. By how much will the population of the town increase over the next 8 months?

**54.** An object is moving so that its speed after $t$ minutes is $5 + 2t + 3t^2$ m/min. How far does the object travel during the second minute?

**55.** It is estimated that the demand for oil is increasing exponentially at the rate of 10%/yr. If the demand for oil is currently 30 billion barrels per year, how much oil will be consumed during the next 10 years?

## Level 3

**56.** The administrators of a town estimate that the fraction of people who will still be residing in the town $t$ years after they arrive is given by the function

$$f(t) = e^{-0.04t}$$

If the current population is 20,000 people and new townspeople arrive at the rate of 500/yr, what will the population be ten years from now? *Hint*: Divide the interval $0 \le t \le 10$ into $n$ equal subintervals and form a Riemann sum.

**57.** The operators of a new computer dating service estimate that the fraction of people who will retain their membership in the service for at least $t$ months is given by the function $f(t) = e^{-t/10}$. There are 8,000 charter members, and the operators expect to attract 200 new members/mo. How many members will the service have ten months from now? *Hint:* Divide the interval $0 \le t \le 10$ into $n$ equal subintervals and form a Riemann sum.

**58. Modeling Problem** A certain nuclear power plant produces radioactive waste in the form of strontium-90 at the constant rate of 500 pounds/yr. The waste decays exponentially with a half-life of 28 years. How much of the radioactive waste from the nuclear plant will be present after 1400 years. *Hint*: Think of this as a survival and renewal problem and form a Riemann sum.

**59. Modeling Problem** A certain investment generates income continuously over a period of $N$ years. After $t$ years, the investment will be generating income at the rate of $f(t)$ dollars/yr. The prevailing annual rate of interest, compounded continuous, is $r$. Derive the formula

$$\int_0^N f(t)e^{-rt}\,dt$$

for the present value of the income flow by completing these steps.

**a.** Subdivide the time interval $[0, N]$ into $n$ equal subintervals, the $k$th of which is $[t_{k-1}, t_k]$ for $k = 1, 2, \cdots, n$. Assuming that the income for the entire $k$th subinterval is generated at a constant rate of $f(t_k)$, what is the present value of this income?

**b.** Approximate the present value of the total income flow by an appropriate sum.

**c.** Note that the sum in part **b** is a Riemann sum. Make the approximation precise by taking a limit to obtain the required integral formula for present value.

**60. Modeling Problem** A manufacturer receives $N$ units of a certain raw material that are initially placed in storage and then withdrawn and used at a constant rate until the supply is exhausted one year later. Suppose storage costs remain fixed at $p$ dollars per unit per year. Use definite integration to find an expression for the total storage costs the manufacturer will pay during the year. *Hint*: Let $Q(t)$ denote the number of units in storage after $t$ years and find an expression for $Q(t)$. Then subdivide the interval $0 \le t \le 1$ into $n$ equal subintervals and express the total storage cost as the limit of a sum.

# CHAPTER 6 REVIEW

*Nature laughs at the difficulties of integration.*

Pierre-Simon de Laplace
Quoted in *The Armchair Science Reader, 1959*

## Proficiency Examination

### Concept Problems

**1.** Describe the process for finding the area between two curves.
**2.** Describe the process for finding volumes of solids with a known cross-sectional area.
**3.** Compare and contrast the methods of finding volumes by disks, washers, and shells.
**4.** What are the formulas for changing from polar to rectangular form? From polar to rectangular form?
**5.** Draw one example of each of the given polar-form curves.
**a.** cardioid **b.** limaçon **c.** lemniscate **d.** rose curve with three petals
**6.** Identify and then sketch each of the given curves.
**a.** $r = 2\cos\theta$ **b.** $r = \cos\theta + 1$ **c.** $r = \theta, r \geq 0$ **d.** $r^2 = \cos 2\theta$
**7.** What is the procedure for finding the intersection of polar-form curves?
**8.** Outline a procedure for finding the area bounded by a polar graph.
**9.** What is the formula for arc length of a graph?
**10.** What is the formula for area of a surface of revolution?
**11.** How do you find the work done by a variable force?
**12.** How do you find the force exerted by a liquid on an object submerged vertically?
**13.** What are the formulas for the mass and centroid of a homogeneous lamina?
**14.** State the volume theorem of Pappus.
**15.** What formulas are used to compute future value and present value of an income flow?
**16.** How do you find the net change of a function $f$ whose derivative $f'$ is given?
**17.** Describe the process for finding consumer's and producer's surplus.
**18.** Describe a typical survival and renewal problem.
**19.** What formula is used for computing the flow of blood through an artery?

### Practice Problems

**20. Journal Problem** (*Mathematics Teacher**) *Use the integrals A-G to answer the given questions.*

**A.** $\pi\int_a^b [f(x)]^2\,dx$ **B.** $\pi\int_c^d [f(y)]^2\,dy$ **C.** $\int_a^b A(x)\,dx$ **D.** $\int_c^d A(y)\,dy$

**E.** $\pi\int_a^b [f(x) - g(x)]\,dx$ **F.** $\pi\int_a^b \{[f(x)]^2 - [g(x)]^2\}\,dx$ **G.** $\pi\int_c^d \{[f(y)]^2 - [g(y)]^2\}\,dy$

**a.** Which formula does not seem to belong to this list?
**b.** Which expressions represent volumes of solids of revolution?
**c.** Which expressions represent volumes of solids containing holes?
**d.** Which expressions represent volumes of a solid with cross sections that are perpendicular to an axis?
**e.** Which expressions represent volumes of solids of revolution revolved about the $x$-axis?
**f.** Which expressions represent volumes of solids of revolution revolved about the $y$-axis?

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*Vol. 83, No. 9, December 1990, p. 695.

**21.** Find the area above the curve $f(x) = 3x^2 - 6$ and below the $x$-axis.
**22.** Find the area between the curves $y = x^2$ and $y^3 = x$.
**23.** The base of a solid in the $xy$-plane is the circular region given by $x^2 + y^2 = 4$. Every cross section perpendicular to the $x$-axis is a rectangle whose height is twice the length of the side that lies in the $xy$-plane. Find the volume of this solid.
**24.** Let $R$ be a region bounded by the graphs of the equations $y = (x-2)^2$ and $y = 4$.
  **a.** Sketch the graph of $R$ and find its area.
  **b.** Find the volume of a solid of revolution of $R$ about the $x$-axis.
  **c.** Find the volume of a solid of revolution of $R$ about the $y$-axis.
**25.** Find the area of the intersection of the circles $r = 2a\cos\theta$ and $r = 2a\sin\theta$, where $a > 0$ is constant.
**26.** Find the arc length of the curve $y = 2 - x^{3/2}$ from $(0, 2)$ to $(1, 1)$. Give the exact value.
**27.** Find the surface area obtained by revolving the parabolic arc $y^2 = x$ from $(0, 0)$ to $(1, 1)$, about the $x$-axis. Give the exact value.
**28.** Find the centroid of the homogeneous region between $y = x^2 - x$ and $y = x - x^3$ on $[0, 1]$.
**29.** An observation porthole on a ship is circular with diameter of 2 feet and is located so its center is 4 feet below the waterline. Find the fluid force on the porthole, assuming that the boat is in seawater (weight density is 64 lb/ft$^3$). Set up the integral only, with the origin at the bottom of the porthole.
**30.** Money is transferred continuously into an account at the constant rate of \$1,200/yr. The account earns interest at the annual rate of 8% compounded continuously. How much money will be in the account at the end of 5 years?

## Supplementary Problems*

*Name and sketch the curves whose equations are given in Problems* 1-8.

**1.** $r = 2\sin\theta - 3\cos\theta$ **2.** $r = -2\theta,\ \theta \geq 0$ **3.** $r = -\csc\theta$ **4.** $r = \sin 3\theta$

**5.** $r = \cos(\theta - \frac{\pi}{3})$ **6.** $r^2 = 3\cos 2\theta$ **7.** $r = \dfrac{4}{1 - \cos\theta}$ **8.** $r = \dfrac{5}{2 + 3\sin\theta}$

*Find the points of intersection of the polar-form curves in Problems* 9-12.

**9.** $\begin{cases} r = 2\cos\theta \\ r = 1 + \cos\theta \end{cases}$ **10.** $\begin{cases} r = 1 + \sin\theta \\ r = 1 + \cos\theta \end{cases}$

**11.** $\begin{cases} r\cos\theta + 2r\sin\theta = 4 \\ r = 2\sec\theta \end{cases}$ **12.** $\begin{cases} r = 2\sin 2\theta \\ r = 1 \text{ for } 0 \leq \theta \leq \frac{\pi}{2} \end{cases}$

*Find the area of the regions bounded by the curves and lines in Problems* 13-21.

**13.** $4y^2 = x$ and $2y = x - 2$ **14.** $y = x^3$, $y = x^4$
**15.** $x = y^{2/3}$, $x = y^2$ **16.** $y = \sqrt{3}\sin x$, $y = \cos x$, $x = 0$, and $x = \frac{\pi}{2}$
**17.** the region bounded by the polar curve $r = 4\cos 2\theta$
**18.** the region inside both the circles $r = 1$ and $r = 2\sin\theta$
**19.** the region inside one loop of the lemniscate $r^2 = \cos 2\theta$
**20.** the region inside one petal of the rose curve $r = \sin 2\theta$
**21.** the region inside the circle $r = 2a\sin\theta$ and outside the circle $r = a$

*Find the volume of a solid of revolution obtained by revolving the region R about the indicated axis in Problems* 22-29.
**22.** $R$ is bounded by $y = \sqrt{x}$, $x = 9$, $y = 0$; about the $x$-axis
**23.** $R$ is bounded by $y = x^2$, $y = 9 - x^2$, and $x = 0$; about the $y$-axis
**24.** $R$ is bounded by $y = x^2$ and $y = x^4$; about the $x$-axis
**25.** $R$ is bounded by $y = \sqrt{\cos x}$, $x = \frac{\pi}{4}$, $x = \frac{\pi}{3}$, and $y = 0$; about the $x$-axis.
**26.** $R$ is bounded by $y = x$, $y = 2x$, $x = 1$; about the line $y = -1$.

*The supplementary probelms are presented in a somewhat random order, not necessarily in order of difficulty.

**27.** $R$ is bounded by $y = x$, $y = 2x$, $x = 1$; around the line $y = 4$.
**28.** $R$ is bounded by $y = x^2$ and $y = \sqrt{x}$; about the $x$-axis by using washers.
**29.** $R$ is bounded by $y = x^2$ and $y = \sqrt{x}$; about the $x$-axis by using shells.

*In Problems 30-31, find the volume of the solid formed when the region described is revolved about the $x$-axis using washers or disks.*

**30.** the region under the curve $y = x^2 + x^3$ on the interval $0 \le x \le \pi$; approximate your answer to the nearest unit
**31.** the region under the curve $y = \sqrt{2 \sin x}$ on the interval $0 \le x \le \pi$
**32.** Find the centroid of a homogeneous lamina covering the region $R$ that is bounded by the parabola $y = 2 - x^2$ and the line $y = x$.
**33.** Estimate the area of the surface obtained by revolving $y = \tan x$ about the $x$-axis on the interval $[0, 1]$.
**34.** The base of a solid is the region in the $xy$-plane that is bounded by the curve $y = 4 - x^2$ and the $x$-axis. Every cross section of the solid perpendicular to the $y$-axis is a rectangle whose height is twice the length of the side that lies in the $xy$ plane. Find the volume of the solid.
**35.** The base of a solid is the region $R$ in the $xy$ plane bounded by the curve $y^2 = x$ and the line $y = x$. Find the volume of the solid if each cross section perpendicular to the $x$-axis is a semicircle with a diameter in the $xy$ plane.
**36.** Use the method of cross sections to show that the volume of a regular tetrahedron of side $a$ is $\frac{1}{12}\sqrt{2}a^3$.
**37.** A force of 100 lb is required to compress a spring from its natural length of 10 in. to a length of 8 in. Find the work required to compress it to a length of 7 in. from its natural length.
**38.** A bowl is formed by revolving the region bounded by the curve $y = 2x^2$ and the lines $x = 0$ and $y = 1$ about the $y$-axis (units are in ft). The bowl is filled with water to a height of 9 in.
**a.** How much work is done in pumping the water to the top of the bowl?
**b.** How much work is done in pumping the water to a point 0.5 ft above the top of the bowl?
**39.** Two solids $S_1$ and $S_2$ have as their base the region in the $xy$-plane bounded by the parabola $y = x^2$ and the line $y = 1$. For solid $S_1$, each cross section perpendicular to the $x$-axis is a square, and for $S_2$, each cross section perpendicular to the $y$-axis is a square. Which solid has the greater volume?
**40.** Use calculus to find the lateral surface area of a right circular cylinder with radius $r$ and height $h$.
**41.** Use calculus to find the surface area of a right circular cone with (top) radius $r$ and height $h$.
**42.** A 20-lb bucket in a well is attached to the end of a 60-ft chain that weighs 16 lb. If the bucket is filled with 50 lb of water, how much work is done in lifting the bucket and chain to the top of the well?
**43.** A swimming pool is 3 ft deep at one end and 8 ft deep at the other, as shown in Figure 6.80. The pool is 30 ft long and 25 ft wide with vertical sides. Set up and analyze a model to find the fluid force against one of the 30-ft sides.
**44.** The ends of a container filled with water have the shape of the region $R$ bounded by the curve $y = x^4$ and the line $y = 16$. Find the fluid force (to two decimal places) exerted by the water on one end of the container.

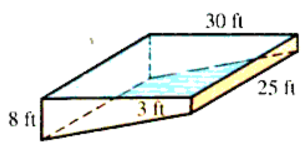

**Figure 6.80** Cross section of a swimming pool

**45.** A conical tank whose vertex points down is 12 ft high with (top) diameter 6 ft. The tank is filled with a fluid of weight density $\delta = 22$ lb/ft$^3$. If the tank is filled to a depth of 6 ft, find the work required to pump all the fluid to a height of 2 ft above the top of the tank.
**46.** Each end of a container filled with water has the shape of an isosceles triangle, as shown in Figure 6.81. Find the force exerted by the water on an end of the container.

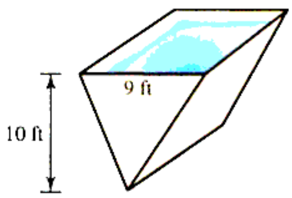

**Figure 6.81** Water pressure in a tank

**47.** A container has the shape of the solid formed by rotating the region bounded by the curve $y = x^3$ and the lines $x = 0$ and $y = 1$ about the $y$-axis, where $x$ and $y$ are measured in feet. If the container is filled with water, how much work is required to pump all the water out of the container?

**48.** A gate in a water main has the shape of a circle with diameter 10 ft. If water stands 2 ft high in the main, what is the force exerted by the water on the gate?

*Suppose a particle of mass $m$ moves along a number line under the influence of a variable force $F$ which always acts in a direction parallel to the line of motion. At a given time the particle is said to have* ***kinetic energy***

$$K = \frac{1}{2}mv^2$$

*where $v$ is the particle's velocity, and it has* ***potential energy***

$$P = -\int_{s_0}^{s} F\,dx$$

*relative to the point $s$ on the line of motion. Use these definitions in Problems* 49-50.

**49.** A spring at rest has potential energy $P = 0$. If the spring constant is 30 lb/ft, how far must the spring be stretched so that its potential energy is 20 ft-lb?

**50.** A particle with mass $m$ falls freely near the earth's surface. At time $t = 0$, the particle is $s_0$ units above the ground and is falling with velocity $v_0$.

**a.** Find the potential energy $P$ of the particle in terms of position, $s$, if the potential energy is 0 when $s = s_0$.

**b.** Find the potential energy $P$ and the kinetic energy $K$ of the particle in terms of time $t$.

**51.** An object weighs $w = 1{,}000r^{-2}$ grams, where $r$ is the object's distance from the center of the earth. What work is required to lift the object from the surface of the earth ($r = 6{,}400$ km) to a point 3,600 km above the surface of the earth?

**52. Modeling Problem** A certain species has population $P(t)$ at time $t$ whose growth rate is affected by seasonal variations in the food supply. Suppose the population may be modeled by the differential equation

$$\frac{dP}{dt} = k(P - A)\cos t$$

where $k$ and $A$ are physical constants. Solve the differential equation and express $P(t)$ in terms of $k, A$, and $P_0$, the initial population.

**53.** Find the center of mass of a thin plate of constant density if the plate occupies the region bounded by the lines $2y = x, x + 3y = 5$, and the $x$-axis.

**54.** The portion of the ellipse (Figure 6.82)

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

that lies in the first quadrant between $x = r$ and $x = a$, for $0 < r < a$, is revolved about the $x$-axis to form a solid $S$. Find the volume of $S$.

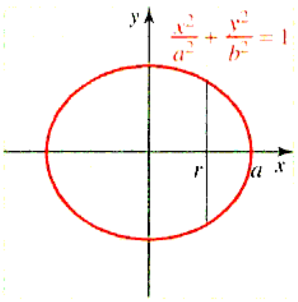


**Figure 6.82** Ellipse

**55.** Find the volume of the football-shaped solid (called an *ellipsoid*) formed by revolving the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

about the $x$-axis. What is the volume if the ellipse is revolved about the $y$-axis?

**56.** When viewed from above, a swimming pool has the shape of the ellipse

$$\frac{x^2}{900} + \frac{y^2}{400} = 1$$

The cross sections of the pool perpendicular to the ground and parallel to the $y$-axis are squares. If the units are in feet, what is the volume of the pool?

**57.** The portion of the hyperbola (Figure 6.83)

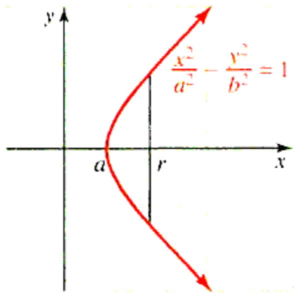


**Figure 6.83** Hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

that lies in the first quadrant between $x = a$ and $x = r$, for $0 < a < r$, is revolved about the $x$-axis to form a solid $S$. Find the volume of $S$.

**58.** The base of the solid $S$ is the hyperbola

$$\frac{x^2}{4} - \frac{y^2}{9} = 1$$

for $2 \leq x \leq 5$, and the cross sections perpendicular to the $x$-axis are squares. Find the volume of $S$.

**59.** What is the present value of an investment that will generate income continuously at a constant rate of \$1,000/yr for 10 years if the prevailing annual interest rate remains fixed at 7% compounded continuously?

**60.** In a certain community the fraction of the homes placed on the market that remain unsold for at least $t$ weeks is approximately $f(t) = e^{-0.2t}$. If 200 homes are currently on the market and if additional homes are placed on the market at the rate of 8 per week, approximately how many homes will be on the market 10 weeks from now?

**61.** The population density $r$ miles from the center of a certain city is $D(r) = 5{,}000e^{-0.02r^2}$ people/mi$^2$. How many people live within 5 miles of the center of the city?

**62.** A certain oil well that yields 900 barrels of crude oil per month will run dry in 3 years. The price of crude oil is currently \$110/barrel and is expected to rise at the constant rate of \$1 per barrel per month. If the oil is sold as soon as it is extracted from the ground, what will be the total future revenue from the well?

**63.** Suppose that the consumer's demand function for a certain commodity is $D(q) = 50 - 3q - q^2$ dollars per unit.

**a.** Find the number of units that will be bought if the market price is \$32/unit.

**b.** Compute the consumer's surplus when the market price is \$32/unit.

**c.** Graph the demand curve and interpret the consumer's surplus as an area in relation to this curve.

**64.** Let $R$ be the part of the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

that lies in the first quadrant. Use the theorem of Pappus to find the coordinates of the centroid of $R$. You may use the facts that $R$ has area $A = \frac{1}{4}\pi ab$, and the semiellipsoid it generates when it is revolved about the $x$-axis has volume $V = \frac{2}{3}\pi ab^2$.

**65. Modeling Problem** A container of a certain fluid is raised from the ground on a pulley. While resting on the ground, the container and fluid together weigh 200 lb. However, as the container is raised, fluid leaks out in such a way that at height $x$ feet above the ground, the total weight of the container and fluid is $200 - 0.5x$ lb. Set up and analyze a model to find the work done in raising the container 100 ft.

**66.** A bag of sand is being lifted vertically upward at the rate of 31 ft/min. If the bag originally weighed 40 lb and leaks at the rate of 0.2 lb/s, how much work is done in raising it until it is empty?

**67.** The vertical end of a trough is an equilateral triangle, 3 ft on a side. Assuming the trough is filled with a fluid of weight density $\delta = 40$ lb/ft$^3$, find the force on the end of the trough.

**68.** The radius of a circular water main is 2 ft. Assuming the main is half full of water, find the total force exerted by the water on a gate that crosses the main at one end (see Figure 6.84).

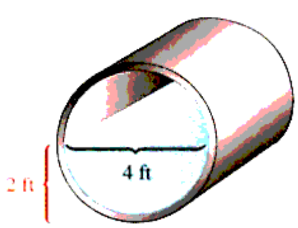


**Figure 6.84** Circular water main

**69.** The force acting on an object moving along a straight line is known to be a linear function of its position. Find the force function $F$ if it is known that 13 ft-lb of work is required to move the object 6 ft and 44 ft-lb of work is required to move it 12 ft.

**70.** A tank has the shape of a rectangular parallelepiped with length 10 ft, width 6 ft, and height 8 ft. The tank contains a liquid of weight density $\delta = 20$ lb/ft$^3$. Find the level of the liquid in the tank if it will require 2,400 ft-lb of work to move all the liquid in the tank to the top.

**71.** A tank is constructed by placing a right circular cylinder of height 10 ft and radius 2 ft on top of a hemisphere with the same radius (see Figure 6.85).

If the tank is filled with a fluid of weight density $\delta = 24$ lb/ft$^3$, how much work is done in bringing all the fluid to the top of the tank?

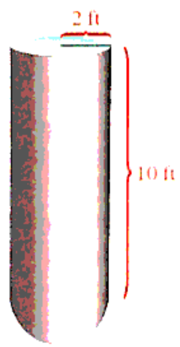


**Figure 6.85** Cylindrical tank

**72. Modeling Problem** A piston in a cylinder causes a gas either to expand or to contract. Assume the pressure $P$ and the volume of gas $V$ in the cylinder satisfy the adiabatic law (no exchange of heat)

$$P(V^{1.4}) = C$$

where $C$ is a constant.

**a.** If the gas goes from volume $V_1$ to volume $V_2$, show that the work done may be modeled by

$$W = \int_{V_1}^{V_2} CV^{-1.4}\,dV$$

**b.** How much work is done if 0.6 m$^3$ of steam at a pressure of 2,500 Newtons/m$^2$ expands 50%?

**73.** A thin sheet of tin is in the shape of a square, 16 cm on a side. A rectangular corner of area 156 $cm^2$ is cut from the square, and the centroid of the portion that remains is found to be 4.88 cm from one side of the original square, as shown in Figure 6.86. How far is it from the other three sides?

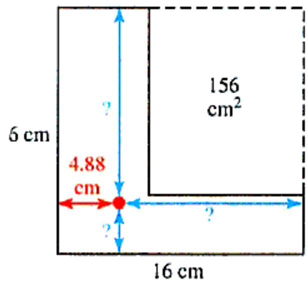


**Figure 6.86** Centroid of a region

**74.** Let $R$ be the region bounded by the semicircle $y = \sqrt{4 - x^2}$ and the $x$-axis, and let $S$ be the solid with base $R$ and trapezoidal cross sections perpendicular to the $x$-axis. Assume that the two slant sides and the shorter parallel side are all half the length that lies in the base, as shown in Figure 6.87. Set up and evaluate an integral for the volume of $S$.

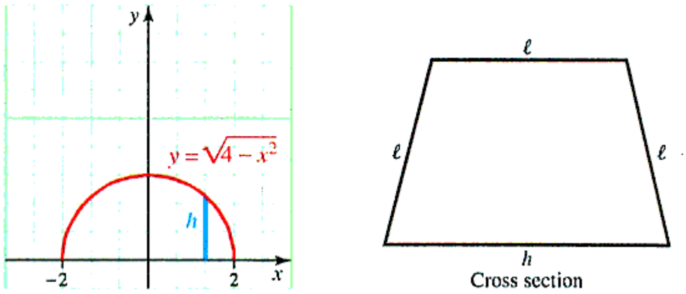


**Figure 6.87** Volume of a region $S$

**75.** A child is building a sand structure at the beach.

**a.** Find the work done if the weight density of the sand is $\delta = 140$ lb/$ft^3$ and the structure is a cone of height 1 ft and diameter 2 ft.

**b.** What if the structure is a tower (cylinder) of height 1 ft and radius 4 in.

**76.** Let $S$ be the solid generated by revolving about the $x$-axis the region bounded by the parabola $y = x^2$ and the line $y = x$. Find the number $A$ so that the plane perpendicular to the $x$-axis at $x = A$ divides $S$ in half. That is, the volume on one side of $A$ equals the volume on the other side.

**77.** A plate in the shape of an isosceles triangle is submerged vertically in water, as shown in Figure 6.88.

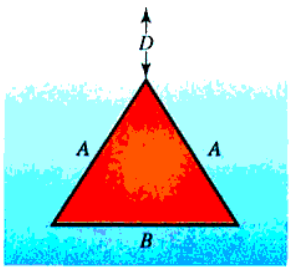


**Figure 6.88** Hydrostatic force

**a.** Find the force on one side of the plate if the two equal sides have length 3, the third side has length 5, and the top vertex of the triangle is 4 units below the surface.

**b.** Find the fluid force for the general case where the triangle has equal sides of length $A$, the third side has length $B$, and the top vertex is $D$ units below the surface.

**78.** Find the arc length of the curve

$$y = Ax^3 + \frac{B}{x}$$

on the interval $[a, b]$, where $A$ and $B$ are positive constants with $AB = \frac{1}{12}$.

**79.** **Think Tank Problem** When the line segment $y = x - 2$ between $x = 1$ and $x = 3$ is revolved about the $x$-axis, it generates part of a cone. The formula obtained in Section 6.4 says that the surface area of the cone is

$$\int_1^3 2\pi(x-2)\sqrt{2}\,dx$$

but this integral is equal to 0. What is wrong? What is the actual surface area?

**80.** *Historical Quest* *Emmy Noether has been called the greatest woman mathematician in the history of mathematics. Once again, we see a woman overcoming great obstacles in achieving her education. At the time she received her Ph.D. at the University of Erlangen (in 1907), the Academic Senate declared that the admission of women students would "overthrow all academic order." Emmy Noether is famous for her work in physics and mathematics. In mathematics, she opened up entire areas for study in the theory of ideals, and in 1921 she published a paper on rings, which have since been called Noetherian rings (rings and ideals are objects of study in modern algebra). She was not only a renowned mathematician but also an excellent teacher. It has been reported that she often lectured by "working it out as she went." Nevertheless, her students flocked around her "as if following the Pied Piper." In the fall of 1934, pressure from the Nazi regime forced her to immigrate to Bryn Mawr College in Pennsylvania. In the years that followed, there were mass dismissals of "racially undesirable" professors, and many noted European Jewish mathematicians immigrated to American universities.* For this *Quest* compile a list of notable mathematicians who immigrated to the United States from 1933 to 1940.


Karl Smith library

**Amalie ("Emmy") Noether (1882–1935)**

**81.** **Modeling Problem** An object below the surface of the earth is attracted by a gravitational force that is directly proportional to the square of the distance from the object to the center of the earth. Set up and analyze a model to find the work done in lifting an object weighing $P$ lb from a depth of $s$ feet (below the surface) to the surface. (Assume the earth's radius is 4,000 mi. and that $P$ is constant between $s$ and the surface.)

*In Problems 82-83, consider a continuous curve $y = f(x)$ that passes through* $(-1,4)$, $(0,3)$, $(1,4)$, $(2,6)$, $(3,5)$, $(4,6)$, $(5,7)$, $(6,6)$, $(7,4)$, $(8,3)$, *and* $(9,4)$.

**82.** Use Simpson's rule to estimate the volume of the solid formed by revolving this curve about the $x$-axis for $[-1, 9]$.

**83.** Use the trapezoidal rule to estimate the volume of the solid formed by revolving this curve about the line $y = -1$ on $[-1, 9]$.

**84.** **Modeling Problem** In this computational problem we revisit a problem similar to the Wine Barrel Capacity group research project at the end of Chapter 4, but add a couple of complications. Imagine a cylindrical fuel tank lying on its side; its ends are elliptically shaped and are defined by the equation

$$\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 = 1$$

We are concerned with the amount of fuel in the tank for a given level.

**a.** Explain why the area of an end of the tank can be modeled by

$$A = 2a \int_{-b}^{b} \sqrt{1 - \left(\frac{y}{b}\right)^2}\, dy$$

**b.** By making the right substitution in the integral in part **a**, we can obtain an integral representing half the area inside the circle (with unit radius). Do this and show that the area inside the above ellipse is $\pi ab$.

**c.** Explain why the volume of the fuel at level $h$ $(-b \le h \le b)$ may be modeled by

$$V(h) = 2La \int_{-b}^{h} \sqrt{1 - \left(\frac{y}{b}\right)^2}\, dy$$

**d.** For $a = 5$, $b = 4$, and $L = 20$, numerically compute $V(h)$ for $h = -3, -2, \cdots, 3, 4$. *Note:* $V(0)$ and $V(4)$ will serve as a check on your work.

**85.** We think of the last part of Problem 84 as the "direct" problem: Given a value of $h$, compute $V(h)$ by a formula. Now we turn to the "inverse" problem: For a set value of $V$, say $V_0$, find $h$ such that $V(h) = V_0$. Typically, the inverse problem is more difficult than the direct problem and requires some sort of iterative method to approximate the solution (for example, Newton-Raphson's method). For example, if $V = 500$ ft$^3$, we could define $f(h) = V(h) - 500$ and seek the relevant zero of $f$. Set up the necessary computer program to do this and find the $h$ values, to two decimal place accuracy, corresponding to these values of $V$:

**a.** 500 **b.** 800

*Hint:* A key to this problem is thinking about $V'(h)$, where $V(h)$ is expressed in Problem 84**c**. Also, for starting values $h_0$, refer to your work in Problem 84**d**.

**86.** Find the centroid of a right triangle whose legs have lengths $a$ and $b$. *Hint*: Use the volume theorem of Pappus.

**87.** The largest of the pyramids at Giza in Egypt (see Problem 49, p. 445) is roughly 480 ft high and has a square base 750 ft on a side. How much work was required to lift the stones to build this pyramid if we assume that the rock in the pyramid had an average weight density of 160 lb/ft$^3$?

**88.** Find the length of the curve $y = e^x$ on the interval $[0, 1]$.

**89.** Find the volume of the solid generated when the region under the curve

$$y = \frac{1}{\sqrt{9 - x^2}}$$

on the interval $[0, 2]$ is revolved about the $y$-axis.

**90.** Find the volume of the solid formed by revolving about the $x$-axis the region bounded by the curve

$$y = \frac{2}{\sqrt{3x - 2}}$$

the $x$-axis, and the lines $x = 1$ and $x = 2$.

**91.** Find the area between the curves $y = \tan^2 x$ and $y = \sec^2 x$ on the interval $\left[0, \frac{\pi}{4}\right]$.

**92.** Find the surface area of the solid generated by revolving the region bounded by the $x$-axis and the curve

$$y = e^x + \frac{1}{4}e^{-x}$$

on $[0, 1]$ about the $x$-axis.

**93.** Find the volume of the solid whose base is the region $R$ bounded by the curve $y = e^x$ and the lines $y = 0$, $x = 0$, $x = 1$, if cross sections perpendicular to the $x$-axis are equilateral triangles.

**94.** Find the length of the curve

$$x = \frac{ay^2}{b^2} - \frac{b^2}{8a} \ln \frac{y}{b}$$

between $y = b$ and $y = 2b$.

**95. Putnam Examination Problem** Find the length of the curve $y^2 = x^3$ from the origin to the point where the tangent makes an angle of 45° with the $x$-axis.

**96. Putnam Examination Problem** A solid is bounded by two bases in the horizontal planes $z = \frac{h}{2}$ and $z = -\frac{h}{2}$, and by a surface with the property that every horizontal cross section has area given by an expression of the form

$$A = a_0z^3 + a_1z^2 + a_2z + a_3$$

**a.** Show that the solid has volume $V = \frac{1}{6}h(B_1 + B_2 + 4M)$, where $B_1$ and $B_2$ are the area of the bases and $M$ is the area of the middle horizontal cross section at $z = 0$.

**b.** Show that the formulas for the volume of a cone and a sphere can be obtained from the formula in part **a** for certain special choices of $a_0$, $a_1$, $a_2$, and $a_3$.

**97. Putnam Examination Problem** The horizontal line $y = c$ intersects the curve $y = 2x - 3x^3$ in the first quadrant as shown in Figure 6.89. Find $c$ so that the areas of the two shaded regions are equal.

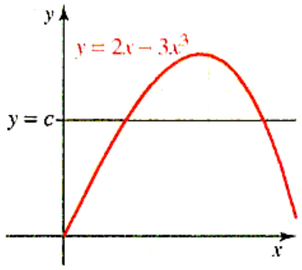


**Figure 6.89** Putnam Problem 97

**98. Putnam Examination Problem** Find the volume of the region of points $(x, y, z)$ such that $(x^2 + y^2 + z^2 + 8)^2 \leq 36(x^2 + y^2)$. *Hint*: Let $r = \sqrt{x^2 + y^2}$ to generate a torus $(r - 3)^2 + z^2 \leq 1$, the area being rotated is the disk $(x - 3)^2 + z^2 \leq 1$ in the $xz$-plane. Use the volume theorem of Pappus.

**99. Book Report** Eli Maor, a native of Israel, has a long-standing interest in the relations between mathematics and the arts. Read the fascinating book *To Infinity and Beyond, A Cultural History of the Infinite* (Boston: Birkhäuser, 1987), and prepare a book report.

# CHAPTER 6 GROUP RESEARCH PROJECT*

*Working in small groups is typical of most work environments, and this book seeks to develop skills with group activities. We present a group project at the end of each chapter. These projects are to be done in groups of three or four students.*

Harry Houdini (born Ehrich Weiss) was a famous escape artist. In this project we relive a trick of his that challenged his mathematical prowess, as well as his skill and bravery. It may challenge these qualities in you as well.

Houdini had his feet shackled to the top of a concrete block which was placed on the bottom of a giant laboratory flask. The cross-sectional radius of the flask, measured in feet, was given as a function of height $z$ from the ground by the formula $r(z) = 10z^{-1/2}$, with the bottom of the flask at $z = 1$ ft. The flask was then filled with water at a steady rate of $22\pi$ ft$^3$/min. Houdini's job was to escape the shackles before he was drowned by the rising water in the flask.

Now Houdini knew it would take him exactly 10 minutes to escape the shackles. For dramatic impact, he wanted to time his escape so it was completed precisely at the moment the water level reached the top of his head. Houdini was exactly 6 ft tall. In the design of the apparatus, he was allowed to specify only one thing: the height of the concrete block he stood on.

**Houdini's Escape**



Harry Houdini (1874–1926)

**Extended paper for further study**. Your paper is not limited to the following questions but should include these concerns: How high should the block be? (You can neglect Houdini's volume and the volume of the block.) How fast is the water level changing when the flask first starts to fill? How fast is the water level changing at the instant when the water reaches the top of his head? You might also help Houdini with any size flask by generalizing the derivation: Consider a flask with cross-sectional radius $r(z)$, an arbitrary function of $z$ with a constant inflow rate of $\frac{dV}{dt} = A$. Can you find $\frac{dh}{dt}$ as a function of $h(t)$?

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*MAA Notes* 17 (1991), "Priming the Calculus Pump: Innovations and Resources," by Marcus S. Cohen, Edward D. Gaughan, R. Arthur Knoebel, Douglas S. Kurtz, and David J. Pengelley.